Math2221
Higher Theory and Applications of Differential Equations

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Contents

Part I

Linear ODEs

Introduction

In first year, you studied second-order linear ODEs with constant coefficients. We will see that the techniques you learned can be extended to handle higher-order linear ODEs with variable coefficients. A key idea, used repeatedly throughout the remainder of the course, is linear superposition.

Let's recall some basic facts that will be helpful to know in what follows.

We begin with a brief review of methods for first-order equations. Recall that a first-order ODE has, in its most general realisation, the form

$$y'(t) = f\big(t, y(t)\big).$$

A special case is the equation

$$a(t)y'(t) + b(t)y(t) = f(t),$$

with $a(t) \neq 0$ on some interval $I \subseteq \mathbb{R}$. This special first-order ODE is called a **linear first-order ODE**. Another special case is

$$y'(t) = f(t)g(y),$$

which is known as a **separable first-order ODE**.

For a separable equation the solution is found (at least, implicitly) by writing

$$\int \frac{1}{g(y)}\ dy = \int f(t)\ dt.$$

This, admittedly bizarre looking computation, is justified by supposing that F and G are, respectively, antiderivatives of f and g, and then observing that if we have the implicit relation

$$G\big(y(t)\big) = F(t) + C,$$

then upon differentiating both sides of this relation we obtain

$$G'\big(y(t)\big)y'(t) = F'(t)$$

so that

$$\frac{1}{g\big(y(t)\big)}y'(t) = f(t)$$

so that

$$y'(t) = f(t)g\big(y(t)\big) \equiv f(t)g(y).$$

Thus, we conclude that whenever $y' = f(t)g(y)$, it follows that $G(y) = F(t) + C$ is an implicit family of solutions.

Example

Consider $y' = t^2 y$, $y(0) = 3$. This is separable with $f(t) = t^2$ and $g(y) = y$. Then

$$\int \frac{1}{y}\, dy = \int t^2\, dt$$

so that

$$\ln \big|y(t)\big| = \frac{1}{3}t^3 + C.$$

Now apply e^t to both sides to obtain

$$\big|y(t)\big| = e^{\frac{1}{3}t^3 + C} = e^C e^{\frac{1}{3}t^3}.$$

Finally, it is customary to replace e^C by A (taking $A = -e^C$ if necessary). Thus, a general solution of the equation is

$$y(t) = Ae^{\frac{1}{3}t^3}.$$

Since $y(0) = 3$, we see that the unique (?) solution is $y(t) = 3e^{\frac{1}{3}t^3}$.

In the case of a linear first-order equation, i.e. $y' + a(t)y = f(t)$, a useful solution method is the integrating factor technique; we'll use it later. The idea is to find a function μ so that when we multiple both sides of the equation $y' + a(t)y = f(t)$ with μ we find that

$$\big[\mu y\big](t)' = \mu(t)\big(y' + a(t)y\big) = \mu(t)f(t),$$

for if this happens, then the general solution of the ODE should be

$$y(t) = \frac{1}{\mu(t)} \int \mu(t)f(t)\ dt + \frac{C}{\mu(t)}.$$

But what should $\mu(t)$ equal?

Begin with what we want to occur – i.e., we want it to be the case that

$$\big[\mu y\big](t)' = \mu(t)\big(y' + a(t)y\big).$$

So, expanding the left-hand side (by means of the product rule) we obtain

$$\mu'(t)y(t) + \mu(t)y'(t) = \mu(t)y'(t) + \mu(t)a(t)y(t).$$

Thus,

$$\mu'(t)y(t) = \mu(t)a(t)y(t).$$

Divide out the $y(t)$ to get

$$\left(\frac{\mu'}{\mu}\right)(t) = a(t).$$

But then

$$\big(\ln|\mu(t)|\big)' = a(t)$$

so that

$$\mu(t) = e^{\int a(t)\ dt}.$$

And this is how μ must be chosen in order for the desired reduction to occur!

Here's an example.

Example

Let's solve the problem $y' - 2ty = 3t$. (We won't worry about an initial condition.) We pick

$$\mu(t) = e^{\int -2t \ dt} = e^{-t^2}.$$

Then

$$\left(e^{-t^2} y\right)' = 3te^{-t^2}$$

so that

$$e^{-t^2} y = \int 3te^{-t^2} \ dt = -\frac{3}{2} e^{-t^2} + C$$

so that

$$y(t) = -\frac{3}{2} + Ce^{t^2}.$$

Here's a little curiosity to observe. The function $y_h(t) = Ce^{t^2}$ is a general solution of the problem $y' - 2ty = 0$, whereas the function $y_p(t) \equiv -\frac{3}{2}$ is one particular solution of the problem $y' - 2ty = 3t$. A coincidence?

A simple application of this method is to the **annuity problem**. Suppose you deposit d dollars per year in an account earning $r\%$ interest compounded continuously. Then the amount of money, $A(t)$, you have at time t can be modeled by

$$A' - rA = d.$$

The integrating factor for this problem is

$$\mu(t) = e^{-rt}$$

so that the general solution of the problem is found by writing

$$e^{-rt}A(t) = \int de^{-rt}\ dt = -\frac{d}{r}e^{-rt} + C$$

so that

$$A(t) = Ce^{rt} - \frac{d}{r}.$$

Example

Consider the second-order constant coefficient linear homogeneous ODE as follows.

$$ay'' + by' + cy = 0.$$

A suitable *ansatz* is (for a constant r to be determined)

$$y(t) = Me^{rt}.$$

Indeed, since $y'(t) = Mre^{rt}$ and $y''(t) = Mr^2e^{rt}$ we see that this function will solve the ODE provided that

$$a \cdot Mr^2e^{rt} + b \cdot Mre^{rt} + c \cdot Me^{rt} = 0.$$

Factoring the left-hand side yields

$$Me^{rt}\left(ar^2 + br + c\right) = 0.$$

Example

So, this means that if

$$ar^2 + br + c = 0,$$

then the function

$$y(t) = Me^{rt}$$

is a solution of

$$ay'' + by' + cy = 0.$$

Note this is true for any $M \in \mathbb{R}$.

A very important realization, which we'll expand on further in what follows, is that if y_1 and y_2 are any two solutions of the problem

$$ay'' + by' + cy = 0,$$

then $y(t) := c_1 y_1(t) + c_2 y_2(t)$ is a solution for any c_1, $c_2 \in \mathbb{R}$. To see that this is true simply notice that

$$\begin{aligned} ay'' + by' + cy &= a\big(c_1 y_1'' + c_2 y_2''\big) + b\big(c_1 y_1' + c_2 y_2'\big) + c\big(c_1 y_1 + c_2 y_2\big) \\ &= c_1 \underbrace{\big(ay_1'' + by_1' + cy_1\big)}_{=0} + c_2 \underbrace{\big(ay_1'' + by_1' + cy_1\big)}_{=0} \\ &= 0. \end{aligned}$$

This is called the **superposition principle**. We'll explore this extensively in what follows.

Linear differential operators

With that brief review dispatched let's begin looking at things in more generality.

In linear algebra, you have seen the compact notation $A\boldsymbol{x} = \boldsymbol{b}$ for a system of linear equations. A similar notation when dealing with a linear ordinary differential equations is

$$Lu = f.$$

Here, L is an operator (or transformation) that acts on a function u to create a new function Lu.

Notation

Given coefficients $a_0(x)$, $a_1(x)$, ..., $a_m(x)$ we define the linear differential operator L of order m,

$$\begin{aligned} Lu(x) &= \sum_{j=0}^{m} a_j(x) D^j u(x) \\ &= a_m D^m u + a_{m-1} D^{m-1} u + \cdots + a_0 u, \end{aligned} \tag{1}$$

where $D^j u = d^j u/dx^j$ (with $D^0 u = u$).

We refer to a_m as the leading coefficient of L. For simplicity, we assume that each $a_j(x)$ is a smooth function of x.

The ODE $Lu = f$ is said to be singular with respect to an interval $[a, b]$ if the leading coefficient $a_m(x)$ vanishes for any $x \in [a, b]$. (Also say L is singular on $[a, b]$.)

Linearity

An operator of the form (1) is indeed linear: for any constants c_1 and c_2 and any (m-times differentiable) functions u_1 and u_2,

$$L(c_1u_1 + c_2u_2) = c_1Lu_1 + c_2Lu_2.$$

Example

$Lu = (x-3)u''' - (1+\cos x)u' + 6u$ is a linear differential operator of order 3, with leading coefficient $x-3$. Thus, L is singular on $[1, 4]$, but not singular on $[0, 2]$.

Example

$N(u) = u'' + u^2u' - u$ is a *non*linear differential operator of order 2.

Homogeneous or forced?

Ordinary differential equations of the form

$$Lu = 0$$

are known as homogeneous. Those of the form

$$Lu = f$$

are known as inhomogeneous or non-homogeneous. In physical systems the inhomogeneity is often described as a forcing term.

Example

Equation for an oscillating mass (m) subject to an external force (f):

$$m\,x'' + r\,x' + k\,x = f(t),$$

where x is position, rx' is a frictional force and kx is a restoring force.

Homogeneous solutions form a vector space

Let u_1, u_2,...u_k be the solutions to the linear homogeneous differential equation ($Lu = 0$). Because L is linear, the linear combination

$$u = c_1 u_1 + c_2 u_2 + ... + c_k u_k$$

is also a solution (linear superposition).

Hence, the set of solutions forms a vector space with the trivial solution ($u = 0$) its zero vector.

Initial-value and boundary-value problems

When the solution to a differential equation (linear or non-linear, homogeneous or not) is prescribed at a particular point $x = x_0$, that is

$$u(x_0) = \nu_0, \quad u'(x_0) = \nu_1, \quad \ldots, \quad u^{(m-1)}(x_0) = \nu_{m-1},$$

we call it an initial value problem. Why initial? Because typically x is time and $x_0 = 0$.

Where a differential equation is order 2 or greater, solutions at 2 or more locations can be prescribed. Such problems are called boundary value problems.

Unique solution to the linear initial problem

Consider a general mth-order linear differential operator

$$Lu = \sum_{j=0}^{m} a_j(x) D^j u.$$

Given $f(x)$ and m initial values ν_0, ν_1, ..., ν_{m-1} we seek $u = u(x)$ satisfying

$$Lu = f \quad \text{on } [a, b], \tag{2}$$

with

$$u(a) = \nu_0, \quad u'(a) = \nu_1, \quad \ldots, \quad u^{(m-1)}(a) = \nu_{m-1}. \tag{3}$$

Theorem

For an ODE $Lu = f$ which is not singular with respect to $[a, b]$, with f continuous on $[a, b]$, the IVP (2) and (3) has a unique solution.

See discussion of uniqueness in Technical Proofs.

Unique solution: special case

Consider the linear inhomogeneous initial value problem

$$a_2u'' + a_1u' + a_0u = f(x)$$
$$u(0) = \nu_0 \;\; ; \;\; u'(0) = \nu_1$$

with a_2, a_1 and a_0 positive constants and f continuous for all x.

Assume two solutions u_a and u_b. Let $U = u_a - u_b$. Hence

$$U(0) = u_a(0) - u_b(0) = \nu_0 - \nu_0 = 0,$$
$$U'(0) = u_a'(0) - u_b'(0) = \nu_1 - \nu_1 = 0$$
$$and$$
$$a_2U'' + a_1U' + a_0U = f(x) - f(x) = 0.$$

Unique solution: special case

Multiplying the homogeneous equation by U' and integrating over $0 \leq x \leq t$,

$$a_2 \int_0^t U''U'dx + a_1 \int_0^t (U')^2 dx + a_0 \int_0^t UU'dx = 0,$$

$$\frac{a_2}{2}[U'(t)^2 - U'(0)^2] + a_1 \int_0^t (U')^2 dx + \frac{a_0}{2}[U(t)^2 - U(0)^2] = 0.$$

Above, $U'(0) = 0 = U(0)$ and since each of the remaining terms is ≥ 0 we find that $U = 0$ is the only permitted solution. Therefore the two solutions to the inhomogeneous problem, u_a and u_b, are identical.

Solution to mth order problem has dimension m

Theorem

Assume that the linear, mth-order differential operator L is not singular on $[a, b]$. Then the set of all solutions to the homogeneous equation $Lu = 0$ on $[a, b]$ is a vector space of dimension m.

Proof.

Let $V = \{\, u : Lu = 0 \text{ on } [a, b] \,\}$ and define the linear transformation $\Theta : V \to \mathbb{R}^m$ by

$$\Theta u = [u(a), u'(a), \ldots, u^{(m-1)}(a)]^T.$$

Uniqueness of solutions means that Θ is one-to-one, and existence means that Θ is onto. Hence, Θ is an isomorphism, and therefore the vector space V has dimension m. □

General solution

If $\{u_1, u_2, \ldots, u_m\}$ is any basis for the solution space of $Lu = 0$, then every solution can be written in a unique way as

$$u(x) = c_1 u_1(x) + c_2 u_2(x) + \cdots + c_m u_m(x) \quad \text{for } a \le x \le b. \tag{4}$$

We refer to (4) as the general solution of the homogeneous equation $Lu = 0$ on $[a, b]$.

Linear superposition refers to this technique of constructing a new solution out of a linear combination of old ones. Of course, this trick works only because L is linear.

Example

The general solution to $u'' - u' - 2u = 0$ is

$$u(x) = c_1 e^{-x} + c_2 e^{2x}.$$

Inhomogeneous problem

Consider the *in*homogeneous equation $Lu = f$ on $[a, b]$, and fix a particular solution u_{P}.

For *any* solution u, the difference $u - u_{\mathrm{P}}$ is a solution of the *homogeneous* equation because

$$L(u - u_{\mathrm{P}}) = Lu - Lu_{\mathrm{P}} = f - f = 0 \quad \text{on } [a, b].$$

Hence, $u(x) - u_{\mathrm{P}}(x) = c_1 u_1(x) + \cdots + c_m u_m(x)$ for some constants c_1, . . . , c_m, and so

$$u(x) = u_{\mathrm{P}}(x) + \underbrace{c_1 u_1(x) + \cdots + c_m u_m(x)}_{u_{\mathrm{H}}(x)}, \qquad a \leq x \leq b,$$

is the general solution of the *in*homogeneous equation $Lu = f$.

Inhomogeneous problem

Example

The inhomogeneous ODE

$$u'' - u' - 2u = -2e^x$$

has the particular solution

$$u_{\mathrm{P}}(x) = e^x.$$

The general solution for its homogeneous counterpart is

$$u_{\mathrm{H}}(x) = c_1 e^{-x} + c_2 e^{2x}.$$

So the general solution of the inhomogeneous ODE is

$$u(x) = u_{\mathrm{P}}(x) + u_{\mathrm{H}}(x) = e^x + c_1 e^{-x} + c_2 e^{2x}.$$

Reduction of order

Theorem

For $u = u_1(x) \neq 0$, a solution to the ODE

$$u'' + p(x)u' + q(x)u = 0,$$

on some interval I, then a second solution is

$$u = u_1(x) \int \frac{1}{u_1^2 \exp(\int p\,dx)}\,dx$$

Substitute $u = u_1(x)v(x)$ into the ODE and rearrange to obtain

$$\underbrace{(u_1'' + pu_1' + qu_1)}_{=0}v + u_1 v'' + (2u_1' + pu_1)v' = 0.$$

This is just a *first*-order, linear ODE for the derivative of the unknown factor $v(x)$: put $w = v'$ then

$$u_1 w' + (2u_1' + pu_1)w = 0.$$

Reduction of order (continued)

Writing the ODE for w in the standard form

$$w' + (2u_1' u_1^{-1} + p)w = 0,$$

and noting that this equation is first-order linear homogeneous, we seek an integrating factor, namely
$A(x) = \exp\bigl(\int (2u_1' u_1^{-1} + p)\, dx\bigr) = u_1^2 \exp(\int p\, dx)$, so that

$$\frac{d}{dx}(Aw) = Aw' + A'w = A\bigl(w' + (2u_1' u_1^{-1} + p)w\bigr) = 0.$$

Then $Aw = C$ for some constant C, and so

$$v = \int \frac{C}{A(x)}\, dx.$$

Example

For the ODE $u'' - 6u' + 9u = 0$, take $u_1 = e^{3x}$ and find v.

Differential operators with constant coefficients

If L has constant coefficients, then the problem of solving $Lu = 0$ reduces to that of factorizing the polynomial having the same coefficients. Some complications occur if the polynomial has any repeated roots.

Characteristic polynomial

Suppose that a_j is constant for $0 \le j \le m$, with $a_m \ne 0$. We define the associated polynomial of degree m,

$$p(z) = \sum_{j=0}^{m} a_j z^j = a_m z^m + a_{m-1} z^{m-1} + \cdots + a_1 z + a_0,$$

so that, formally, $L = p(D)$.

Since $D^j e^{\lambda x} = \lambda^j e^{\lambda x}$ we have

$$p(D)e^{\lambda x} = \big(a_m \lambda^m + a_{m-1}\lambda^{m-1} + \cdots + a_1 \lambda + a_0\big)e^{\lambda x} = p(\lambda)e^{\lambda x},$$

and so

$$p(D)e^{\lambda x} = 0 \iff p(\lambda) = 0.$$

Factorization

By the fundamental theorem of algebra,

$$p(z) = a_m(z-\lambda_1)^{k_1}(z-\lambda_2)^{k_2}\cdots(z-\lambda_r)^{k_r}$$

where λ_1, λ_2, ..., λ_r are the distinct roots of p, with corresponding multiplicities k_1, k_2, ..., k_r satisfying

$$k_1 + k_2 + \cdots + k_r = m.$$

Lemma

$(D-\lambda)x^j e^{\lambda x} = jx^{j-1}e^{\lambda x}$ *for* $j \geq 0$.

Lemma

$(D-\lambda)^k x^j e^{\lambda x} = 0$ *for* $j = 0, 1, \ldots, k-1$.

Proof

An elementary calculation gives

$$(D-\lambda)x^j e^{\lambda x} = jx^{j-1}e^{\lambda x} + x^j\lambda e^{\lambda x} - \lambda x^j e^{\lambda x} = jx^{j-1}e^{\lambda x},$$

as claimed, and then

$$\begin{aligned}
(D-\lambda)^2 x^j e^{\lambda x} &= (D-\lambda)jx^{j-1}e^{\lambda x} = j(j-1)x^{j-2}e^{\lambda x},\\
&\vdots\\
(D-\lambda)^j x^j e^{\lambda x} &= j!e^{\lambda x},\\
(D-\lambda)^{j+1} x^j e^{\lambda x} &= j!(D-\lambda)e^{\lambda x} = 0,
\end{aligned}$$

so $(D-\lambda)^k x^j e^{\lambda x} = 0$ for all $k \geq j+1$, that is, for $j \leq k-1$.

Basic solutions

Lemma

If $(z-\lambda)^k$ is a factor of $p(z)$ then the function $u(x) = x^j e^{\lambda x}$ is a solution of $Lu = 0$ for $0 \leq j \leq k-1$.

Proof.

Write $p(z) = (z-\lambda)^k q(z)$, so that $q(z)$ is a polynomial of degree $m-k$. For any two polynomials p_1 and p_2 we have $p_1(D)p_2(D) = p_2(D)p_1(D)$. It follows that

$$p(D) = (D-\lambda)^k q(D) = q(D)(D-\lambda)^k$$

and so for $0 \leq j \leq k-1$,

$$p(D)x^j e^{\lambda x} = q(D)(D-\lambda)^k x^j e^{\lambda x} = q(D)0 = 0.$$

$\square$

General solution

Theorem

For the constant-coefficient case, the general solution of the homogeneous equation $Lu = 0$ is

$$u(x) = \sum_{q=1}^{r} \sum_{l=0}^{k_q-1} c_{ql} x^l e^{\lambda_q x},$$

where the c_{ql} are arbitrary constants.

Since $(z - \lambda_q)^{k_q}$ is a factor of $p(z)$,

$$Lu = \sum_{q=1}^{r} \sum_{l=0}^{k_q-1} c_{ql} L x^l e^{\lambda_q x} = \sum_{q=1}^{r} \sum_{l=0}^{k_l-1} c_{ql} \times 0 = 0.$$

Linear independence is shown in the Technical Proofs.

Distinct real roots

Example

From the factorization

$$D^4 - 2D^3 - 11D^2 + 12D = (D+3)D(D-1)(D-4)$$

we see that the general solution of

$$u'''' - 2u''' - 11u'' + 12u' = 0$$

is

$$u = c_1 e^{-3x} + c_2 + c_3 e^{x} + c_4 e^{4x}.$$

Repeated real root

Example

From the factorization

$$D^4 + 6D^3 + 9D^2 - 4D - 12 = (D-1)(D+2)^2(D+3)$$

we see that the general solution of

$$u'''' + 6u''' + 9u'' - 4u' - 12u = 0$$

is

$$u = c_1 e^{x} + c_2 e^{-2x} + c_3 x e^{-2x} + c_4 e^{-3x}.$$

Complex root

Example

From the factorization

$$\begin{aligned} D^3 - 7D^2 + 17D - 15 &= (D^2 - 4D + 5)(D-3) \\ &= (D-2-i)(D-2+i)(D-3) \end{aligned}$$

we see that the general solution of

$$u''' - 7u'' + 17u' - 15u = 0$$

is

$$\begin{aligned} u(x) &= c_1 e^{(2+i)x} + c_2 e^{(2-i)x} + c_3 e^{3x} \\ &= c_4 e^{2x} \cos x + c_5 e^{2x} \sin x + c_3 e^{3x}. \end{aligned}$$

Simple oscillator

Second-order ODEs arise naturally in classical mechanics.

Consider a particle of mass m that moves along the x-axis with velocity $v = \dot{x} = dx/dt$ under the influence of

$$\begin{aligned}\text{an external applied force} &= f(t),\\ \text{a frictional resistance force} &= -r(v)v,\\ \text{a restoring force} &= -k(x)x.\end{aligned}$$

Newton's second law,

$$m\,\ddot{x} = m\,\frac{d^2x}{dt^2} = f(t) - r(v)v - k(x)x,$$

leads to a second-order differential equation

$$m\,\ddot{x} + r(\dot{x})\,\dot{x} + k(x)x = f(t).$$

Simplest case is when $r(v) = r_0 > 0$ and $k(x) = k_0 > 0$ are constant, giving a linear ODE with constant coefficients:

$$m\,\ddot{x} + r_0\,\dot{x} + k_0 x = f(t).$$

Typically interested in the case when the applied force is periodic with frequency ω; for example,

$$f(t) = F \sin \omega t.$$

The general solution $x(t) = x_{\mathrm{H}}(t) + x_{\mathrm{P}}(t)$.

We now show that $x_{\mathrm{H}}(t) \to 0$ as $t \to \infty$ and that $x_{\mathrm{P}}(t)$ exists. Since $x_{\mathrm{P}}(t + T) = x_{\mathrm{P}}(t)$ for $T = 2\pi/\omega$, it follows that the solution $x(t)$ always tends to a periodic function with frequency ω, regardless of the initial values $x(0)$ and $\dot{x}(0)$.

The characteristic equation is

$$m\lambda^2 + r_0\lambda + k_0 = (\lambda - \lambda_+)(\lambda - \lambda_-),$$

and the roots are

$$\lambda_\pm = \frac{-r_0 \pm \sqrt{\Delta}}{2m} \quad \text{where} \quad \Delta = r_0^2 - 4mk_0.$$

If $\Delta > 0$, then $\lambda_- < \lambda_+ < 0$ so, for any constants A and B,

$$x_\mathrm{H}(t) = Ae^{\lambda_+ t} + Be^{\lambda_- t} \to 0 \quad \text{as } t \to \infty.$$

If $\Delta = 0$ then $\lambda_- = \lambda_+ < 0$ so again $x_\mathrm{H}(t) \to 0$.

If $\Delta < 0$ then $\sqrt{\Delta} = i\sqrt{|\Delta|}$ so $\operatorname{Re}\lambda_+ = \operatorname{Re}\lambda_- < 0$, and again $x_\mathrm{H}(t) \to 0$.

Since $\operatorname{Re}\lambda_\pm \neq 0$ the particular solution has the form

$$x_\mathrm{P}(t) = C\cos\omega t + E\sin\omega t,$$

and we find that

$$\begin{aligned} m\ddot{x}_\mathrm{P} + r_0\dot{x}_\mathrm{P} + k_0 x_\mathrm{P} = {}& \bigl(-m\omega^2 C + r_0\omega E + k_0 C\bigr)\cos\omega t \\ &+ \bigl(-m\omega^2 E - r_0\omega C + k_0 E\bigr)\sin\omega t, \end{aligned}$$

which equals $F\sin\omega t$ iff

$$\begin{aligned} (k_0 - m\omega^2)C + r_0\omega E &= 0, \\ -r_0\omega C + (k_0 - m\omega^2)E &= F. \end{aligned}$$

This 2×2 system has a unique solution since its determinant is

$$(k_0 - m\omega^2)^2 + (r_0\omega)^2 > 0.$$

Wronskians and linear independence

We introduce a function, called the Wronskian, that provides us with a way of testing whether a family of solutions to $Lu = 0$ is linearly independent. The Wronskian also turns out to have several other uses.

Linear independence

Let $u_1(x), u_2(x), \ldots, u_m(x)$ be functions defined on an interval $I \subset \mathbb{R}$. The functions $u_1, \ldots, u_m$ are called linearly dependent if there exist constants $a_1, a_2, \ldots, a_m$ not all zero such that

$$a_1u_1(x) + a_2u_2(x) + \cdots + a_mu_m(x) = 0 \quad \forall x \in I.$$

If the above equation only holds for

$$a_i = 0, \qquad i = 1, 2, \ldots, m$$

then the functions are linearly independent.

Example

$u_1 = sin\,2x$ and $u_2 = sin\,x\,cos\,x$ are linearly dependent.
$u_1 = sin\,x$ and $u_2 = cos\,x$ are linearly independent.

Wronskian

The Wronskian of the functions u_1, u_2, ..., u_m is the $m \times m$ determinant

$$W(x) = W(x; u_1, u_2, \ldots, u_m) = \det[D^{i-1} u_j].$$

For instance, if $m = 2$ then

$$W(x) = \begin{vmatrix} u_1 & u_2 \\ u_1' & u_2' \end{vmatrix} = u_1 u_2' - u_2 u_1',$$

and when $m = 3$,

$$W(x) = \begin{vmatrix} u_1 & u_2 & u_3 \\ u_1' & u_2' & u_3' \\ u_1'' & u_2'' & u_3'' \end{vmatrix}.$$

Of course, $W(x)$ is defined only when the functions are differentiable $m - 1$ times.

Wronskian

Example

The Wronskian of the functions $u_1 = e^{2x}$, $u_2 = xe^{2x}$ and $u_3 = e^{-x}$ is

$$W = \begin{vmatrix} e^{2x} & xe^{2x} & e^{-x} \\ 2e^{2x} & e^{2x} + 2xe^{2x} & -e^{-x} \\ 4e^{2x} & 4e^{2x} + 4xe^{2x} & e^{-x} \end{vmatrix} = 9e^{3x}.$$

Example

The Wronskian of the functions $u_1 = e^x \cos 3x$ and $u_2 = e^x \sin 3x$ is

$$W = \begin{vmatrix} e^x \cos 3x & e^x \sin 3x \\ e^x \cos 3x - 3e^x \sin 3x & e^x \sin 3x + 3e^x \cos 3x \end{vmatrix} = 3e^{2x}.$$

Linearly dependent functions

Lemma

If $u_1, \ldots, u_m$ are linearly dependent over an interval $[a,b]$ then $W(x; u_1, \ldots, u_m) = 0$ for $a \le x \le b$.

Example

The functions $u_1 = \cosh x$, $u_2 = \sinh x$ and $u_3 = e^x$ are linearly dependent because

$$\cosh x + \sinh x = \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2} = e^x$$

for any interval $[a,b]$. Their Wronskian is

$$W = \begin{vmatrix} \cosh x & \sinh x & e^x \\ \sinh x & \cosh x & e^x \\ \cosh x & \sinh x & e^x \end{vmatrix} = 0$$

for all $x \in \mathbb{R}$.

Proof (for $m = 3$)

Assume that u_1, u_2, u_3 are linearly dependent on the interval $[a, b]$, that is, there exist constants c_1, c_2, c_3, not all zero, such that

$$c_1 u_1(x) + c_2 u_2(x) + c_3 u_3(x) = 0 \quad \text{for } a \leq x \leq b.$$

Differentiating, it follows that

$$\begin{aligned} c_1 u_1'(x) + c_2 u_2'(x) + c_3 u_3'(x) &= 0, \\ c_1 u_1''(x) + c_2 u_2''(x) + c_3 u_3''(x) &= 0, \end{aligned}$$

so

$$\begin{bmatrix} u_1(x) & u_2(x) & u_3(x) \\ u_1'(x) & u_2'(x) & u_3'(x) \\ u_1''(x) & u_2''(x) & u_3''(x) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{for } a \leq x \leq b.$$

This 3×3 matrix must be singular and thus $W(x) = 0$.

Wronskian satisfies a first-order ODE

Lemma

If u_1*,* u_2*, . . . ,* u_m *are solutions of* $Lu = 0$ *on the interval* $[a, b]$ *then their Wronskian satisfies*

$$a_m(x)W'(x) + a_{m-1}(x)W(x) = 0, \qquad a \le x \le b.$$

Example

The second-order ODE

$$u'' + 3u' - 4u = 0$$

has solutions $u_1 = e^x$ and $u_2 = e^{-4x}$. Their Wronskian is

$$\begin{vmatrix} e^x & e^{-4x} \\ e^x & -4e^{-4x} \end{vmatrix} = -5e^{-3x},$$

and satisfies $W' + 3W = 0$.

Proof (for $m = 2$)

Differentiating $W = u_1u_2' - u_1'u_2$ we have

$$W' = (u_1'u_2' + u_1u_2'') - (u_1''u_2 + u_1'u_2') = u_1u_2'' - u_1''u_2,$$

so

$$\begin{aligned} a_2W' + a_1W &= a_2(u_1u_2'' - u_1''u_2) + a_1(u_1u_2' - u_1'u_2) \\ &\qquad + a_0\underbrace{(u_1u_2 - u_1u_2)}_{=0} \\ &= u_1(a_2u_2'' + a_1u_2' + a_0u_2) \\ &\qquad - (a_2u_1'' + a_1u_1' + a_0u_1)u_2 \\ &= u_1(Lu_2) - (Lu_1)u_2 = 0. \end{aligned}$$

Linear independence of solutions

Theorem

Let u_1, u_2, ..., u_m be solutions of a non-singular, linear, homogeneous, mth-order ODE $Lu = 0$ on the interval $[a, b]$.
Either

> $W(x) = 0$ *for* $a \le x \le b$ *and the* m *solutions are linearly dependent,*

or else

> $W(x) \neq 0$ *for* $a \le x \le b$ *and the* m *solutions are linearly independent.*

Proof

The Wronskian satisfies

$$W' + pW = 0 \quad \text{for } a \le x \le b, \quad \text{where} \quad p = \frac{a_{m-1}}{a_m}.$$

Define an integrating factor

$$I(x) = \exp\left(\int p(x)\,dx\right) \ne 0,$$

so that $I' = Ip$ and hence

$$(IW)' = IW' + IpW = I(W' + pW) = 0.$$

Thus, $I(x)W(x) = C$ for some constant C.

Either $C = 0$ in which case $W(x) = 0$ for all $x \in [a, b]$, or else $C \ne 0$ in which case $W(x)$ is never zero for $x \in [a, b]$.

(Assume now that $m = 3$.) We already know that

$$u_1,\ u_2,\ u_3 \text{ linearly dependent } \implies W \equiv 0.$$

Hence, to complete the proof it suffices to show

$$W(a) = 0 \implies u_1,\ u_2,\ u_3 \text{ linearly dependent.}$$

If $W(a) = 0$, then there exist c_1, c_2, c_3, not all zero, such that

$$\begin{bmatrix} u_1(x) & u_2(x) & u_3(x) \\ u_1'(x) & u_2'(x) & u_3'(x) \\ u_1''(x) & u_2''(x) & u_3''(x) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{at } x = a,$$

so the function $u(x) = c_1u_1(x) + c_2u_2(x) + c_3u_3(x)$ satisfies

$$Lu = 0 \text{ for } a \le x \le b, \text{ with } u(a) = u'(a) = u''(a) = 0.$$

The solution of this initial-value problem is unique, so $u(x) \equiv 0$ and thus u_1, u_2, u_3 are linearly dependent.

Methods for inhomogeneous equations

In first year, you learned the method of undetermined coefficients for constructing a particular solution u_{P} to an inhomogeneous second-order linear ODE $Lu = f$ in some simple cases. We will study this method systematically for higher-order linear ODEs with constant coefficients.

We also discuss variation of parameters, a technique that applies for general L and f, but which requires the evaluation of possibly very difficult integrals.

Superposition of solutions

We now consider methods for finding a particular solution u_{P} satisfying $Lu_{\mathrm{P}} = f$.

First note that if

$$f(x) = c_1 f_1(x) + c_2 f_2(x)$$

and if we know u_{P1} and u_{P2} satisfying

$$Lu_{\mathrm{P1}} = f_1 \quad \text{and} \quad Lu_{\mathrm{P2}} = f_2,$$

then we can put

$$u_{\mathrm{P}}(x) = c_1 u_{\mathrm{P1}}(x) + c_2 u_{\mathrm{P2}}(x),$$

because by the linearity of L,

$$Lu_{\mathrm{P}} = c_1 Lu_{\mathrm{P1}} + c_2 Lu_{\mathrm{P2}} = c_1 f_1 + c_2 f_2 = f.$$

Example

Suppose that u_1 solves $Lu = e^{3x}$, and u_2 solves $Lu = \sin x$, where L is a linear differential operator. Then a solution of

$$Lu = e^{3x} + \sin x$$

is

$$u(x) = u_1(x) + u_2(x).$$

And a solution of

$$Lu = \frac{1}{2}e^{3x} - 5 \sin x$$

is

$$u(x) = \frac{1}{2}u_1(x) - 5u_2(x).$$

Now we want to investigate some methods for finding particular solutions – i.e., finding a solution of $Lu = f$. One such method is the *method of undetermined coefficients* – a.k.a., *the method of judicious guessing*. It relies on selecting a suitable *ansatz* u_p from knowledge of f. For example:

1. If f is a polynomial, then guess that u_p is a polynomial.
2. If f is exponential, then guess that u_p is exponential.
3. If f is a sine or cosine, then guess that u_p is a combination of such functions.

One problem with this method: it will only work for the types of functions identified above.

Example

Suppose that $u'' - u' = e^{5t}$. Note that

$$u_h(t) = c_1 + c_2 e^t.$$

So, guess

$$u_p(t) = Ae^{5t}.$$

Then

$$e^{5t} = u_p'' - u_p' = 25Ae^{5t} - 5Ae^{5t} = 20Ae^{5t}$$

so that

$$A = \frac{1}{20}.$$

Therefore,

$$u(t) = u_h(t) + u_p(t) = c_1 + c_2 e^t + \frac{1}{20}e^{5t}.$$

Example

Suppose that $u'' - u' = t^2 + 2t$. Note, as before, that

$$u_h(t) = c_1 + c_2 e^t.$$

So, guess

$$u_p(t) = At^3 + Bt^2 + Ct + D.$$

Then

$$t^2 + 2t = u_p'' - u_p' = -3At^2 + (6A - 2B)t + (2B - C).$$

So, equating coefficients of like power terms, we see that

$$A = -\frac{1}{3},\ B = -2,\ C = -4, \text{ and } D \text{ is unrestricted.}$$

Therefore, reabsorbing D into c_1, we see that

$$u(t) = u_h(t) + u_p(t) = c_1 + c_2 e^t - \frac{1}{3}t^3 - 2t^2 - 4t.$$

Example

Suppose that $u'' - u' = e^t$. Note that

$$u_h(t) = c_1 + c_2 e^t.$$

We can't guess e^t since it lives in the kernel of L. So, instead, guess

$$u_p(t) = Ate^t.$$

Then

$$e^t = u_p'' - u_p' = (2Ae^t + Ate^t) - (Ae^t + Ate^t) = Ae^t$$

so that

$$A = 1.$$

Therefore,

$$u(t) = u_h(t) + u_p(t) = c_1 + c_2 e^t + te^t.$$

Now we look at this idea of judicious guessing in a more systematic way.

Polynomial solutions

Let $L = p(D)$ be a linear differential operator of order m with constant coefficients.

Theorem

Assume that $a_0 = p(0) \neq 0$. For any integer $r \geq 0$, there exists a unique polynomial u_P of degree r such that $Lu_\mathrm{P} = x^r$.

For simplicity, we prove the result only for the case $m = 2$. Thus,

$$Lu = a_2 u'' + a_1 u' + a_0 u,$$

where a_0, a_1, a_2 are constants with $a_2 \neq 0$ and $a_0 \neq 0$.

Look for u_P in the form

$$u_\mathrm{P}(x) = \sum_{j=0}^{r} c_j \frac{x^j}{j!}.$$

We find that

$$
\begin{aligned}
Lu_{\mathrm{P}} &= a_2 \sum_{j=2}^{r} c_j \frac{x^{j-2}}{(j-2)!} + a_1 \sum_{j=1}^{r} c_j \frac{x^{j-1}}{(j-1)!} + a_0 \sum_{j=0}^{r} c_j \frac{x^j}{j!} \\
&= a_0 c_r \frac{x^r}{r!} + \left(a_0 c_{r-1} + a_1 c_r\right) \frac{x^{r-1}}{(r-1)!} \\
&\quad + \sum_{j=0}^{r-2} \left(a_2 c_{j+2} + a_1 c_{j+1} + a_0 c_j\right) \frac{x^j}{j!},
\end{aligned}
$$

which equals x^r if and only if

$$
\begin{aligned}
a_0 c_j + a_1 c_{j+1} + a_2 c_{j+2} &= 0, \quad 0 \le j \le r-2, \\
a_0 c_{r-1} + a_1 c_r &= 0, \\
a_0 c_r &= r!.
\end{aligned}
$$

This upper triangular system is uniquely solvable because $a_0 \neq 0$.

An example

Let $Lu = 3u'' - u' + 2u$ and suppose we want a particular solution to $Lu = 8x^3$. The theorem ensures that

$$u_{\mathrm{P}}(x) = C + Ex + Fx^2 + Gx^3$$

works for some C, E, F, G. In fact,

$$Lu_{\mathrm{P}} = (2C - E + 6F) + (2E - 2F + 18G)x + (2F - 3G)x^2 + 2Gx^3$$

so

$$\begin{aligned} 2C - \quad E + 6F \qquad &= 0, \\ 2E - 2F + 18G &= 0, \\ 2F - \quad 3G &= 0, \\ 2G &= 8 \end{aligned}$$

and back substitution gives $u_{\mathrm{P}} = -33 - 30x + 6x^2 + 4x^3$.

Exponential solutions

Theorem

Let $L = p(D)$, $M \in \mathbb{R}$, and $\mu \in \mathbb{C}$. If $p(\mu) \neq 0$, then the function

$$u_{\mathrm{P}}(x) = \frac{Me^{\mu x}}{p(\mu)}$$

satisfies $Lu_{\mathrm{P}} = Me^{\mu x}$.

Proof.

Follows at once because $p(D)e^{\mu x} = p(\mu)e^{\mu x}$. □

Example

A particular solution of $u'' + 4u' - 3u = 3e^{2x}$ is $u_{\mathrm{P}} = e^{2x}/3$.

Product of polynomial and exponential

Theorem

Let $L = p(D)$ and assume that $p(\mu) \neq 0$. For any integer $r \geq 0$, there exists a unique polynomial v of degree r such that $u_{\mathrm{P}} = v(x)e^{\mu x}$ satisfies $Lu_{\mathrm{P}} = x^r e^{\mu x}$.

Proof.

Again, for simplicity, we prove the result only for $m = 2$.
Put $v = e^{-\mu x}u$ so that $u = ve^{\mu x}$, and observe that

$$\begin{aligned} Lu &= \big[a_2(v'' + 2\mu v' + \mu^2 v) + a_1(v' + \mu v) + a_0 v\big]e^{\mu x} \\ &= \big[a_2 v'' + (a_1 + 2a_2\mu)v' + (a_2\mu^2 + a_1\mu + a_0)v\big]e^{\mu x}. \end{aligned}$$

Thus, $Lu = e^{\mu x}q(D)v$ where $q(z) = a_2 z^2 + (a_1 + 2a_2\mu)z + p(\mu)$, and our earlier result yields the desired v satisfying $q(D)v = x^r$ because $q(D)1 = q(0) = p(\mu) \neq 0$. □

An example

Consider

$$2u'' + u' - 3u = 9xe^{-2x}.$$

Here,

$$p(z) = 2z^2 + z - 3$$

so $p(-2) = 3 \neq 0$ and a particular solution $u_{\mathrm{P}} = (Cx + E)e^{-2x}$ exists. In fact, we find that

$$p(D)u_{\mathrm{P}} = (3Cx - 7C + 3E)e^{-2x}$$

so

$$3C = 9 \quad \text{and} \quad -7C + 3E = 0.$$

Thus, $C = 3$ and $E = 7$, giving

$$u_{\mathrm{P}} = (3x + 7)e^{-2x}.$$

Annihilator method

In the previous cases we proposed a solution $u = u_P$ and showed that it satisfied $Lu = f$. The following is a method to derive a particular solution given $Lu = f$. If f(x) is differentiable at least n times and

$$[a_n D^n + a_{n-1} D^{n-1} + \cdots + a_1 D^1 + a_0] f(x) = 0$$

then $[a_n D^n + a_{n-1} D^{n-1} + \cdots + a_1 D^1 + a_0]$ annihilates f.

Example

D^n annihilates x^{m-1} for $m \leq n$.

Example

$(D - \alpha)^n$ annihilates $x^{m-1} e^{\alpha x}$ for $m \leq n$.

Annihilator method: Three simple examples (1)

Given $Lu = f$ we can apply the appropriate annihilator to both sides and solve the resulting homogeneous DE.
Let $Lu = u' - u$ and suppose we want a solution such that $Lu = x^2$.
Annihilating both sides we have

$$D^3(u' - u) = u'''' - u''' = 0.$$

Setting $w = u'''$, clearly $w = Ce^x$ is the general solution. Integrating three times yields

$$u = Ce^x + Ex^2 + Fx + G.$$

Clearly $u_h = Ae^x$ and the form of the particular solution is $u_p = Ex^2 + Fx + G$ (a polynomial of degree 2, as expected). Substituting find $E = -1$, $F = 2$ and $G = 1$.

Annihilator method: Three simple examples (2)

Let $Lu = u'' - u'$ and suppose we want a solution such that $Lu = x^2$ (note that $p(0) \neq 0$ for $L = p(D)$ so we are not yet able to solve this). Annihilating both sides we have

$$D^3(u'' - u') = u^{(5)} - u^{(4)} = 0.$$

Setting $w = u^{(4)}$, $w = Ce^x$ is the general solution. Integrating four times yields

$$u = Ce^x + Ex^3 + Fx^2 + Gx + H.$$

Here $u_h = Ae^x + H$ is the homogeneous solution and the particular solution is $u_p = x(Ex^2 + Fx + G)$. Substituting find $E = -1/3$, $F = -1$, $G = -2$.

Annihilator method: Three simple examples (3)

Let $Lu = u' - u$ and suppose we want a solution such that $Lu = e^x$ (note that $\mu = 1$ and $p(\mu) = 0$ for $L = p(D)$ so we are not yet able to solve this). Annihilating both sides we have

$$(D-1)(u'-u) = u'' - 2u' + u = 0.$$

The characteristic polynomial has a repeated root so $u = Ae^x + Bxe^x$ is the general solution. Here $u_h = Ae^x$ is the homogeneous solution and so the particular solution is $u_p = Bxe^x$. Substituting we find that $B = 1$.

Polynomial solutions: the remaining case

Theorem

Let $L = p(D)$ *and assume* $p(0) = p'(0) = \cdots = p^{(k-1)}(0) = 0$ *but* $p^{(k)}(0) \neq 0$ *where* $1 \leq k \leq m-1$. *For any integer* $r \geq 0$, *there exists a unique polynomial* v *of degree* r *such that* $u_p(x) = x^k v(x)$ *satisfies* $Lu_{\mathrm{P}} = x^r$.

Example

Let $Lu = u''' + 2u''$ and seek a particular solution to $Lu = 12x^2$. The theorem ensures that

$$u_{\mathrm{P}} = x^2(C + Ex + Fx^2) = Cx^2 + Ex^3 + Fx^4$$

works for some C, E, F. In fact,

$$Lu_{\mathrm{P}} = (4C + 6E) + (12E + 24F)x + 24Fx^2$$

so

$$\begin{aligned} 4C + 6\ \ E \qquad\quad &= 0, \\ 12E + 24F &= 0, \\ 24F &= 12 \end{aligned}$$

and back substitution gives

$$u_{\mathrm{P}} = \frac{x^2}{2}(3 - 2x + x^2).$$

Exponential times polynomial: remaining case

Lemma

If $u(x) = w(x)e^{\mu x}$ *then*

$$p(D)u = e^{\mu x} q(D)w \quad \textit{where} \quad q(z) = \sum_{j=0}^{m} p^{(j)}(\mu)\,\frac{z^j}{j!}.$$

Theorem

Let $L = p(D)$ *and assume* $p(\mu) = p'(\mu) = \cdots = p^{(k-1)}(\mu) = 0$ *but* $p^{(k)}(\mu) \neq 0$, *where* $1 \leq k \leq m-1$. *For any integer* $r \geq 0$, *there exists a unique polynomial* v *of degree* r *such that* $u_{\mathrm{P}}(x) = x^k v(x) e^{\mu x}$ *satisfies* $Lu_{\mathrm{P}} = x^r e^{\mu x}$.

Exponential times polynomial: remaining case

Proof.

Since $q^{(j)}(0) = p^{(j)}(\mu)$ for all j, there is a unique polynomial v of degree r such that $w(x) = x^k v(x)$ satisfies $q(D)w = x^r$ and hence

$$p(D)u_{\mathrm{P}} = e^{\mu x} q(D)w = e^{\mu x} x^r.$$

$\square$

Proof of Lemma

$$
\begin{aligned}
p(D)we^{\mu x} &= \sum_{k=0}^{m} a_k D^k \left(we^{\mu x}\right) = \sum_{k=0}^{m} a_k \sum_{j=0}^{k} \binom{k}{j} D^j w \, D^{k-j} e^{\mu x} \\
&= \sum_{k=0}^{m} a_k \sum_{j=0}^{k} \frac{k!}{j!(k-j)!} D^j w \, D^{k-j} e^{\mu x} \\
&= \sum_{j=0}^{m} \frac{D^j w}{j!} \sum_{k=j}^{m} a_k k(k-1)(k-2)\cdots(k-j+1)\mu^{k-j} e^{\mu x} \\
&= \sum_{j=0}^{m} \frac{D^j w}{j!} \sum_{k=j}^{m} a_k k(k-1)(k-2)\cdots(k-j+1)\mu^{k-j} e^{\mu x} \\
&= \sum_{j=0}^{m} \frac{D^j w}{j!} p^{(j)}(\mu) e^{\mu x} = e^{\mu x} \sum_{j=0}^{m} p^{(j)}(\mu) \, \frac{D^j w}{j!} \\
&= e^{\mu x} q(D) w.
\end{aligned}
$$

An example

Consider the ODE

$$Lu = 12e^{2x} \quad \text{where} \quad Lu = u''' - 4u'' + 4u'.$$

Here, $L = p(D)$ for $p(z) = z^3 - 4z^2 + 4z = z(z-2)^2$, so $p(2) = p'(2) = 0$ but $p''(2) \neq 0$. Thus, we try

$$u_{\mathrm{P}} = Cx^2e^{2x}$$

and find

$$u_{\mathrm{P}}' = C(2x + 2x^2)e^{2x}, \qquad u_{\mathrm{P}}'' = C(2 + 8x + 4x^2)e^{2x},$$
$$u_{\mathrm{P}}''' = C(12 + 24x + 8x^2)e^{2x},$$

so $Lu_{\mathrm{P}} = 4Ce^{2x}$ and we require $4C = 12$. Therefore, a particular solution is

$$u_{\mathrm{P}} = 3x^2e^{2x}.$$

Complex conjugate roots

Consider

$$Lu \equiv p(D) \equiv u''' + u' + 10u = 13e^x \sin 2x.$$

Here, $p(z) = z^3 + z + 10 = [(z-1)^2 + 4](z+2)$ so $p(1 \pm 2i) = 0$, and

$$e^x \sin 2x = e^x \, \frac{e^{2ix} - e^{-2ix}}{2i}$$

is a linear combination of $e^{(1+2i)x}$ and $e^{(1-2i)x}$. Therefore put

$$u_{\mathrm{P}}(x) = Cxe^{(1+2i)x} + Exe^{(1-2i)x} = xe^x\bigl(F\cos 2x + G\sin 2x\bigr).$$

We find that if $F = -3/4$ and $G = -1/2$ then

$$Lu_{\mathrm{P}} = (-8F + 12G)e^x\cos 2x + (-12F - 8G)e^x \sin 2x = 13e^x \sin 2x,$$

so

$$u_{\mathrm{P}} = -\frac{xe^x}{4}\bigl(3\cos 2x + 2\sin 2x\bigr).$$

Variation of parameters

What if f is not a polynomial times an exponential, or if L does not have constant coefficients?

Consider a linear, second-order, inhomogeneous ODE with leading coefficient 1:

$$Lu = u''(x) + p(x)u'(x) + q(x)u(x) = f(x). \tag{5}$$

Let $u_1(x)$ and $u_2(x)$ be linearly independent solutions to the homogeneous equation and let $W(x) = W(x; u_1, u_2)$ denote their Wronskian. Thus,

$$Lu_1 = 0, \quad Lu_2 = 0, \quad W \neq 0.$$

We seek v_1 and v_2 such that

$$u(x) = v_1(x)u_1(x) + v_2(x)u_2(x)$$

is a (particular) solution to $Lu = f$.

Variation of parameters (continued)

To simplify the expression

$$u' = v_1' u_1 + v_1 u_1' + v_2' u_2 + v_2 u_2'$$

we impose the condition $v_1' u_1 + v_2' u_2 = 0$, then (as if v_1 and v_2 were constant parameters)

$$u' = v_1 u_1' + v_2 u_2'.$$

A short calculation now shows

$$Lu = v_1 Lu_1 + v_2 Lu_2 + v_1' u_1' + v_2' u_2' = v_1' u_1' + v_2' u_2',$$

since by assumption $Lu_1 = 0 = Lu_2$.

Conclusion: $u = v_1 u_1 + v_2 u_2$ satisfies $Lu = f$ if

$$\begin{aligned} v_1' u_1 + v_2' u_2 &= 0, \\ v_1' u_1' + v_2' u_2' &= f. \end{aligned}$$

Thus, we have a pair of equations for the unknown v_1' and v_2'. In matrix form

$$\begin{bmatrix} u_1(x) & u_2(x) \\ u_1'(x) & u_2'(x) \end{bmatrix} \begin{bmatrix} v_1'(x) \\ v_2'(x) \end{bmatrix} = \begin{bmatrix} 0 \\ f(x) \end{bmatrix},$$

so

$$\begin{bmatrix} v_1'(x) \\ v_2'(x) \end{bmatrix} = \frac{1}{W(x)} \begin{bmatrix} u_2'(x) & -u_2(x) \\ -u_1'(x) & u_1(x) \end{bmatrix} \begin{bmatrix} 0 \\ f(x) \end{bmatrix},$$

or in other words,

$$v_1'(x) = \frac{-u_2(x)f(x)}{W(x)} \quad \text{and} \quad v_2'(x) = \frac{u_1(x)f(x)}{W(x)}.$$

Example

Find the general solution to

$$u'' - 4u' + 4u = (x+1)\exp 2x.$$

Solution via power series

If L has variable coefficients, then we cannot expect in general that the solution of $Lu = 0$ is expressible in terms of elementary functions like polynomials, trigonometric functions, exponentials etc. Power series provide a flexible way to represent u in this case.

Constructing a series solution

Consider the initial-value problem

$$Lu = (1 - x^2)u'' - 5xu' - 4u = 0, \quad u(0) = 1, \quad u'(0) = 2.$$

Look for a solution in the form of a power series

$$u(x) = \sum_{k=0}^{\infty} A_k x^k = A_0 + A_1 x + A_2 x^2 + \cdots .$$

Formal calculations show that

$$Lu = \sum_{k=0}^{\infty} (k+2)[(k+1)A_{k+2} - (k+2)A_k]x^k,$$

and the initial conditions imply $A_0 = 1$ and $A_1 = 2$.

Convergence?

Since Lu is identically zero iff the coefficient of x^k vanishes for every k, we obtain the recurrence relation

$$A_{k+2} = \frac{k+2}{k+1}\,A_k \quad \text{for } k = 0,\ 1,\ 2,\ \ldots.$$

Thus,

$$A_0 = 1, \quad A_1 = 2, \quad A_2 = 2, \quad A_3 = 3, \quad \ldots,$$

giving

$$u(x) = 1 + 2x + 2x^2 + 3x^3 + \cdots .$$

Since

$$\lim_{k\to\infty} \frac{A_{k+2}x^{k+2}}{A_k x^k} = \lim_{k\to\infty} \frac{k+2}{k+1}\,x^2 = x^2,$$

the ratio test shows that $\sum_{j=0}^{\infty} A_{2j}x^{2j}$ and $\sum_{j=0}^{\infty} A_{2j+1}x^{2j+1}$ converge for $x^2 < 1$ but diverge for $x^2 > 1$.

General case

Consider a general second-order, linear, homogeneous ODE

$$Lu = a_2(x)u'' + a_1(x)u' + a_0(x)u = 0.$$

Equivalently,

$$u'' + p(x)u' + q(x)u = 0,$$

where

$$p(x) = \frac{a_1(x)}{a_2(x)} \quad \text{and} \quad q(x) = \frac{a_0(x)}{a_2(x)}.$$

Assume that a_j is analytic at 0 for $0 \leq j \leq 2$, and that $a_2(0) \neq 0$. Then p and q are analytic at 0, that is, they admit power series expansions

$$p(z) = \sum_{k=0}^{\infty} p_k z^k \quad \text{and} \quad q(z) = \sum_{k=0}^{\infty} q_k z^k \quad \text{for } |z| < \rho,$$

for some $\rho > 0$.

Formal expansions

If

$$u(z) = \sum_{k=0}^{\infty} A_k z^k$$

then we find that

$$\begin{aligned} Lu(z) = (2A_2 + p_0 A_1 + q_0 A_0) \\ + (6A_3 + 2p_0 A_2 + p_1 A_1 + q_0 A_1 + q_1 A_0)z + \cdots , \end{aligned}$$

where, on the RHS, the coefficient of z^{n-1} for a general $n \geq 1$ is

$$(n+1)nA_{n+1} + \sum_{j=0}^{n-1} \big[(n-j)p_j A_{n-j} + q_j A_{n-1-j}\big].$$

Convergence theorem

Given $u(0)$ and $u'(0)$, we put

$$A_0 = u(0) \quad \text{and} \quad A_1 = u'(0),$$

and compute recursively

$$A_{n+1} = \frac{-1}{n(n+1)} \sum_{j=0}^{n-1} \big[(n-j)p_j A_{n-j} + q_j A_{n-1-j}\big], \quad n \geq 1.$$

Theorem

If the coefficients $p(z)$ and $q(z)$ are analytic for $|z| < \rho$, then the formal power series for the solution $u(z)$, constructed above, is also analytic for $|z| < \rho$.

Previous example

Earlier we considered

$$Lu = (1 - x^2)u'' - 5xu' - 4u = 0, \quad u(0) = 1, \quad u'(0) = 2.$$

In this case,

$$p(z) = \frac{-5z}{1 - z^2} = -5 \sum_{k=0}^{\infty} z^{2k+1}$$

and

$$q(z) = \frac{-4}{1 - z^2} = -4 \sum_{k=0}^{\infty} z^{2k}$$

are analytic for $|z| < 1$, so the theorem guarantees that $u(z)$, given by the formal power series, is also analytic for $|z| < 1$.

Expansion about a point other than 0

Suppose we want a power series expansion about a point $c \neq 0$, for instance because the initial conditions are given at $x = c$.

A simple change of the independent variable allows us to write

$$u = \sum_{k=0}^{\infty} A_k (z-c)^k = \sum_{k=0}^{\infty} A_k Z^k \quad \text{where } Z = z - c.$$

Since $du/dz = du/dZ$ and $d^2u/dz^2 = d^2u/dZ^2$, we obtain the translated equation

$$\frac{d^2u}{dZ^2} + p(Z+c)\,\frac{du}{dZ} + q(Z+c)u = 0.$$

Now compute the A_k using the series expansions of $p(Z+c)$ and $q(Z+c)$ in powers of Z.

Example

We construct a power series solution about $z = 1$ to Airy's equation:

$$u'' - zu = 0.$$

Put $Z = z - 1$ and find that $u'' - zu = u'' - (Z + 1)u$ equals

$$\sum_{k=0}^{\infty} k(k-1)A_k Z^{k-2} - \sum_{k=0}^{\infty} A_k Z^{k+1} - \sum_{k=0}^{\infty} A_k Z^k$$

$$= \sum_{k=0}^{\infty} (k+2)(k+1)A_{k+2} Z^k - \sum_{k=1}^{\infty} A_{k-1} Z^k - \sum_{k=0}^{\infty} A_k Z^k$$

$$= (2A_2 - A_0) + \sum_{k=1}^{\infty} \big[(k+2)(k+1)A_{k+2} - A_{k-1} - A_k\big] Z^k.$$

Thus, the coefficients must satisfy $2A_2 - A_0 = 0$ and

$$(k+2)(k+1)A_{k+2} - A_{k-1} - A_k = 0 \quad \text{for all } k \geq 1,$$

so

$$A_2 = \frac{A_0}{2} \quad \text{and} \quad A_{k+2} = \frac{A_{k-1} + A_k}{(k+2)(k+1)} \quad \text{for } k \geq 1.$$

We find that

$$u(z) = A_0\left(1 + \frac{(z-1)^2}{2} + \frac{(z-1)^3}{6} + \frac{(z-1)^4}{24} + \cdots\right) + A_1\left((z-1) + \frac{(z-1)^3}{6} + \frac{(z-1)^4}{12} + \cdots\right).$$

Singular ODEs

Recall that our basic existence and uniqueness theorem for $Lu = f$ assumes that L is not singular, that is, the leading coeffecient of L does not vanish on the interval of interest. However, some important applications lead to singular ODEs so we must now address this case.

Singular ODEs of second order

Consider

$$a_2(x)u'' + a_1(x)u' + a_0(x)u = 0 \quad \text{for } a \le x \le b,$$

and suppose that $a_2(x_0) = 0$ for some x_0 with $a < x_0 < b$, but $a_2(x) \neq 0$ if $x \neq x_0$. Put

$$b_j(y) = a_j(x) \quad \text{and} \quad v(y) = u(x) \quad \text{where } y = x - x_0,$$

so that, with $c = a - x_0 < 0 < d = b - x_0$,

$$b_2(y)v'' + b_1(y)v' + b_0(y)v = 0 \quad \text{for } c \le y \le d.$$

Since $y = 0$ when $x = x_0$, we have $b_2(0) = 0$.

In this way, it suffices to consider the case when the leading coefficient vanishes at the origin.

Cauchy–Euler ODE

A second-order Cauchy–Euler ODE has the form

$$Lu = ax^2u'' + bxu' + cu = f(x),$$

where a, b and c are constants, with $a \neq 0$. This ODE is singular at $x = 0$.

Noticing that

$$Lx^r = \bigl[ar(r-1) + br + c\bigr]x^r,$$

we see that $u = x^r$ is a solution of the homogeneous equation ($f = 0$) iff

$$ar(r-1) + br + c = 0.$$

Factorization

Suppose $ar(r-1)+br+c = a(r-r_1)(r-r_2)$. If $r_1 \neq r_2$ then the general solution of the homogeneous equation $Lu = 0$ is

$$u(x) = C_1 x^{r_1} + C_2 x^{r_2}, \qquad x > 0.$$

Lemma

If $r_1 = r_2$ then the general solution of the homogeneous Cauchy–Euler equation $Lu = 0$ is

$$u(x) = C_1 x^{r_1} + C_2 x^{r_1} \ln x, \qquad x > 0.$$

Example

Solve $x^2 u'' - xu' + u = 0$.

Example

Solve $2(x-2)^2 u'' - 3(x-2)u' - 3u = 0$.

Proof of the lemma

Since $r_1 = r_2$ the function $F(x, r) = x^r$ satisfies

$$ax^2 \frac{\partial^2 F}{\partial x^2} + bx \frac{\partial F}{\partial x} + cF = a(r - r_1)^2 x^r,$$

Put $v(x) = \frac{\partial F}{\partial r}(x, r_1)$ where

$$\frac{\partial F}{\partial r}(x, r) = \frac{\partial}{\partial r} e^{r \ln x} = e^{r \ln x} \ln x = x^r \ln x$$

and observe that

$$\begin{aligned}
ax^2 \frac{\partial^2 v}{\partial x^2} + bx \frac{\partial v}{\partial x} + cv &= \left(ax^2 \frac{\partial^2}{\partial x^2} \frac{\partial F}{\partial r} + bx \frac{\partial}{\partial x} \frac{\partial F}{\partial r} + c \frac{\partial F}{\partial r} \right) \Bigg|_{r=r_1} \\
&= \frac{\partial}{\partial r} \left(ax^2 \frac{\partial^2 F}{\partial x^2} + bx \frac{\partial F}{\partial x} + cF \right) \Bigg|_{r=r_1} \\
&= \frac{\partial}{\partial r} \big(a(r - r_1)^2 x^r \big) \Big|_{r=r_1} \\
&= \big(2a(r - r_1) x^r + (r - r_1)^2 x^r \ln x \big) \Big|_{r=r_1} = 0.
\end{aligned}$$

More general singular ODEs

A number of important applications lead to ODEs that can be written in the Frobenious normal form

$$z^2u'' + zP(z)u' + Q(z)u = 0,$$

where $P(z)$ and $Q(z)$ are analytic at $z = 0$:

$$P(z) = \sum_{k=0}^{\infty} P_k z^k \quad \text{and} \quad Q(z) = \sum_{k=0}^{\infty} Q_k z^k, \qquad |z| < \rho. \tag{6}$$

Notice that $u'' + p(z)u' + q(z)u = 0$ but $p(z) = z^{-1}P(z)$ and $q(z) = z^{-2}Q(z)$ are not analytic at $z = 0$ unless $P(0) = 0$ and $Q(0) = Q'(0) = 0$.

So in general we cannot expect a solution $u(z)$ to be analytic at $z = 0$.

A clue

We can think of an ODE in Frobenius normal form as a Cauchy–Euler ODE with variable coefficients.

For z near 0 we have $P(z) \approx P_0$ and $Q(z) \approx Q_0$ so we might expect $u(z)$ to behave like a solution of

$$z^2 u'' + P_0 z u' + Q_0 u = 0.$$

We therefore consider the indicial polynomial

$$I(r) = r(r-1) + P_0 r + Q_0 = (r - r_1)(r - r_2).$$

If $r_1 \neq r_2$ then the approximating Cauchy–Euler ODE has the general solution $c_1 z^{r_1} + c_2 z^{r_2}$, so it is natural to seek a solution in the form

$$u(z) = z^r \sum_{k=0}^{\infty} A_k z^k = \sum_{k=0}^{\infty} A_k z^{k+r}, \quad |z| < \rho, \quad \text{with } A_0 \neq 0.$$

An example

Consider

$$Lu = 2z^2u'' + 7zu' - (z^2 + 3)u = 0.$$

Here, $P(z) = 7/2$ and $Q(z) = -(z^2 + 3)/2$ are trivially analytic at $z = 0$ (since they are polynomials).

The approximations $P(z) \approx P_0 = 7/2$ and $Q(z) \approx Q_0 = -3/2$ lead to the approximating Cauchy–Euler equation $z^2u'' + (7/2)zu' - (3/2)u = 0$ or

$$2z^2u'' + 7zu' - 3u = 0.$$

Thus, the indicial polynomial is

$$2r(r-1) + 7r - 3 = 2r^2 + 5r - 3 = (2r-1)(r+3)$$

so $r_1 = 1/2$ and $r_2 = -3$.

Using

$$u = \sum_{k=0}^{\infty} A_k z^{k+r}, \qquad u' = \sum_{k=0}^{\infty} (k+r) A_k z^{k+r-1},$$

$$u'' = \sum_{k=0}^{\infty} (k+r)(k+r-1) A_k z^{k+r-2},$$

we find that

$$\begin{aligned} Lu &= (2z^2 u'' + 7zu' - 3u) - z^2 u \\ &= \sum_{k=0}^{\infty} \bigl[2(k+r)(k+r-1) + 7(k+r) - 3\bigr] A_k z^{k+r} \\ &\qquad - \sum_{k=0}^{\infty} A_k z^{k+r+2}. \end{aligned}$$

Since

$$
\begin{aligned}
2(k+r)(k+r-1)+7(k+r)-3 &= 2(k+r)^2+5(k+r)-3 \\
&= [2(k+r)-1][(k+r)+3]
\end{aligned}
$$

and

$$\sum_{k=0}^{\infty} A_k z^{k+r+2} = \sum_{k=2}^{\infty} A_{k-2} z^{k+r}$$

it follows that

$$
\begin{aligned}
Lu = (2r-1)(r+3)A_0 z^r &+ (2r+1)(r+4)A_1 z^{r+1} \\
&+ \sum_{k=2}^{\infty} \big[(2k+2r-1)(k+r+3)A_k - A_{k-2}\big] z^{k+r}.
\end{aligned}
$$

Conclusion: u is a solution if $r \in \{1/2, -3\}$ with

$$A_1 = 0, \qquad A_k = \frac{A_{k-2}}{(2k+2r-1)(k+r+3)} \quad \text{for all } k \geq 2.$$

First solution: $r = 1/2$ with

$$A_1 = 0, \qquad A_k = \frac{A_{k-2}}{k(2k+7)} \quad \text{for all } k \geq 2,$$

so

$$A_2 = \frac{A_0}{22}, \quad A_3 = \frac{A_1}{39} = 0, \quad A_4 = \frac{A_2}{60} = \frac{A_0}{1320}, \quad \ldots$$

and

$$u(z) = A_0 z^{1/2}\left(1 + \frac{z^2}{22} + \frac{z^4}{1320} + \cdots\right).$$

Second solution: $r = -3$ with

$$A_1 = 0, \qquad A_k = \frac{A_{k-2}}{k(2k-7)} \quad \text{for all } k \geq 2,$$

so

$$A_2 = -\frac{A_0}{6}, \quad A_3 = -\frac{A_1}{3} = 0, \quad A_4 = -\frac{A_2}{4} = \frac{A_0}{24}, \quad \ldots$$

and

$$u(z) = A_0 z^{-3}\left(1 - \frac{z^2}{6} + \frac{z^4}{24} + \cdots\right).$$

General solution of $Lu = 0$:

$$u(z) = Az^{1/2}\left(1 + \frac{z^2}{22} + \frac{z^4}{1320} + \cdots\right) + Bz^{-3}\left(1 - \frac{z^2}{6} + \frac{z^4}{24} + \cdots\right).$$

General case

Now consider

$$z^2u'' + zP(z)u' + Q(z)u = 0$$

for $P(z)$ and $Q(z)$ satisfying (6). Formal manipulations show that $Lu(z)$ equals

$$I(r)A_0z^r + \sum_{k=1}^{\infty}\left(I(k+r)A_k + \sum_{j=0}^{k-1}[(j+r)P_{k-j} + Q_{k-j}]A_j\right)z^{k+r},$$

so we define $A_0(r) = 1$ and

$$A_k(r) = \frac{-1}{I(k+r)}\sum_{j=0}^{k-1}[(j+r)P_{k-j} + Q_{k-j}]A_j(r), \quad k \geq 1,$$

provided $I(k+r) \neq 0$ for all $k \geq 1$.

Choice of exponent

The preceding calculations show that the series

$$F(z,r) = \sum_{k=0}^{\infty} A_k(r) z^{k+r}$$

satisfies

$$z^2 \frac{\partial^2 F}{\partial z^2} + zP(z)\frac{\partial F}{\partial z} + Q(z)F = I(r)z^r,$$

with

$$I(r) = r(r-1) + P_0 r + Q_0 = (r - r_1)(r - r_2).$$

Assume, with no loss of generality, that $\operatorname{Re} r_1 \geq \operatorname{Re} r_2$. It follows that $I(k + r_1) \neq 0$ for all integers $k \geq 1$, and therefore $u_1(z) = F(z, r_1)$ is (formally) a solution.

If $r_1 - r_2$ is not a whole number, then a second, linearly independent solution is $u_2(z) = F(z, r_2)$.

Roots differing by an integer

Suppose that $r_1 = r_2$. In this case, $I(r) = (r - r_1)^2$ and so

$$z^2 \frac{\partial^2 F}{\partial z^2} + zP(z)\frac{\partial F}{\partial z} + Q(z)F = (r - r_1)^2 z^r.$$

The function $v = \partial F / \partial r$ satisfies

$$z^2 \frac{\partial^2 v}{\partial z^2} + zP(z)\frac{\partial v}{\partial z} + Q(z)v = 2(r - r_1)z^r + (r - r_1)^2 z^r \ln z,$$

and the RHS is zero if $r = r_1$, so a second, linearly independent solution is

$$u_2(z) = \frac{\partial F}{\partial r}(z, r_1) = \sum_{k=0}^{\infty} \frac{\partial A_k}{\partial r}\bigg|_{r=r_1} z^{k+r_1} + \underbrace{\sum_{k=0}^{\infty} A_k(r_1) z^{k+r_1} \ln z}_{u_1(z)\ln z}.$$

Even worse complications if $r_1 = r_2 + n$ for an integer $n \geq 1$.

Bessel and Legendre equations

We wrap up this part of the course with two particularly important examples of second-order, linear ODEs with variable coefficients. Both occur several times later in the course.

Bessel equation

The Bessel equation with parameter ν is

$$z^2 u'' + zu' + (z^2 - \nu^2)u = 0.$$

This ODE is in Frobenius normal form, with indicial polynomial

$$I(r) = (r + \nu)(r - \nu),$$

and we seek a series solution

$$u(z) = \sum_{k=0}^{\infty} A_k z^{k+r}.$$

We assume $\operatorname{Re} \nu \geq 0$, so $r_1 = \nu$ and $r_2 = -\nu$.

Recurrence relation

We find that if

$$(r+1+\nu)(r+1-\nu)A_1 = 0,$$
$$(k+r+\nu)(k+r-\nu)A_k + A_{k-2} = 0, \quad k \geq 2.$$

then

$$z^2u'' + zu' + (z^2-\nu^2)u = (r+\nu)(r-\nu)A_0 z^r.$$

Taking $r = \nu$ gives

$$A_k = \frac{-A_{k-2}}{k(k+2\nu)} \quad \text{for } k \geq 2,$$

so with A_0 arbitrary and $A_1 = 0$ we obtain

$$u(z) = A_0 z^\nu \Bigg[1 - \frac{(z/2)^2}{1+\nu} + \frac{(z/2)^4}{2(2+\nu)(1+\nu)} - \frac{(z/2)^6}{3!(3+\nu)(2+\nu)(1+\nu)} + \cdots \Bigg].$$

Bessel function

With the normalisation

$$A_0 = \frac{1}{2^\nu \Gamma(1+\nu)}$$

the series solution is called the Bessel function of order ν and is denoted

$$J_\nu(z) = \frac{(z/2)^\nu}{\Gamma(1+\nu)} \left[1 - \frac{(z/2)^2}{1+\nu} + \frac{(z/2)^4}{2!(1+\nu)(2+\nu)} - \cdots \right].$$

From the functional equation $\Gamma(1+z) = z\Gamma(z)$ we see that

$$J_\nu(z) = \frac{(z/2)^\nu}{\Gamma(1+\nu)} - \frac{(z/2)^{\nu+2}}{\Gamma(2+\nu)} + \frac{(z/2)^{\nu+4}}{2!\Gamma(3+\nu)} - \frac{(z/2)^{\nu+6}}{3!\Gamma(4+\nu)} + \cdots$$

and so

$$J_\nu(z) = \sum_{k=0}^{\infty} \frac{(-1)^k (z/2)^{2k+\nu}}{k!\Gamma(k+1+\nu)}.$$

Bessel function of negative order

If ν is not an integer, then a second, linearly independent, solution is

$$J_{-\nu}(z) = \sum_{k=0}^{\infty} \frac{(-1)^k (z/2)^{2k-\nu}}{k!\Gamma(k+1-\nu)}.$$

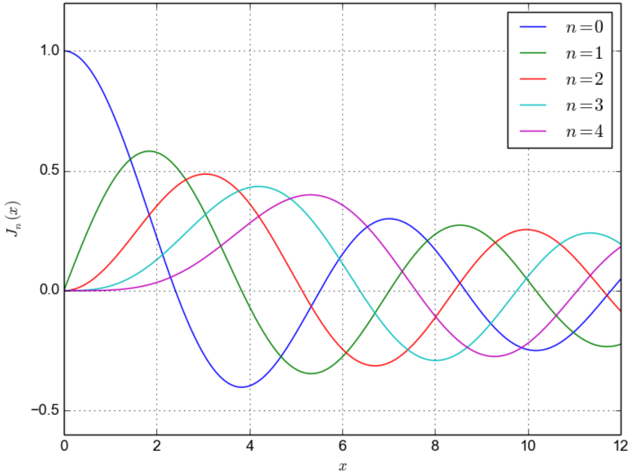
For an integer $\nu = n \in \mathbb{Z}$, since $\Gamma(n+1) = n!$ we have

$$J_n(z) = \sum_{k=0}^{\infty} \frac{(-1)^k (z/2)^{2k+n}}{k!(k+n)!}.$$

Also, since $1/\Gamma(z) = 0$ for $z = 0, -1, -2, \ldots$, we find that J_n and J_{-n} are linearly dependent; in fact,

$$J_{-n}(z) = (-1)^n J_n(z).$$

Bessel functions of integer order



Neumann function

The Neumann function (or Bessel function of the second kind) is

$$Y_\nu(z) = \frac{J_\nu(z)\cos\nu\pi - J_{-\nu}(z)}{\sin\nu\pi}, \quad \text{if } \nu \notin \mathbb{Z}.$$

For $n \in \mathbb{Z}$, L'Hospital's rule shows that if $\nu \to n$ then $Y_\nu(z)$ tends to a finite limit

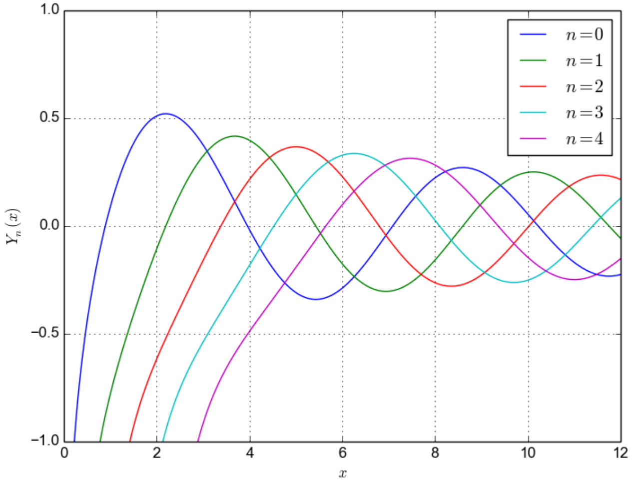
$$Y_n(z) = \lim_{\nu\to n} Y_\nu(z).$$

The functions J_ν and Y_ν are linearly independent solutions of Bessel's equation for all complex ν.
As $z \to 0$ with ν fixed,

$$J_\nu(z) \sim \frac{(z/2)^\nu}{\Gamma(\nu+1)}, \quad \nu \notin \{-1,-2,-3,\ldots\},$$

$$Y_0(z) \sim \frac{2}{\pi}\ln z, \qquad Y_\nu(z) \sim \frac{-\Gamma(\nu)}{\pi(z/2)^\nu}, \quad \operatorname{Re}\nu > 0.$$

Neumann functions of integer order



Legendre equation

The Legendre equation with parameter ν is

$$(1-z^2)u'' - 2zu' + \nu(\nu+1)u = 0.$$

This ODE is not singular at $z = 0$ so the solution has an ordinary Taylor series expansion

$$u = \sum_{k=0}^{\infty} A_k z^k.$$

The A_k must satisfy

$$(k+1)(k+2)A_{k+2} - [k(k+1) - \nu(\nu+1)]A_k = 0$$

for $k \geq 0$, and since

$$k(k+1) - \nu(\nu+1) = (k-\nu)(k+\nu+1),$$

the recurrence relation is

$$A_{k+2} = \frac{(k-\nu)(k+\nu+1)}{(k+1)(k+2)} A_k \quad \text{for } k \geq 0.$$

General solution

We have

$$u(z) = A_0 u_0(z) + A_1 u_1(z)$$

where

$$u_0(z) = 1 - \frac{\nu(\nu+1)}{2!}\, z^2 + \frac{(\nu-2)\nu(\nu+1)(\nu+3)}{4!}\, z^4 - \cdots$$

and

$$u_1(z) = z - \frac{(\nu-1)(\nu+2)}{3!}\, z^3 + \frac{(\nu-3)(\nu-1)(\nu+2)(\nu+4)}{5!}\, z^5 - \cdots .$$

Suppose now that $\nu = n$ is a non-negative integer. If n is even then the series for $u_0(z)$ terminates, whereas if n is odd then the series for $u_1(z)$ terminates.

Legendre polynomial

The terminating solution is called the Legendre polynomial of degree n and is denoted by $P_n(z)$ with the normalization

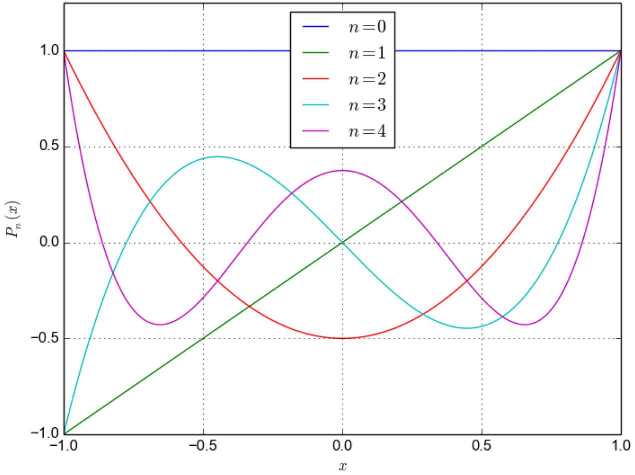
$$P_n(1) = 1.$$

The first few Legendre polynomials are

$$\begin{aligned} P_0(z) &= 1, & P_3(z) &= \tfrac{1}{2}(5z^3 - 3z), \\ P_1(z) &= z, & P_4(z) &= \tfrac{1}{8}(35z^4 - 30z^2 + 3), \\ P_2(z) &= \tfrac{1}{2}(3z^2 - 1), & P_5(z) &= \tfrac{1}{8}(63z^5 - 70z^3 + 15z). \end{aligned}$$

Notice that P_n is an even or odd function according to whether n is even or odd.

Behaviour of Legendre polynomials



Part II

Dynamical Systems

Introduction

In this part of the course, we survey several topics in the theory of differential equations (DEs). None is discussed in any depth. Instead, the aim is to introduce you to a few of the key themes in this subject, some of which are developed further in our third-year mathematics courses.

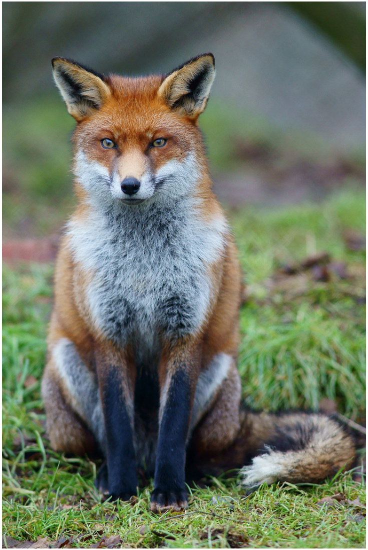
Examples and terminology

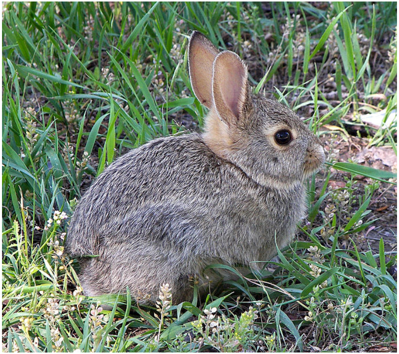
We begin with some examples of how systems of differential equations arise in applications, and see how all such problems can be formulated as a first-order system

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{F}(\boldsymbol{x}).$$

Such a formulation leads to a natural geometric interpretation of a solution.

Predator and prey populations





Lotka–Volterra equations

Simplified ecology with two species:

$$\begin{aligned} F(t) &= \text{number of foxes at time } t, \\ R(t) &= \text{number of rabbits at time } t. \end{aligned}$$

Assume populations large enough that F and R can be treated as smoothly varying in time.

In the 1920s, Alfred Lotka and Vito Volterra independently proposed the preditor–prey model

$$\begin{aligned} \frac{dF}{dt} &= -aF + \alpha FR, & F(0) &= F_0, \\ \frac{dR}{dt} &= bR - \beta FR, & R(0) &= R_0. \end{aligned}$$

Here, a, α, b and β are non-negative constants.

Zero predation case

If $\alpha = 0 = \beta$ then the system uncouples to give

$$\frac{dF}{dt} = -aF \quad \text{and} \quad \frac{dR}{dt} = bR$$

so

$$F(t) = F_0 e^{-at} \quad \text{and} \quad R(t) = R_0 e^{bt}.$$

Interpretation: if the foxes fail to catch any rabbits, then the foxes starve $(F \to 0)$ and the rabbits multiply without limit $(R \to \infty)$.

Interaction terms

Rewrite equations as

$$\frac{1}{F}\frac{dF}{dt} = -a + \alpha R,$$
$$\frac{1}{R}\frac{dR}{dt} = b - \beta F.$$

So

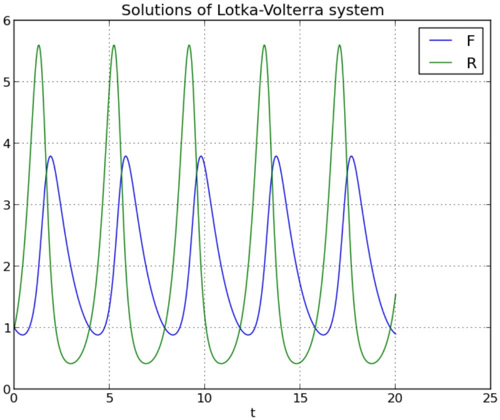
$$\alpha R = \begin{pmatrix}\text{relative rate of increase in fox popula-} \\ \text{tion due to predation on rabbits}\end{pmatrix}$$

and

$$\beta F = \begin{pmatrix}\text{relative rate of decrease in rabbit pop-} \\ \text{ulation due to predation by foxes}\end{pmatrix}.$$

Periodic solutions

Example with $a = 1.0$, $\alpha = 0.5$, $b = 1.5$ and $\beta = 0.75$.



Vector field

Any first-order system of N ODEs in the form

$$
\begin{aligned}
\frac{dx}{dt} &= F_1(x, y, \ldots, x_N), & x(0) &= x_{10},\\
\frac{dy}{dt} &= F_2(x, y, \ldots, x_N), & y(0) &= x_{20},\\
&\vdots & &\vdots\\
\frac{dx_N}{dt} &= F_N(x, y, \ldots, x_N), & x_N(0) &= x_{N0},
\end{aligned}
$$

can be written in vector notation as

$$
\frac{d\boldsymbol{x}}{dt} = \boldsymbol{F}(\boldsymbol{x}), \qquad \boldsymbol{x}(0) = \boldsymbol{x}_0.
$$

The system of ODEs is determined by the vector field $\boldsymbol{F} : \mathbb{R}^N \to \mathbb{R}^N$.

Predator–prey example

Recall

$$\frac{dF}{dt} = -aF + \alpha FR, \qquad F(0) = F_0,$$
$$\frac{dR}{dt} = bR - \beta FR, \qquad R(0) = R_0.$$

So defining

$$\boldsymbol{x} = \begin{bmatrix} F \\ R \end{bmatrix}, \qquad \boldsymbol{F}(\boldsymbol{x}) = \begin{bmatrix} -aF + \alpha FR \\ bR - \beta FR \end{bmatrix}, \qquad \boldsymbol{x}_0 = \begin{bmatrix} F_0 \\ R_0 \end{bmatrix}$$

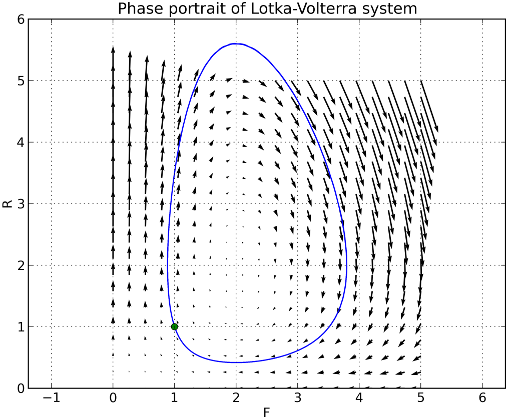
we have

$$\frac{d\boldsymbol{x}}{dt} = \begin{bmatrix} dF/dt \\ dR/dt \end{bmatrix} = \begin{bmatrix} -aF + \alpha FR \\ bR - \beta FR \end{bmatrix} = \boldsymbol{F}(\boldsymbol{x}),$$

with $\boldsymbol{x}(0) = \boldsymbol{x}_0$.

Geometric viewpoint

Think of the trajectory $\boldsymbol{x}(t)$ as a parametric curve in the phase space $\mathbb{R}^N$. Then $d\boldsymbol{x}/dt = \boldsymbol{F}(\boldsymbol{x})$ means that along any trajectory, $\boldsymbol{F}\big(\boldsymbol{x}(t)\big)$ always points in the forward tangent direction to the trajectory, with the "speed" of $\boldsymbol{x}(t)$ equal to the magnitude of $\boldsymbol{F}$.



Non-autonomous ODEs

A system of ODEs of the form

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{F}(\boldsymbol{x}).$$

is said to be autonomous.

For example, we have just seen that the Lotka–Volterra equations are autonomous.

In a non-autonomous system, $\boldsymbol{F}$ will depend explicitly on t:

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{F}(\boldsymbol{x}, t). \tag{7}$$

Equivalent autonomous system

Given a non-autonomous system (7), let

$$\boldsymbol{y} = \begin{bmatrix} \boldsymbol{x} \\ t \end{bmatrix} \quad \text{and} \quad \boldsymbol{G}(\boldsymbol{y}) = \begin{bmatrix} \boldsymbol{F}(\boldsymbol{x}, t) \\ 1 \end{bmatrix}.$$

If $\boldsymbol{x} = \boldsymbol{x}(t)$ is a solution of (7), then $\boldsymbol{y} = \boldsymbol{y}(t)$ is a solution of the autonomous system

$$\frac{d\boldsymbol{y}}{dt} = \begin{bmatrix} d\boldsymbol{x}/dt \\ dt/dt \end{bmatrix} = \begin{bmatrix} \boldsymbol{F}(\boldsymbol{x}, t) \\ 1 \end{bmatrix} = \boldsymbol{G}(\boldsymbol{y}),$$

and vice versa.

Therefore, sufficient (in principle) to develop theory for the autonomous case.

Second-order ODE

Consider an initial-value problem for a general (possibly non-autonomous) second-order ODE

$$\frac{d^2x}{dt^2} = f\left(x, \frac{dx}{dt}, t\right), \quad \text{with } x = x_0 \text{ and } \frac{dx}{dt} = y_0 \text{ at } t = 0.$$

If $x = x(t)$ is a solution, and if we let $y = dx/dt$, then

$$\frac{dy}{dt} = \frac{d^2x}{dt^2} = f\left(x, \frac{dx}{dt}, t\right) = f(x, y, t),$$

that is, (x, y) is a solution of the first-order system

$$\begin{aligned} \frac{dx}{dt} &= y, & x(0) &= x_0, \\ \frac{dy}{dt} &= f(x, y, t), & y(0) &= y_0. \end{aligned}$$

Simple oscillator as first-order system

The second-order ODE

$$m\ddot{x} + r_0\dot{x} + k_0 x = f(t)$$

is equivalent to the (non-autonomous) pair of first-order ODEs

$$\begin{aligned}\dot{x} &= v,\\ \dot{v} &= \frac{1}{m}\big(f(t) - r_0 v - k_0 x\big).\end{aligned}$$

That is,

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{F}(\boldsymbol{x}, t),$$

where

$$\boldsymbol{x} = \begin{bmatrix} x \\ v \end{bmatrix} \quad \text{and} \quad \boldsymbol{F}(\boldsymbol{x}, t) = \begin{bmatrix} v \\ m^{-1}\big(f(t) - r_0 v - k_0 x\big) \end{bmatrix}.$$

Existence and uniqueness

The most fundamental question about a dynamical system

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{F}(\boldsymbol{x}, t)$$

is

> *For a given initial value $\boldsymbol{x}_0$, does a solution $\boldsymbol{x}(t)$ satisfying $\boldsymbol{x}(0) = \boldsymbol{x}_0$ exist, and if so is this solution unique?*

Answer is yes, whenever the vector field $\boldsymbol{F}$ is Lipschitz.

Lipschitz constant

Definition

The number L is a Lipschitz constant for a function $f : [a, b] \to \mathbb{R}$ if

$$|f(x) - f(y)| \le L|x - y| \quad \text{for all } x,\ y \in [a, b].$$

Example

Consider $f(x) = 2x^2 - x + 1$ for $0 \le x \le 1$. Since

$$\begin{aligned} f(x) - f(y) &= 2(x^2 - y^2) - (x - y) = 2(x + y)(x - y) - (x - y) \\ &= (2x + 2y - 1)(x - y) \end{aligned}$$

we have $|f(x) - f(y)| = |2x + 2y - 1|\,|x - y|$ so a Lipschitz constant is

$$L = \max_{x,y \in [0,1]} |2x + 2y - 1| = 3.$$

Lipschitz implies (uniformly) continuous

We say that the function $f : [a, b] \to \mathbb{R}$ is Lipschitz if a Lipschitz constant for f exists.

Theorem

If f is Lipschitz then f is (uniformly) continuous.

Proof.

Suppose L is a Lipschitz constant for $f : [a, b] \to \mathbb{R}$. Given $\epsilon > 0$, if $\delta = \epsilon/L$ then

$$|f(x) - f(y)| \le L|x - y| < \epsilon \quad \text{whenever } |x - y| < \delta,$$

so f is (uniformly) continuous on $[a, b]$. $\square$

Continuous does not imply Lipschitz

Example

Consider the (uniformly) continuous function

$$f(x) = 3 + \sqrt{x} \quad \text{for } 0 \leq x \leq 4.$$

In this case, if x, $y \in (0, 4]$ then

$$\begin{aligned} f(x) - f(y) = \sqrt{x} - \sqrt{y} &= \left(\sqrt{x} - \sqrt{y}\right) \times \frac{\sqrt{x} + \sqrt{y}}{\sqrt{x} + \sqrt{y}} \\ &= \frac{x - y}{\sqrt{x} + \sqrt{y}} \end{aligned}$$

so if a Lipschitz constant L exists then

$$L \geq \frac{|f(x) - f(y)|}{|x - y|} = \frac{1}{\sqrt{x} + \sqrt{y}}$$

for arbitrarily small x and y, a contradiction.

Continuously differentiable implies Lipschitz

Definition

A function $f : I \to \mathbb{R}$ is C^k if f, f', f'', $\ldots$, $f^{(k)}$ all exist and are continous on the interval I.

Theorem

For any closed and bounded interval $I = [a, b]$, if f is C^1 on I then $L = \max_{x\in I} |f'(x)|$ is a Lipschitz constant for f on I.

Proof.

Given $a \leq x < y \leq b$, the Mean Value Theorem says that there exists a number c (depending on x and y) such that

$$f(x) - f(y) = f'(c)(x - y) \quad \text{with } x < c < y,$$

so $|f(x) - f(y)| = |f'(c)|\,|x - y| \leq L|x - y|$. □

Lipschitz vector field

A vector field $\boldsymbol{F} : S \to \mathbb{R}^N$ is Lipschitz on $S \subseteq \mathbb{R}^N$ if

$$\|\boldsymbol{F}(\boldsymbol{x}) - \boldsymbol{F}(\boldsymbol{y})\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\| \quad \text{for all } \boldsymbol{x},\ \boldsymbol{y} \in S.$$

Here,

$$\|\boldsymbol{x}\| = \left(\sum_{j=1}^{N} x_j^2\right)^{1/2}$$

denotes the Euclidean norm of the vector $\boldsymbol{x} \in \mathbb{R}^N$.

We say that $\boldsymbol{F}(\boldsymbol{x}, t)$ is Lipschitz in $\boldsymbol{x}$ if

$$\|\boldsymbol{F}(\boldsymbol{x}, t) - \boldsymbol{F}(\boldsymbol{y}, t)\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|.$$

Local existence and uniqueness

Theorem

Let $\boldsymbol{x}_0 \in \mathbb{R}^N$*, fix* $r > 0$ *and* $\tau > 0$*, and put*

$$S = \{\, (\boldsymbol{x}, t) \in \mathbb{R}^N \times \mathbb{R} : \|\boldsymbol{x} - \boldsymbol{x}_0\| \le r \text{ and } |t| \le \tau \,\}.$$

If $\boldsymbol{F}(\boldsymbol{x}, t)$ *is Lipschitz in* $\boldsymbol{x}$ *for* $(\boldsymbol{x}, t) \in S$*, and if*

$$\|\boldsymbol{F}(\boldsymbol{x}, t)\| \le M \quad \text{for } (\boldsymbol{x}, t) \in S,$$

then there exists a unique C^1 *function* $\boldsymbol{x}(t)$ *satisfying*

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{F}(\boldsymbol{x}, t) \quad \text{for } |t| \le \min(r/M, \tau), \text{ with } \boldsymbol{x}(0) = \boldsymbol{x}_0.$$

Proof.

See *Technical Proofs* handout. □

Example of non-uniqueness

The initial-value problem

$$\frac{dx}{dt} = 3x^{2/3} \quad \text{for } t > 0, \text{ with } x(0) = 0,$$

has infinitely many solutions, namely, for any $a > 0$,

$$x(t) = \begin{cases} 0, & 0 \le t \le a, \\ (t-a)^3, & t > a. \end{cases}$$

In this case, $f(x) = 3x^{2/3}$ is not Lipschitz on any neighbourhood of 0 because

$$\frac{|f(x) - f(0)|}{|x - 0|} = \frac{3x^{2/3} - 0}{x} = 3x^{-1/3} \to \infty \quad \text{as } x \to 0.$$

Example of local, but not global, existence

The (separable) initial-value problem

$$\frac{dx}{dt} = 1 + x^2 \quad \text{for } t > 0\text{, with } x(0) = 1,$$

has a unique solution

$$x(t) = \tan\left(t + \frac{\pi}{4}\right) \quad \text{for } 0 < t < \frac{\pi}{4}.$$

Applying the theorem with $f(x) = 1 + x^2$, $x_0 = 1$, $r > 0$ and $\tau = \infty$, we find that

$$|f(x)| \le M = r^2 + 2r + 2 \quad \text{for } |x - x_0| \le r,$$

and $\min(r/M, \tau)$ equals

$$\frac{r}{r^2 + 2r + 2} \le \left.\frac{r}{r^2 + 2r + 2}\right|_{r=\sqrt{2}} = \frac{\sqrt{2} - 1}{2} < \frac{\pi}{4}.$$

Distinct trajectories cannot intersect

Suppose that the trajectories of two solutions $\boldsymbol{x}_1(t)$ and $\boldsymbol{x}_2(t)$ intersect, that is,

$$\boldsymbol{x}_1(t_1) = \boldsymbol{x}_2(t_2)$$

for some t_1 and t_2. If we define

$$\boldsymbol{y}_1(t) = \boldsymbol{x}_1(t_1 + t) \quad \text{and} \quad \boldsymbol{y}_2(t) = \boldsymbol{x}_2(t_2 + t),$$

then, for $j \in \{1, 2\}$,

$$\dot{\boldsymbol{y}}_j(t) = \dot{\boldsymbol{x}}_j(t_j + t) = \boldsymbol{F}\big(\boldsymbol{x}_j(t_j + t)\big) = \boldsymbol{F}\big(\boldsymbol{y}_j(t)\big).$$

Since $\boldsymbol{y}_1(0) = \boldsymbol{x}_1(t_1) = \boldsymbol{x}_2(t_2) = \boldsymbol{y}_2(0)$, uniqueness implies that $\boldsymbol{y}_1(t) = \boldsymbol{y}_2(t)$ for all t, or in other words

$$\boldsymbol{x}_1(t_1 + t) = \boldsymbol{x}_2(t_2 + t) \quad \text{for all } t.$$

Therefore, the two solutions trace out the same trajectory in phase space.

Linear dynamical systems

Linear differential equations are generally much easier to solve than nonlinear ones. Hence, most of this course is devoted to linear problems. Fortunately, linear DEs suffice for describing many important applications.

Also, understanding how the solutions of linear ODEs behave can help us understand nonlinear ODEs.

First-order, linear systems of ODEs

We say that the $N \times N$, first-order system of ODEs

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{F}(\boldsymbol{x}, t)$$

is linear if the RHS has the form

$$\boldsymbol{F}(\boldsymbol{x}, t) = A(t)\boldsymbol{x} + \boldsymbol{b}(t)$$

for some $N \times N$ matrix-valued function $A(t) = [a_{ij}(t)]$ and a vector-valued function $\boldsymbol{b}(t) = [b_i(t)]$.

The system is autonomous precisely when A and $\boldsymbol{b}$ are constant.

Global existence and uniqueness

We have a stronger existence result in the linear case.

Theorem

If the elements of $A(t)$ and components of $\boldsymbol{b}(t)$ are continuous for $0 \leq t \leq T$, then the linear initial-value problem

$$\frac{d\boldsymbol{x}}{dt} = A(t)\boldsymbol{x} + \boldsymbol{b}(t) \quad \text{for } 0 \leq t \leq T, \quad \text{with } \boldsymbol{x}(0) = \boldsymbol{x}_0,$$

has a unique solution $\boldsymbol{x}(t)$ for $0 \leq t \leq T$.

We now investigate the special case when A is constant and $\boldsymbol{b}(t) \equiv \mathbf{0}$:

$$\frac{d\boldsymbol{x}}{dt} = A\boldsymbol{x}.$$

General solution via eigensystem

If $\boldsymbol{v}$ is a constant vector and $A\boldsymbol{v} = \lambda\boldsymbol{v}$, we define $\boldsymbol{x}(t) = e^{\lambda t}\boldsymbol{v}$. Then

$$\frac{d\boldsymbol{x}}{dt} = \lambda e^{\lambda t}\boldsymbol{v} = e^{\lambda t}(\lambda\boldsymbol{v}) = e^{\lambda t}(A\boldsymbol{v}) = A(e^{\lambda t}\boldsymbol{v}) = A\boldsymbol{x}$$

that is, $\boldsymbol{x}$ is a solution of $d\boldsymbol{x}/dt = A\boldsymbol{x}$.
If $A\boldsymbol{v}_j = \lambda_j\boldsymbol{v}_j$ for $1 \leq j \leq N$, then the linear combination

$$\boldsymbol{x}(t) = \sum_{j=1}^{N} c_j e^{\lambda_j t}\boldsymbol{v}_j \tag{8}$$

is also a solution because the ODE is linear and homogeneous.
Provided the $\boldsymbol{v}_j$ are linearly independent, (8) is the general solution because given any $\boldsymbol{x}_0 \in \mathbb{R}^N$ there exist unique c_j such that

$$\boldsymbol{x}(0) = \sum_{j=1}^{N} c_j\boldsymbol{v}_j = \boldsymbol{x}_0.$$

Example

Consider

$$\frac{dx}{dt} = -5x + 2y, \qquad\qquad x(0) = 5,$$
$$\frac{dy}{dt} = -6x + 3y, \qquad\qquad y(0) = 7.$$

In this case,

$$A = \begin{bmatrix} -5 & 2 \\ -6 & 3 \end{bmatrix}, \quad \lambda_1 = -3, \quad \boldsymbol{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \lambda_2 = 1, \quad \boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix},$$

so the general solution is

$$\boldsymbol{x}(t) = c_1 e^{-3t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{t} \begin{bmatrix} 1 \\ 3 \end{bmatrix},$$

and the initial conditions imply $c_1 = 4$ and $c_2 = 1$, so

$$\boldsymbol{x}(t) = 4e^{-3t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + e^{t} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \qquad \text{or} \qquad \begin{aligned} x &= 4e^{-3t} + e^{t}, \\ y &= 4e^{-3t} + 3e^{t}. \end{aligned}$$

Exponential of a matrix

Recall that the Taylor series

$$e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots$$

converges for all $z \in \mathbb{C}$ (Math2621).

By analogy, given $A \in \mathbb{C}^{N\times N}$, we define (Math2601)

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots .$$

This series always converges because, for any matrix norm (),

$$\|e^A\| \le \sum_{k=0}^{\infty} \frac{\|A^k\|}{k!} \le \sum_{k=0}^{\infty} \frac{\|A\|^k}{k!} = e^{\|A\|}.$$

Term-by-term differentiation

Given A and $\boldsymbol{x}_0$, let

$$\boldsymbol{x}(t) = e^{tA}\boldsymbol{x}_0 = \left(I + tA + \frac{t^2A^2}{2!} + \cdots + \frac{t^kA^k}{k!} + \cdots\right)\boldsymbol{x}_0.$$

Since

$$\begin{aligned}\frac{d}{dt}\,e^{tA} &= 0 + A + tA^2 + \cdots + \frac{t^{k-1}A^k}{(k-1)!} + \cdots \\ &= A\left(I + tA + \cdots + \frac{t^{k-1}A^{k-1}}{(k-1)!} + \cdots\right) = Ae^{tA},\end{aligned}$$

we have

$$\frac{d\boldsymbol{x}}{dt} = Ae^{tA}\boldsymbol{x}_0 = A\boldsymbol{x} \quad \text{and} \quad \boldsymbol{x}(0) = I\boldsymbol{x}_0 = \boldsymbol{x}_0.$$

But how can we calculate e^{tA} practically?

Diagonalising a matrix

Definition

A square matrix $A \in \mathbb{C}^{N\times N}$ is diagonalisable if there exists a non-singular matrix $Q \in \mathbb{C}^{N\times N}$ such that $Q^{-1}AQ$ is diagonal.

Theorem

A square matrix $A \in \mathbb{C}^{N\times N}$ is diagonalisable if and only if there exists a basis $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_N\}$ for $\mathbb{C}^N$ consisting of eigenvectors of A. Indeed, if

$$A\boldsymbol{v}_j = \lambda_j \boldsymbol{v}_j \quad \textit{for } j = 1,\, 2,\, \ldots,\, N,$$

and we put $Q = [\boldsymbol{v}_1 \quad \boldsymbol{v}_2 \quad \cdots \quad \boldsymbol{v}_N]$, then $Q^{-1}AQ = \Lambda$ where

$$\Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_N \end{bmatrix}.$$

Matrix powers

Consider a diagonalisable matrix A. Since $Q^{-1}AQ = \Lambda$, it follows that A has an eigenvalue decomposition

$$A = Q\Lambda Q^{-1}.$$

Thus

$$\begin{aligned} A^2 &= (Q\Lambda Q^{-1})(Q\Lambda Q^{-1}) = Q\Lambda(Q^{-1}Q)\Lambda Q^{-1} = Q\Lambda I \Lambda Q^{-1} \\ &= Q\Lambda^2 Q^{-1} \end{aligned}$$

and

$$A^3 = A^2 A = Q\Lambda^2 Q^{-1} Q\Lambda Q^{-1} = Q\Lambda^3 Q^{-1}.$$

In general, we see by induction on k that

$$A^k = Q\Lambda^k Q^{-1} \quad \text{for } k = 0,\ 1,\ 2,\ \ldots.$$

Since Λ is diagonal, so is

$$\Lambda^2 = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_N \end{bmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_N \end{bmatrix}$$

$$= \begin{bmatrix} \lambda_1^2 & & & \\ & \lambda_2^2 & & \\ & & \ddots & \\ & & & \lambda_N^2 \end{bmatrix},$$

and in general, we see by induction on k that

$$\Lambda^k = \begin{bmatrix} \lambda_1^k & & & \\ & \lambda_2^k & & \\ & & \ddots & \\ & & & \lambda_N^k \end{bmatrix} \quad \text{for } k = 0,\ 1,\ 2,\ 3,\ \ldots.$$

Example

If

$$A = \begin{bmatrix} -5 & 2 \\ -6 & 3 \end{bmatrix}$$

then

$$\Lambda = \begin{bmatrix} -3 & 0 \\ 0 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}, \quad Q^{-1} = \frac{1}{2}\begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix}$$

so

$$\begin{aligned} A^k = Q\Lambda^k Q^{-1} &= \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} (-3)^k & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix} \\ &= \frac{1}{2}\begin{bmatrix} (-1)^k \times 3^{k+1} - 1 & (-1)^{k+1} \times 3^k + 1 \\ (-1)^k \times 3^{k+1} - 3 & (-1)^{k+1} \times 3^k + 3 \end{bmatrix}. \end{aligned}$$

Polynomial of a matrix

For any polynomial

$$p(z) = c_0 + c_1 z + c_2 z^2 + \cdots + c_m z^m$$

and any square matrix A, we define

$$p(A) = c_0 I + c_1 A + c_2 A^2 + \cdots + c_m A^m.$$

When A is diagonalisable, $A^k = Q\Lambda^k Q^{-1}$ so

$$\begin{aligned}
p(A) &= c_0 QIQ^{-1} + c_1 Q\Lambda Q^{-1} + c_2 Q\Lambda^2 Q^{-1} + \cdots \\
&\qquad + c_m Q\Lambda^m Q^{-1} \\
&= \big(c_0 QI + c_1 Q\Lambda + c_2 Q\Lambda^2 + \cdots + c_m Q\Lambda^m\big)Q^{-1} \\
&= Q\big(c_0 I + c_1 \Lambda + c_2 \Lambda^2 + \cdots + c_m \Lambda^m\big)Q^{-1} \\
&= Qp(\Lambda)Q^{-1}.
\end{aligned}$$

Lemma

For any polynomial p and any diagonal matrix Λ,

$$p(\Lambda) = \begin{bmatrix} p(\lambda_1) & & & \\ & p(\lambda_2) & & \\ & & \ddots & \\ & & & p(\lambda_N) \end{bmatrix}.$$

Theorem

If two polynomials p and q are equal on the spectrum of a diagonalisable matrix A, that is, if

$$p(\lambda_j) = q(\lambda_j) \quad \text{for } j = 1, 2, \ldots, N,$$

then $p(A) = q(A)$.

Example

Recall that

$$A = \begin{bmatrix} -5 & 2 \\ -6 & 3 \end{bmatrix}$$

has eigenvalues $\lambda_1 = -3$ and $\lambda_2 = 1$. Let

$$p(z) = z^2 - 4 \quad \text{and} \quad q(z) = -2z - 1,$$

and observe

$$p(-3) = 5 = q(-3) \quad \text{and} \quad p(1) = -3 = q(1).$$

We find

$$p(A) = A^2 - 4I = \begin{bmatrix} 9 & -4 \\ 12 & -7 \end{bmatrix} = -2A - I = q(A).$$

Exponential of a diagonalisable matrix

Theorem

If $A = Q\Lambda Q^{-1}$ *is diagonalisable, then*

$$e^A = Qe^{\Lambda}Q^{-1} \quad \textit{and} \quad e^{\Lambda} = \begin{bmatrix} e^{\lambda_1} & & & \\ & e^{\lambda_2} & & \\ & & \ddots & \\ & & & e^{\lambda_N} \end{bmatrix}$$

Proof.

e^A equals

$$\sum_{k=0}^{\infty} \frac{A^k}{k!} = \sum_{k=0}^{\infty} \frac{Q\Lambda^k Q^{-1}}{k!} = Q \sum_{k=0}^{\infty} \frac{\Lambda^k Q^{-1}}{k!} = Q \left(\sum_{k=0}^{\infty} \frac{\Lambda^k}{k!} \right) Q^{-1}.$$

□

Example

Again put

$$A = \begin{bmatrix} -5 & 2 \\ -6 & 3 \end{bmatrix}.$$

We have

$$\begin{aligned} e^A = Qe^{\Lambda}Q^{-1} &= \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} e^{-3} & 0 \\ 0 & e^{1} \end{bmatrix}\begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix} \\ &= \frac{1}{2}\begin{bmatrix} 3e^{-3} - e & -e^{-3} + e \\ 3e^{-3} - 3e & -e^{-3} + 3e \end{bmatrix}. \end{aligned}$$

Notice $tA = Q(t\Lambda)Q^{-1}$ so

$$\begin{aligned} e^{tA} = Qe^{t\Lambda}Q^{-1} &= \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} e^{-3t} & 0 \\ 0 & e^{t} \end{bmatrix}\begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix} \\ &= \frac{1}{2}\begin{bmatrix} 3e^{-3t} - e^{t} & -e^{-3t} + e^{t} \\ 3e^{-3t} - 3e^{t} & -e^{-3t} + 3e^{t} \end{bmatrix}. \end{aligned}$$

Stability

In many applications we are interested to know how the solution $\boldsymbol{x}(t)$ behaves as $t \to \infty$, and might not care much about the precise details of the transient behaviour for finite t. A lot can be discovered without the need to solve the DE, which is just as well because finding $\boldsymbol{x}(t)$ explicitly is often difficult or impossible, especially for nonlinear problems.

Equilibrium points

We say that $\boldsymbol{a} \in \mathbb{R}^N$ is an equilibrium point for the dynamical system $d\boldsymbol{x}/dt = \boldsymbol{F}(\boldsymbol{x})$ if

$$\boldsymbol{F}(\boldsymbol{a}) = \mathbf{0}.$$

Thus the solution of

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{F}(\boldsymbol{x}) \quad \text{for all } t\text{, with } \boldsymbol{x}(0) = \boldsymbol{a},$$

is just the constant function $\boldsymbol{x}(t) = \boldsymbol{a}$.

Stable equilibrium

Definition

An equilibrium point $\boldsymbol{a}$ is stable if for every $\epsilon > 0$ there exists $\delta > 0$ such that whenever $\|\boldsymbol{x}_0 - \boldsymbol{a}\| < \delta$ the solution of

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{F}(\boldsymbol{x}) \quad \text{for } t > 0, \text{ with } \boldsymbol{x}(0) = \boldsymbol{x}_0$$

satisfies

$$\|\boldsymbol{x}(t) - \boldsymbol{a}\| < \epsilon \quad \text{for all } t > 0.$$

(In particular, $\boldsymbol{x}(t)$ must exist for all $t > 0$.)

In this case, if $\boldsymbol{x}_0 \to \boldsymbol{a}$ then $\boldsymbol{x}(t) \to \boldsymbol{a}$, uniformly for $t > 0$.

In particular, a trajectory stays close to a stable equilibrium point $\boldsymbol{a}$ if it starts out sufficiently close to $\boldsymbol{a}$.

Asymptotic stability

Definition

Let D be an open subset of $\mathbb{R}^N$ that contains an equilibrium point $\boldsymbol{a}$. We say that $\boldsymbol{a}$ is asymptotically stable in D if $\boldsymbol{a}$ is stable and, whenever $\boldsymbol{x}_0 \in D$, the solution of

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{F}(\boldsymbol{x}) \quad \text{for } t > 0, \text{ with } \boldsymbol{x}(0) = \boldsymbol{x}_0$$

satisfies

$$\boldsymbol{x}(t) \to \boldsymbol{a} \quad \text{as } t \to \infty.$$

In this case D is called a domain of attraction for $\boldsymbol{a}$.

Linear, constant-coefficient case

Consider

$$\frac{d\boldsymbol{x}}{dt} = A\boldsymbol{x} + \boldsymbol{b} \quad \text{with } \boldsymbol{x}(0) = \boldsymbol{x}_0 \text{ and } det(A) \neq 0 \tag{9}$$

Since

$$A\boldsymbol{x} + \boldsymbol{b} = \boldsymbol{0} \iff \boldsymbol{x} = -A^{-1}\boldsymbol{b},$$

the only equilibrium point is $\boldsymbol{a} = -A^{-1}\boldsymbol{b}$. Moreover,

$$\boldsymbol{x}(t) = \boldsymbol{a} + e^{tA}(\boldsymbol{x}_0 - \boldsymbol{a})$$

because

$$\frac{d\boldsymbol{x}}{dt} = Ae^{tA}(\boldsymbol{x}_0 - \boldsymbol{a}) = A(\boldsymbol{x} - \boldsymbol{a}) = A\boldsymbol{x} - A\boldsymbol{a} = A\boldsymbol{x} + \boldsymbol{b}$$

and

$$\boldsymbol{x}(0) = \boldsymbol{a} + I(\boldsymbol{x}_0 - \boldsymbol{a}) = \boldsymbol{x}_0.$$

Criteria for stability

Theorem

Let A be a diagonalisable matrix with eigenvalues λ_1, λ_2, ..., λ_N. The equilibrium point $\boldsymbol{a} = -A^{-1}\boldsymbol{b}$ is of (9)

1. *stable if and only if $\operatorname{Re}\lambda_j \leq 0$ for all j.*
2. *asymptotically stable if and only if $\operatorname{Re}\lambda_j < 0$ for all j.*

In the second case, the domain of attraction is the whole of $\mathbb{R}^N$.

Proofs

1. Using the eigenvalue decomposition $A = Q\Lambda Q^{-1}$, we have

$$\boldsymbol{x}(t) - \boldsymbol{a} = e^{tA}(\boldsymbol{x}_0 - \boldsymbol{a}) = Qe^{t\Lambda}Q^{-1}(\boldsymbol{x}_0 - \boldsymbol{a}).$$

If $\operatorname{Re}\lambda_j \leq 0$ then $0 < |e^{\lambda_j t}| = e^{(\operatorname{Re}\lambda_j)t} \leq 1$ for all $t \geq 0$, so

$$\|e^{t\Lambda}\boldsymbol{w}\|^2 = \sum_{j=1}^{N}\left|e^{\lambda_j t}w_j\right|^2 \leq \sum_{j=1}^{N}|w_j|^2 = \boldsymbol{w}\cdot\boldsymbol{w} = \|\boldsymbol{w}\|^2,$$

implying that $\boldsymbol{a}$ is stable. Otherwise, $\operatorname{Re}\lambda_j > 0$ for at least one j and $e^{\lambda_j t} \to \infty$ as $t \to \infty$.

2. For asymptotic stability it is necessary and sufficient that $e^{\lambda_j t} \to 0$ as $t \to \infty$, for all j. $\square$

Classification of 2d linear systems with $\det A \neq 0$

- The equilibrium point $\boldsymbol{a} = \mathbf{0}$ may be asymptotically stable, stable or unstable but may also have various other properties.
- Here we discuss some common categories of equilibrium points.

Case 1: Real eigenvalues λ_1 and λ_2 and two linearly independent eigenvectors $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$

General solution:

$$\boldsymbol{x} = c_1 e^{\lambda_1 t} \boldsymbol{v}_1 + c_2 e^{\lambda_2 t} \boldsymbol{v}_2.$$

Canonical form:

$$\Lambda = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}.$$

Stable node example ($\lambda_2 < \lambda_1 < 0$)

$$\frac{dx}{dt} = -x, \quad \frac{dy}{dt} = -2y, \qquad A = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix}$$

Eigenvalues and eigenvectors:

$$\lambda_1 = -1, \quad \lambda_2 = -2, \quad \boldsymbol{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \boldsymbol{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

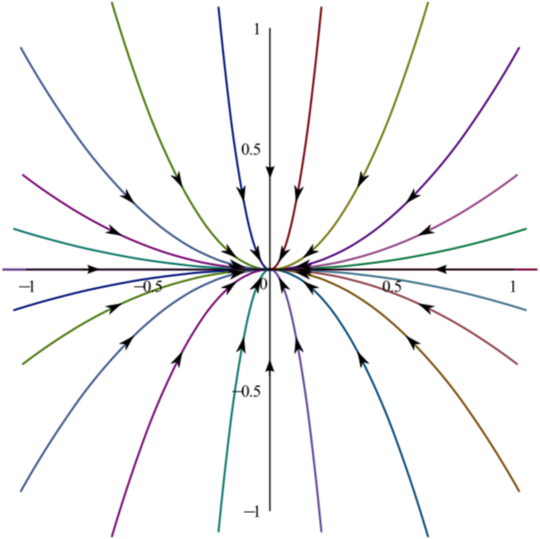
The general solution is

$$\boldsymbol{x} = c_1 e^{-t}\boldsymbol{v}_1 + c_2 e^{-2t}\boldsymbol{v}_2 = \begin{pmatrix} c_1 e^{-t} \\ c_2 e^{-2t} \end{pmatrix}.$$

Solution of initial value problem:

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} x(0)e^{-t} \\ y(0)e^{-2t} \end{pmatrix}.$$

A stable node $a = 0$



Unstable node example $(0 < \lambda_1 < \lambda_2)$

$$\frac{dx}{dt} = x, \quad \frac{dy}{dt} = 2y, \qquad A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

Eigenvalues and eigenvectors:

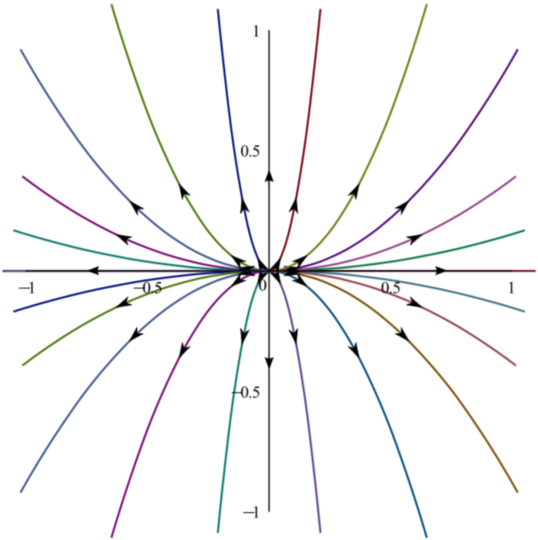
$$\lambda_1 = 1, \quad \lambda_2 = 2, \quad \boldsymbol{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \boldsymbol{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Solution of initial value problem:

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} x(0)e^{t} \\ y(0)e^{2t} \end{pmatrix}.$$

All trajectories (except $\boldsymbol{x}(t) = \mathbf{0}$) are repelled from equilibrium point which is unstable.

An unstable node $a = 0$



(Un)stable stars: $\lambda_1 = \lambda_2 \neq 0$

General form

$$\frac{dx}{dt} = \lambda_1 x, \quad \frac{dy}{dt} = \lambda_1 y, \qquad A = \lambda_1 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

All vectors are eigenvectors.

The general solution is

$$\boldsymbol{x} = e^{\lambda_1 t}\boldsymbol{v}.$$

Solution of initial value problem:

$$\boldsymbol{x}(t) = e^{\lambda_1 t}\boldsymbol{x}(0).$$

All orbits (except $\boldsymbol{x}(t) = \mathbf{0}$) are oriented half-lines which are either attracted ($\lambda_1 < 0$) or repelled ($\lambda_1 > 0$) by the equilibrium point.

Saddle node example (unstable: $\lambda_2 < 0 < \lambda_1$)

$$\frac{dx}{dt} = x + 2y, \quad \frac{dy}{dt} = 3x + 2y, \qquad A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$$

Eigenvalues and eigenvectors:

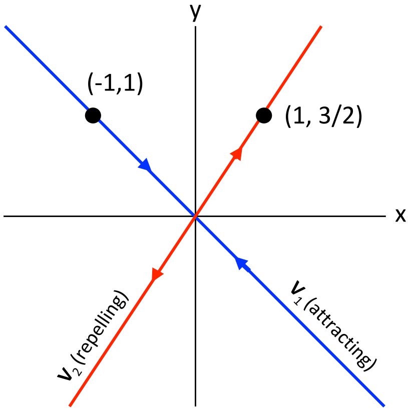
$$\lambda_1 = -1, \quad \lambda_2 = 4, \quad \boldsymbol{v}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \quad \boldsymbol{v}_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

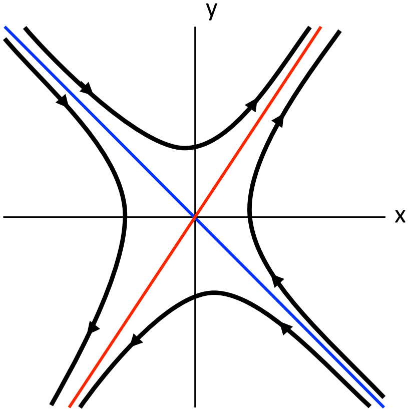
Solution of initial value problem:

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = c_1 e^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + c_2 e^{4t} \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$$

Here the first solution is repelling and the second is attracting, so the solution is unstable.

A saddle node $a = 0$ (unstable)





Classification of 2d linear systems with $\det A \neq 0$

Case 2: Complex conjugate eigenvalues $\lambda_1 = \bar{\lambda}_2 \notin \mathbb{R}$

General solution:

$$\boldsymbol{x} = c_1 \operatorname{Re}(e^{\lambda_1 t}\boldsymbol{v}_1) + c_2 \operatorname{Im}(e^{\lambda_1 t}\boldsymbol{v}_1).$$

Canonical form:

$$A = \begin{pmatrix} \alpha & \beta \\ -\beta & \alpha \end{pmatrix}, \qquad \lambda_1 = \alpha + i\beta.$$

From now on we focus on the canonical dynamical system

$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} \alpha & \beta \\ -\beta & \alpha \end{pmatrix} \boldsymbol{x}.$$

Eigenvector:

$$\mathbf{0} = (A - \lambda_1 I)\boldsymbol{v}_1 = \begin{pmatrix} -i\beta & \beta \\ -\beta & -i\beta \end{pmatrix} \boldsymbol{v}_1 \quad \Rightarrow \quad \boldsymbol{v}_1 = \begin{pmatrix} 1 \\ i \end{pmatrix}$$

Complex solution:

$$e^{\lambda_1 t}\boldsymbol{v}_1 = e^{\alpha t}(\cos\beta t + i\sin\beta t)\begin{pmatrix} 1 \\ i \end{pmatrix}$$

General real solution:

$$\boldsymbol{x} = e^{\alpha t}\left[c_1 \begin{pmatrix} \cos\beta t \\ -\sin\beta t \end{pmatrix} + c_2 \begin{pmatrix} \sin\beta t \\ \cos\beta t \end{pmatrix}\right]$$

Solution of initial value problem:

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^{\alpha t}\left[x(0) \begin{pmatrix} \cos\beta t \\ -\sin\beta t \end{pmatrix} + y(0) \begin{pmatrix} \sin\beta t \\ \cos\beta t \end{pmatrix}\right]$$

Interpretation

$$\boldsymbol{x}(t) = e^{\alpha t} R(t)\boldsymbol{x}(0), \qquad R(t) = \begin{pmatrix} \cos\beta t & \sin\beta t \\ -\sin\beta t & \cos\beta t \end{pmatrix}$$

Thus, the initial vector $\boldsymbol{x}(0)$ is rotated by the rotation matrix $R(t)$ and scaled by the factor $e^{\alpha t}$.

Proof.

$R(t)$ is indeed a rotation matrix since $\det R(t) = 1$ and

$$[R(t)]^T R(t) = \begin{pmatrix} \cos\beta t & -\sin\beta t \\ \sin\beta t & \cos\beta t \end{pmatrix} \begin{pmatrix} \cos\beta t & \sin\beta t \\ -\sin\beta t & \cos\beta t \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

and, hence,

$$|\boldsymbol{x}(t)|^2 = e^{2\alpha t}[\boldsymbol{x}(0)]^T[R(t)]^T R(t)\boldsymbol{x}(0) = e^{2\alpha t}[\boldsymbol{x}(0)]^T\boldsymbol{x}(0) = e^{2\alpha t}|\boldsymbol{x}(0)|^2. \qquad \square$$

Centre example (stable: $\mathrm{Re}(\lambda_1) = 0$

$$\frac{dx}{dt} = -2y, \quad \frac{dy}{dt} = 2x, \qquad A = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$$

Eigenvalues:

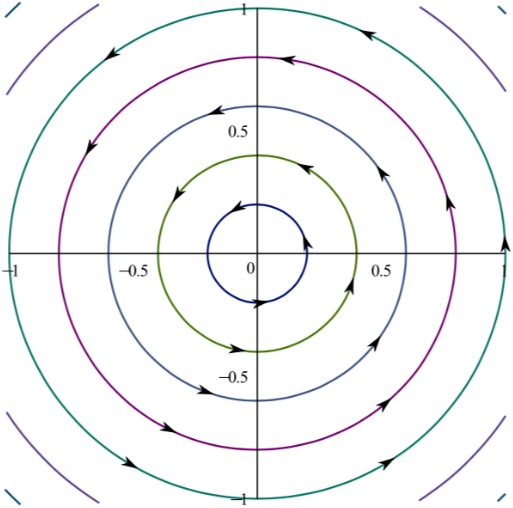
$$\lambda_1 = \bar{\lambda}_2 = -2i$$

Solution of initial value problem:

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} \cos 2t & -\sin 2t \\ \sin 2t & \cos 2t \end{pmatrix} \begin{pmatrix} y(0) \\ y(0) \end{pmatrix}.$$

The solution constitutes orbits which are oriented circles. These are stable (but not asymptotically stable).

A centre $\boldsymbol{a} = \mathbf{0}$ (stable)



Stable foci example $(\mathrm{Re}(\lambda_1) < 0)$

$$\frac{dx}{dt} = -x - 2y, \quad \frac{dy}{dt} = 2x - y, \qquad A = \begin{pmatrix} -1 & -2 \\ 2 & -1 \end{pmatrix}$$

Eigenvalues:

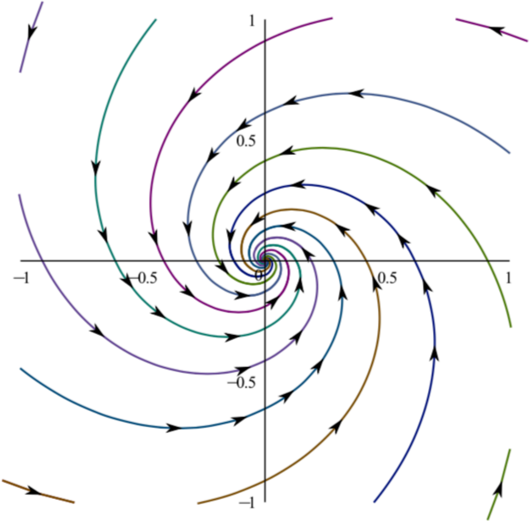
$$\lambda_1 = \bar{\lambda}_2 = -1 - 2i$$

Solution of initial value problem:

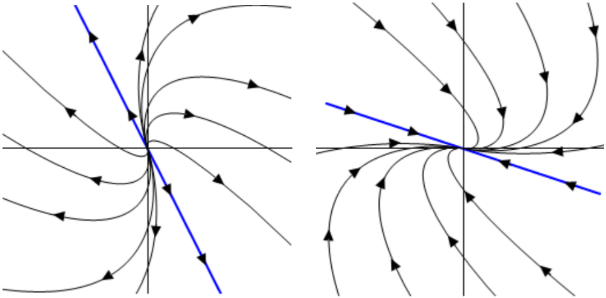
$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^{-t} \begin{pmatrix} \cos 2t & -\sin 2t \\ \sin 2t & \cos 2t \end{pmatrix} \begin{pmatrix} x(0) \\ y(0) \end{pmatrix} \to \mathbf{0} \quad \text{as} \quad t \to \infty.$$

Orbits are oriented spirals which are attracted to the asymptotically stable equilibrium point.

A stable focus $a = 0$



Other cases: E.g. Coinciding eigenvalues $\lambda_1 = \lambda_2$ and only one eigenvector



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(Un)stable Jordan nodes $\boldsymbol{a = 0}$

Stable point summary

An equilibrium point $\boldsymbol{a} = \mathbf{0}$ is

- Asymptotically stable if $\mathrm{Re}(\lambda_i) < 0$ for all eigenvalues. E.g.:
  - Stable nodes and stars; and
  - Stable foci.
- Stable (but not Asymptotically) if $\mathrm{Re}(\lambda_i) \leq 0$ for all eigenvalues. E.g.:
  - Centres.
- Unstable if $\mathrm{Re}(\lambda_i) > 0$ for at least one eigenvalue. E.g.:
  - Unstable nodes and stars;
  - Unstable foci; and
  - Saddle points.

Linearization

Suppose that $\boldsymbol{x}_0$ is close to an equilibrium point $\boldsymbol{a}$. If

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{F}(\boldsymbol{x}) \quad \text{for all } t, \text{ with } \boldsymbol{x}(0) = \boldsymbol{x}_0, \tag{10}$$

then for small t the difference $\boldsymbol{y} = \boldsymbol{x} - \boldsymbol{a}$ is small and satisfies

$$\frac{d\boldsymbol{y}}{dt} = \frac{d\boldsymbol{x}}{dt} = \boldsymbol{F}(\boldsymbol{x}) = \boldsymbol{F}(\boldsymbol{a}+\boldsymbol{y}) \approx \boldsymbol{F}(\boldsymbol{a}) + \boldsymbol{F}'(\boldsymbol{a})\boldsymbol{y}.$$

This suggests that if $\boldsymbol{y}_0 = \boldsymbol{x}_0 - \boldsymbol{a}$ and $\boldsymbol{y}$ is the solution of the *linear* dynamical system

$$\frac{d\boldsymbol{y}}{dt} = \boldsymbol{F}(\boldsymbol{a}) + \boldsymbol{F}'(\boldsymbol{a})\boldsymbol{y} \quad \text{for all } t, \text{ with } \boldsymbol{y}(0) = \boldsymbol{y}_0,$$

then $\boldsymbol{x}(t) \approx \boldsymbol{a} + \boldsymbol{y}(t)$ for small t. In particular, we can infer stability properties of (10) at an equilibrium point $\boldsymbol{a}$ from the eigenvalues of $A = \boldsymbol{F}'(\boldsymbol{a})$.

Damped pendulum

The angular deflection θ of a damped pendulum satisfies

$$m\ddot{\theta} + r\dot{\theta} + k\sin\theta = 0,$$

where $\theta = 0$ is the "down" position of the bob. For simplicity, take $m = 1$, so that we have the equivalent first-order system

$$\begin{aligned}\frac{d\theta}{dt} &= \phi,\\ \frac{d\phi}{dt} &= -k\sin\theta - r\phi.\end{aligned}$$

The equilibrium points are

$$(\theta, \phi) = (n\pi, 0)$$

for $n \in \{\ldots, -2, -1, 0, 1, 2, \ldots\}$.

Writing

$$\frac{d}{dt}\begin{bmatrix}\theta\\ \phi\end{bmatrix} = \boldsymbol{F}(\theta,\phi) = \begin{bmatrix}0 & \phi\\ -k\sin\theta & -r\phi\end{bmatrix}$$

we see that

$$\boldsymbol{F}'(\theta,\phi) = \begin{bmatrix}0 & 1\\ -k\cos\theta & -r\end{bmatrix},$$

and the matrix

$$A = F'(n\pi,0) = \begin{bmatrix}0 & 1\\ (-1)^{n+1}k & -r\end{bmatrix}$$

has eigenvalues given by

$$\det(\lambda I - A) = \begin{vmatrix}\lambda & -1\\ (-1)^n k & \lambda + r\end{vmatrix} = \lambda^2 + r\lambda + (-1)^n k = 0.$$

When n is even, we solve $\lambda^2 + r\lambda + k = 0$ to obtain

$$\lambda = \lambda_\pm = \begin{cases} \frac{1}{2}\left(-r \pm i\sqrt{4k - r^2}\right), & 0 \le r < 2\sqrt{k}, \\ -\sqrt{k}, & r = 2\sqrt{k}, \\ \frac{1}{2}\left(-r \pm \sqrt{r^2 - 4k}\right), & r > 2\sqrt{k}. \end{cases}$$

However, when n is odd, we solve $\lambda^2 + r\lambda - k = 0$ to obtain

$$\lambda = \lambda_\pm = \frac{1}{2}\left(-r \pm \sqrt{r^2 + 4k}\right), \quad r \ge 0.$$

Conclusion: equilibrium point at $(\theta, \phi) = (n\pi, 0)$ classified as follows.

n even	$r = 0$	$\operatorname{Re}\lambda_+ = \operatorname{Re}\lambda_- = 0$	stable
	$0 < r \leq 2\sqrt{k}$	$\operatorname{Re}\lambda_+ = \operatorname{Re}\lambda_- < 0$	asymptot. stable
	$r > 2\sqrt{k}$	$\lambda_- < \lambda_+ < 0$	asymptot. stable
n odd	$r \geq 0$	$\lambda_- < 0 < \lambda_+$	unstable

Logically, $\theta = n\pi$ is the "down" position if n is even, but is the "up" position if n is odd.

Final remarks on nonlinear DEs

We conclude this part of the course by discussing a technique for solving, or at least partially solving, some nonlinear problems. Importantly, this technique allows us to deal with the 2D motion of a particle moving under the influence of a solenoidal (divergence free) force.

First integrals

Definition

A function $G : \mathbb{R}^N \to \mathbb{R}$ is a first integral (or constant of the motion) for the system of ODEs

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{F}(\boldsymbol{x})$$

if $G\big(\boldsymbol{x}(t)\big)$ is constant for every solution $\boldsymbol{x}(t)$.

By the chain rule,

$$\frac{d}{dt}\,G\big(\boldsymbol{x}(t)\big) = \sum_{j=1}^{N} \frac{\partial G}{\partial x_j}\,\frac{dx_j}{dt} = \nabla G(\boldsymbol{x}) \cdot \frac{d\boldsymbol{x}}{dt} = \nabla G(\boldsymbol{x}) \cdot \boldsymbol{F}(\boldsymbol{x}).$$

Geometric Interpretation: G is a first integral iff

$$\nabla G(\boldsymbol{x}) \perp \boldsymbol{F}(\boldsymbol{x}) \quad \text{for all } \boldsymbol{x}.$$

Simple example

The function $G(x, y) = x^2 + y^2$ is a first integral of the linear system of ODEs

$$\frac{dx}{dt} = -y,$$
$$\frac{dy}{dt} = x.$$

In fact, putting

$$\boldsymbol{F}(x, y) = \begin{bmatrix} -y \\ x \end{bmatrix}$$

we have

$$\nabla G \cdot \boldsymbol{F} = \begin{bmatrix} 2x \\ 2y \end{bmatrix} \cdot \begin{bmatrix} -y \\ x \end{bmatrix} = (2x)(-y) + (2y)(x) = 0,$$

or equivalently,

$$\frac{dG}{dt} = \frac{\partial G}{\partial x}\frac{dx}{dt} + \frac{\partial G}{\partial y}\frac{dy}{dt} = (2x)(-y) + (2y)(x) = 0.$$

Partial solutions

In effect, a first integral provides a partial solution of the ODE.

Putting $C = G(\boldsymbol{x}_0)$, we know that $\boldsymbol{x}(t)$ is confined to the surface $G(\boldsymbol{x}) = C$.

If $N = 2$, the equation $G(x, y) = C$ implicitly gives $y = g(x)$, so

$$\frac{dx}{dt} = F_1(x, y) = F_1\bigl(x, g(x)\bigr)$$

and F_1 becomes a known function of x alone. If we can then evaluate

$$t = \int \frac{dx}{F_1}$$

to obtain $t = t(x)$, then implicitly we know $x = x(t)$ and finally $y = g\bigl(x(t)\bigr)$.

A class of second-order ODE

Consider

$$\ddot{x} = f(x) \qquad \text{or equivalently} \qquad \frac{d}{dt}\begin{bmatrix} x \\ \dot{x} \end{bmatrix} = \begin{bmatrix} \dot{x} \\ f(x) \end{bmatrix}, \tag{11}$$

and suppose that $-V(x)$ is an indefinite integral of f, that is, $-dV/dx = f(x)$. Since

$$\frac{d}{dt}\left(\frac{\dot{x}^2}{2}\right) = \dot{x}\,\ddot{x} \quad \text{and} \quad \frac{d}{dt}V(x) = \frac{dV}{dx}\,\frac{dx}{dt} = -f(x)\dot{x}.$$

If $x = x(t)$ is a solution of (11) then

$$\frac{d}{dt}\big[\dot{x}^2/2 + V(x)\big] = \big[\ddot{x} - f(x)\big]\dot{x} = 0,$$

so the function $G(x, \dot{x}) = \frac{1}{2}\dot{x}^2 + V(x)$ is a first integral. When multiplied by mass, this equation gives the total energy, G, in terms of the kinetic energy $\frac{1}{2}\dot{x}^2$ and potential energy V.

Example: undamped pendulum

The angular deflection of an undamped pendulum satisfies

$$m\ddot{\theta} + k \sin\theta = 0, \tag{12}$$

or

$$\ddot{\theta} + \omega^2 \sin\theta = 0, \qquad \omega = \sqrt{\frac{k}{m}},$$

which has the form

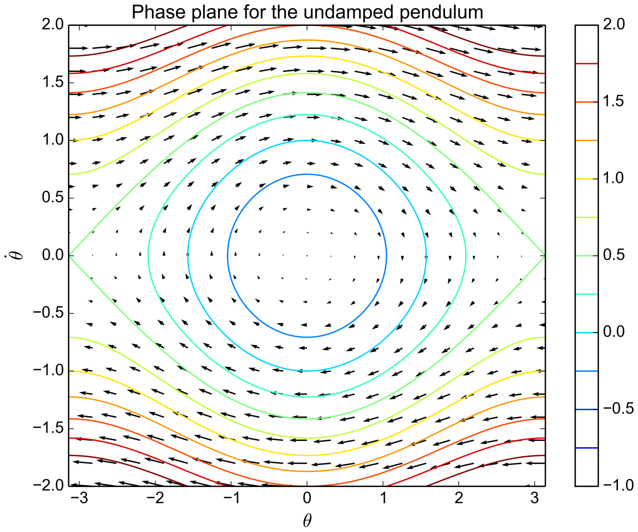
$$\ddot{\theta} = f(\theta) = -\frac{dV}{d\theta}$$

with $f(\theta) = -\omega^2 \sin\theta$ and $V(\theta) = -\omega^2 \cos\theta$. Thus,

$$\frac{d}{dt}\big(\tfrac{1}{2}\dot{\theta}^2 - \omega^2 \cos\theta\big) = \dot{\theta}\ddot{\theta} + \omega^2 \dot{\theta} \sin\theta = \dot{\theta}\big(\ddot{\theta} + \omega^2 \sin\theta\big) = 0$$

and so every solution of (12) satisfies

$$\tfrac{1}{2}\dot{\theta}^2 - \omega^2 \cos\theta = C.$$


Phase plane for the undamped pendulum
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−3
−2
−1
0
1
2
3
θ
$\dot{\theta}$
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0

Partial solutions in higher dimensions

Suppose $N > 2$ and we know several (functionally independent) first integrals G_1, G_2, ..., G_k. Then the solution $\boldsymbol{x}(t)$ must lie on the intersection of the surfaces $G_j(\boldsymbol{x}) = C_j$ for $1 \le j \le k$, where $C_j = G_j(\boldsymbol{x}_0)$.

We necessarily have $k \le N - 1$.

If $k = N - 1$, then the implicit function theorem gives $x_j = g_j(x_1)$ for $2 \le j \le N$, and thus $F_1(\boldsymbol{x})$ becomes a known function of x_1 so, as before

$$t = \int \frac{dx_1}{F_1}.$$

In principle, we then know $x_1 = x_1(t)$ and hence $x_j = g_j\big(x_1(t)\big)$ for $2 \le j \le N$.

However, a system might not have any first integrals.

Lorenz equations

Edward N. Lorenz, *Journal of Atmospheric Sciences* 20:130–141, 1963.

The 3×3 system of ODEs

$$\begin{aligned}\frac{dx}{dt} &= -\sigma x + \sigma y,\\ \frac{dy}{dt} &= rx - y - xz,\\ \frac{dz}{dt} &= xy - bz.\end{aligned}$$

is a very simplified model of convection in a thin layer of fluid heated uniformly from below and cooled uniformly from above.dependent variables give the coefficients of 3 Fourier modes.

Lorenz equations

For $r > 1$, the system has three equilibrium points at

$$(0,0,0), \qquad \left(\sqrt{b(r-1)}, \sqrt{b(r-1)}, r-1\right),$$
$$\left(-\sqrt{b(r-1)}, -\sqrt{b(r-1)}, r-1\right).$$

The parameter choices

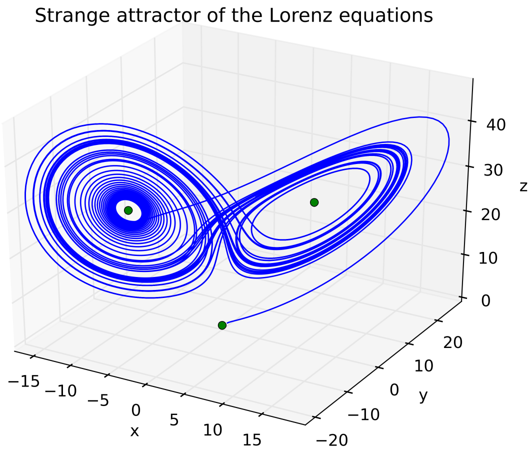
$$r = 28, \qquad \sigma = 10, \qquad b = \frac{8}{3},$$

and initial conditions

$$x_0 = y_0 = z_0 = \frac{1}{2},$$

lead to the trajectory shown on the next page.

Lorenz equations



Lorenz equations

Since the solution does not lie on a smooth surface we conclude that no first integral exists.

The Lorenz equations also exhibit a sensitive dependence on initial conditions, as illustrated in the following video.

`http://www.youtube.com/watch?v=FYE4JKAXSfY`

Part III

Initial-Boundary Value Problems in 1D

Introduction

We have seen that an initial-value problem for a (nonsingular) linear ODE $Lu = f$ always has a unique solution. However, matters are not so simple for a boundary-value problem: a solution might not exist, or if one exists it might not be unique. One aim of this part of the course is to understand when these different outcomes occur with the help of some concepts adapted from linear algebra.

Two-point boundary value problems

In an mth order initial-value problem we specify m initial conditions at the left end of an interval. In an mth order boundary-value problem, we again specify m conditions involving the solution and its derivatives, but some apply at the left end and some at the right end.

We will consider only second-order linear boundary-value problems, so exactly one boundary condition will apply at each end of the interval.

Initial conditions vs boundary conditions

Consider the second-order ODE

$$u'' + u = 0 \quad \text{for } 0 < x < \pi,$$

whose general solution is

$$u(x) = A\cos x + B\sin x.$$

We obtain a unique solution by specifying initial conditions, that is, specifying $u(0)$ and $u'(0)$.

What if we instead specify boundary conditions at $x = 0$ and at $x = \pi$?

Three possibilities

Example

A unique solution $u(x) = \sin x$ exists satisfying

$$u'(0) = 1 \quad \text{and} \quad u(\pi) = 0.$$

Example

No solution exists satisfying

$$u(0) = 0 \quad \text{and} \quad u(\pi) = 1.$$

Example

Infinitely many solutions $u(x) = C \sin x$ exist satisfying

$$u(0) = 0 \quad \text{and} \quad u(\pi) = 0.$$

Linear two-point boundary value problem

We want to solve

$$Lu = f \quad \text{for } a < x < b, \quad \text{with } B_1 u = \alpha_1 \text{ and } B_2 u = \alpha_2, \tag{13}$$

where

$$Lu = a_2 u'' + a_1 u' + a_0 u$$

is a 2nd-order linear differential operator, and the boundary operators have the form

$$\begin{aligned} B_1 u &= b_{11} u'(a) + b_{10} u(a), \\ B_2 u &= b_{21} u'(b) + b_{20} u(b). \end{aligned}$$

Example

$$\begin{aligned} u'' - u &= x - 1 && \text{for } 0 < x < \log 2, \\ u &= 2 && \text{at } x = 0, \\ u' - 2u &= 2\log 2 - 4 && \text{at } x = \log 2. \end{aligned}$$

Existence and uniqueness

Having seen a couple of examples, we now investigate the conditions that determine whether or not a linear boundary-value problem has a unique solution.

Linearity (again)

Since L, B_1 and B_2 are all linear, the solutions of the homogeneous BVP

$$Lu = 0 \quad \text{for } a < x < b, \quad \text{with } B_1 u = 0 \text{ and } B_2 u = 0, \tag{14}$$

form a vector space: if u_1 and u_2 are solutions of (14) then so is $u = c_1 u_1 + c_2 u_2$ for any constants c_1 and c_2.

Any two solutions of the inhomogeneous problem (13) differ by a solution of the homogeneous problem, that is, if u_1 and u_2 satisfy (13) then $u = u_1 - u_2$ satisfies (14).

If u_1 satisfies (13) and u_2 satisfies (14) then $u = u_1 + c u_2$ satisfies (13) for any constant c.

Existence and uniqueness

The preceding facts imply the following result.

Theorem (Uniqueness)

The inhomogeneous BVP (13) *has at most one solution iff the homogeneous BVP* (14) *has only the trivial solution* $u \equiv 0$.

To investigate existence, suppose that the general solution of the homogeneous equation $Lu = 0$ is $u_{\mathrm{H}} = c_1 u_1(x) + c_2 u_2(x)$, and suppose that $u_{\mathrm{P}}(x)$ is a particular solution of the inhomogeneous equation $Lu = f$. The general solution of $Lu = f$ will then be

$$u(x) = u_{\mathrm{H}}(x) + u_{\mathrm{P}}(x) = c_1 u_1(x) + c_2 u_2(x) + u_{\mathrm{P}}(x).$$

Existence and uniqueness (continued)

To satisfy the boundary conditions in (13) we must choose c_1 and c_2 so that

$$B_1(c_1u_1 + c_2u_2 + u_{\mathrm{P}}) = \alpha_1,$$
$$B_2(c_1u_1 + c_2u_2 + u_{\mathrm{P}}) = \alpha_2,$$

Since B_1 and B_2 are linear, the inhomogeneous BVP (13) has at least one solution iff the 2×2 linear system

$$\begin{bmatrix} B_1u_1 & B_1u_2 \\ B_2u_1 & B_2u_2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} \alpha_1 - B_1u_{\mathrm{P}} \\ \alpha_2 - B_2u_{\mathrm{P}} \end{bmatrix} \tag{15}$$

has at least one solution $[c_1, c_2]^T$.

Similarly, $u(x) = c_1u_1(x) + c_2u_2(x)$ is a solution of the homogeneous BVP (14) iff c_1 and c_2 satisfy

$$\begin{bmatrix} B_1u_1 & B_1u_2 \\ B_2u_1 & B_2u_2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The 2×2 matrix on the left is non-singular iff this homogeneous linear system has only the trivial solution $c_1 = c_2 = 0$.

Hence, the following is true.

Theorem

If the homogeneous problem (14) *has only the trivial solution, then for every choice of f, α_1 and α_2 the inhomogeneous problem*

$$Lu = f \quad \text{for } a < x < b, \quad \text{with } B_1u = \alpha_1 \text{ and } B_2u = \alpha_2,$$

has a unique solution.

Inner products and norms of functions

If a homogeneous initial boundary value problem admits nontrivial solutions, then the inhomogeneous problem might or might not have any solutions, depending the forcing term and boundary values (f, α_1 and α_2 etc.).

To formulate a condition that guarantees existence we require a short digression that introduces some ideas from functional analysis.

Inner product and norm

The inner product $\langle f, g \rangle$ of a pair of continuous functions f, $g : [a, b] \to \mathbb{R}$ is defined by

$$\langle f, g \rangle = \int_a^b f(x)g(x)\,dx.$$

The corresponding norm of f is defined by

$$\|f\| = \sqrt{\langle f, f \rangle} = \left(\int_a^b [f(x)]^2\,dx \right)^{1/2}.$$

We say that f and g are orthogonal if $\langle f, g \rangle = 0$.

Example

If

$$[a, b] = [-1, 1], \quad f(x) = x, \quad g(x) = \cos \pi x,$$

then

$$\langle f, g \rangle = 0, \qquad \|f\| = \sqrt{\frac{2}{3}}, \qquad \|g\| = 1.$$

Thus, f and g are orthogonal over the interval $[-1, 1]$.

Example

Later we will show that the Legendre polynomials are orthogonal over the interval $[-1, 1]$:

$$\int_{-1}^{1} P_n(x) P_k(x)\, dx = 0 \quad \text{if } n \neq k.$$

Key properties

The inner product and norm for functions behave like the scalar (dot) product and norm for vectors in $\mathbb{R}^n$.

For any continuous functions f, g, h and any constant α:

1. $\langle f+g, h\rangle = \langle f, h\rangle + \langle g, h\rangle$;
2. $\langle f, g+h\rangle = \langle f, g\rangle + \langle f, h\rangle$;
3. $\langle \alpha f, g\rangle = \alpha\langle f, g\rangle = \langle f, \alpha g\rangle$;
4. $\langle f, g\rangle = \langle g, f\rangle$;
5. $\|f\| = 0$ if and only $f(x) = 0$ for $a < x < b$.

Cauchy–Schwarz inequality

Theorem

$|\langle f, g\rangle| \leq \|f\|\|g\|$.

Proof.

If $f = 0$ or $g = 0$ then $\langle f, g\rangle = 0$ and $\|f\|\|g\| = 0$ so the result reduces to $0 \leq 0$.
Otherwise, $f \neq 0$ and $g \neq 0$ so $\|f\| \neq 0$ and $\|g\| \neq 0$, and we may define $\alpha = 1/\|f\|$ and $\beta = 1/\|g\|$. Since

$$\begin{aligned} 0 \leq \|\alpha f \pm \beta g\|^2 &= \langle \alpha f \pm \beta g, \alpha f \pm \beta g\rangle \\ &= \alpha^2\|f\|^2 \pm 2\alpha\beta\langle f, g\rangle + \beta^2\|g\|^2 = 2\big(1 \pm \alpha\beta\langle f, g\rangle\big), \end{aligned}$$

we see that $-1 \leq \alpha\beta\langle f, g\rangle \leq 1$ so

$$|\langle f, g\rangle| \leq \alpha^{-1}\beta^{-1} = \|f\|\|g\|.$$

$\square$

Triangle inequality

Corollary

$\|f + g\| \leq \|f\| + \|g\|.$

Proof.

$$\begin{aligned}
\|f + g\|^2 &= \langle f + g, f + g \rangle \\
&= \|f\|^2 + 2\langle f, g \rangle + \|g\|^2 \\
&\leq \|f\|^2 + 2\|f\|\|g\| + \|g\|^2 \\
&= (\|f\| + \|g\|)^2.
\end{aligned}$$

$\square$

Self-adjoint differential operators

Having concluded our digression, we now introduce a class of second-order, linear differential operators that possess a number of special properties. Operators of this type are of great importance in applications, as well as enjoying a rich theory that will be developed here and extended later in the course.

Integration by parts

Consider a second-order,linear differential operator

$$Lu = a_2 u'' + a_1 u' + a_0 u.$$

Integrating by parts,

$$\int_a^b (a_1 u') v \, dx = \big[u(a_1 v)\big]_a^b - \int_a^b u(a_1 v)' \, dx,$$
$$\int_a^b (a_2 u'') v \, dx = \big[u'(a_2 v) - u(a_2 v)'\big]_a^b + \int_a^b u(a_2 v)'' \, dx,$$

so

$$\langle Lu, v\rangle = \big[u'(a_2 v) - u(a_2 v)' + u(a_1 v)\big]_a^b + \big\langle u, (a_2 v)'' - (a_1 v)' + a_0 v\big\rangle.$$

Adjoint operator and Lagrange identity

Thus, defining the formal adjoint

$$\begin{aligned} L^*v &= (a_2v)'' - (a_1v)' + a_0v \\ &= a_2v'' + (2a_2' - a_1)v' + (a_2'' - a_1' + a_0)v \end{aligned}$$

and the bilinear concomitant

$$P(u,v) = u'(a_2v) - u(a_2v)' + u(a_1v),$$

we have the Lagrange identity

$$\langle Lu, v\rangle = \langle u, L^*v\rangle + \big[P(u,v)\big]_a^b.$$

The differentiated version of the Lagrange identity is

$$(Lu)v = u\,L^*v + \frac{d}{dx}P(u,v).$$

Example

If

$$Lu = 3xu'' - (\cos x)u' + e^x u$$

then

$$\begin{aligned} L^*v &= (3xv)'' + [(\cos x)v]' + e^x v \\ &= 3xv'' + (6 + \cos x)v' + (e^x - \sin x)v \end{aligned}$$

and

$$\begin{aligned} P(u,v) &= u'(3xv) - u(3xv)' - uv\cos x \\ &= 3x(u'v - uv') - (3 + \cos x)uv. \end{aligned}$$

Formal self-adjointness

The operator L is formally self-adjoint if $L^* = L$.

Theorem

A second-order, linear differential operator L is formally self-adjoint iff it can be written in the form

$$Lu = -(pu')' + qu = -pu'' - p'u' + qu, \tag{16}$$

in which case the Lagrange identity takes the form

$$(Lu)\,v - u\,(Lv) = -\big(p(x)(u'v - uv')\big)', \tag{17}$$

or in other words, the bilinear concomitant is

$$P(u, v) = -p(x)(u'v - uv').$$

Proof

If $L = -(pu')' + qu = -pu'' - p'u' + qu$ then

$$
\begin{aligned}
L^*v &= (-pv)'' + (p'v)' + qv \\
&= -(p''v + 2p'v' + pv'') + (p''v + p'v') + qv \\
&= -(pv'' + p'v') + qv = -(pv')' + qv = Lv,
\end{aligned}
$$

so L is formally self-adjoint. Conversely, if $Lu = a_2u'' + a_1u' + a_0u$ and $L^* = L$, then

$$
\begin{aligned}
&a_2u'' + a_1u' + a_0u \\
&\qquad = a_2u'' + (2a_2' - a_1)u' + (a_2'' - a_1' + a_0)u
\end{aligned}
$$

for all u. Choosing $u \equiv 1$ and then $u \equiv x$, it follows that

$$a_0 = a_2'' - a_1' + a_0 \quad \text{and} \quad a_1 = 2a_2' - a_1.$$

Thus, $a_2' = a_1$, so by putting $p = -a_2$ and $q = a_0$ we have $a_1 = -p'$, implying that L has the form

$$Lu = -pu'' - p'u' + qu = -(pu')' + qu.$$

Moreover, in this case, the Lagrange identity takes the form

$$\begin{aligned}
\langle Lu, v\rangle &= \int_a^b \big(-(pu')'v + quv\big)\, dx \\
&= \big[-(pu')v\big]_a^b + \int_a^b pu'v'\, dx + \int_a^b quv\, dx \\
&= \big[-(pu')v + u(pv')\big]_a^b - \int_a^b u(pv')' + \int_a^b quv\, dx \\
&= \big[-p(u'v - uv')\big]_a^b + \int_a^b u\big(-(pv')' + qv\big)\, dx \\
&= \big[-p(u'v - uv')\big]_a^b + \langle u, Lv\rangle. \quad \square
\end{aligned}$$

Example

Consider the Bessel equation

$$x^2u'' + xu' + (x^2 - \nu^2)u = f(x).$$

Dividing both sides by x gives $Lu = -x^{-1}f(x)$ where

$$Lu = -(xu')' + (\nu^2x^{-1} - x)u.$$

Example

The Legendre equation

$$(1 - x^2)u'' - 2xu' + \nu(\nu + 1)u = f(x)$$

has the form $Lu = -f(x)$ with $Lu = -[(1 - x^2)u']' - \nu(\nu + 1)u$.

Hence, both the Bessel and Legendre differential operators are formally self-adjoint.

Transforming to formally self-adjoint form

If we can evaluate the integrating factor

$$p(x) = \exp\left(\int \frac{a_1(x)}{a_2(x)}\,dx\right),$$

then we can transform an ODE of the form $a_2u'' + a_1u' + a_0u = f(x)$ to formally self-adjoint form:

$$\begin{aligned}-pu'' - \frac{pa_1}{a_2}\,u' - \frac{pa_0}{a_2}\,u &= \frac{-pf(x)}{a_2},\\ -(pu'' + p'u') - \frac{pa_0}{a_2}\,u &= \frac{-pf(x)}{a_2},\\ -(pu')' + qu &= \tilde{f}(x)\end{aligned}$$

where $q = -pa_0/a_2$ and $\tilde{f} = -pf/a_2$.

Example

Write the Cauchy–Euler ODE $ax^2u'' + bxu' + cu = f(x)$ in formally self-adjoint form.

Self-adjointness and boundary operators

Lemma

Any formally self-adjoint operator $L = -(pu')' + qu$ *satisfies the identity*

$$\langle Lu, v\rangle - \langle u, Lv\rangle = \sum_{i=1}^{2} \big(B_i u\, R_i v - R_i u\, B_i v\big), \tag{18}$$

for all u *and* v*, where*

$$R_1 u = \frac{p(a)u(a)}{b_{11}} \qquad \text{or} \qquad R_1 u = -\frac{p(a)u'(a)}{b_{10}}$$

and

$$R_2 u = -\frac{p(b)u(b)}{b_{21}} \qquad \text{or} \qquad R_2 u = \frac{p(b)u'(b)}{b_{20}}.$$

Proof

Suppose $b_{11} \neq 0$ and $b_{21} \neq 0$. At $x = a$,

$$
\begin{aligned}
+p(u'v - uv') &= (b_{11}u' + b_{10}u)\left(\frac{pv}{b_{11}}\right) - \left(\frac{pu}{b_{11}}\right)(b_{11}v' + b_{10}v) \\
&= (B_1u)(R_1v) - (R_1u)(B_1v)
\end{aligned}
$$

and at $x = b$,

$$
\begin{aligned}
-p(u'v - uv') &= (b_{21}u' + b_{20}u)\left(-\frac{pv}{b_{21}}\right) - \left(-\frac{pu}{b_{21}}\right)(b_{21}v' + b_{20}v) \\
&= (B_2u)(R_2v) - (R_2u)(B_2v).
\end{aligned}
$$

A similar argument works if $b_{11} = 0$ or $b_{21} = 0$. □

Necessary condition for existence

If u is a solution of the inhomogeneous BVP (13), and if v is a solution of the homogeneous problem

$$\begin{aligned} Lv &= 0 && \text{for } a < x < b, \\ B_1 v &= 0 && \text{at } x = a, \\ B_2 v &= 0 && \text{at } x = b, \end{aligned} \tag{19}$$

then on the one hand

$$\langle Lu, v\rangle - \langle u, Lv\rangle = \langle f, v\rangle - \langle u, 0\rangle = \langle f, v\rangle$$

and on the other hand, by (18),

$$\langle Lu, v\rangle - \langle u, Lv\rangle = \alpha_1 \, R_1 v - R_1 u \times 0 + \alpha_2 \, R_2 v - R_2 u \times 0.$$

Hence, the data f, α_1 and α_2 must satisfy

$$\langle f, v\rangle = \alpha_1 \, R_1 v + \alpha_2 \, R_2 v. \tag{20}$$

Sufficient condition for existence?

It can be shown that, provided $a_2(x) \neq 0$ for $a \leq x \leq b$, the condition (20) is also *sufficient* for the existence of u, i.e., u exists iff (20) holds for all v satisfying (19).

Example

The 2-point BVP

$$\begin{aligned} u'' + u &= f && \text{for } 0 < x < \pi, \\ u &= \alpha_1 && \text{at } x = 0, \\ u &= \alpha_2 && \text{at } x = \pi, \end{aligned}$$

has a solution iff

$$\int_0^\pi f(x)\sin(x)\,dx = \alpha_1 + \alpha_2.$$

In this case, if u is one solution then every other solution is of the form $u + C\sin x$ for some constant C.

Fredholm alternative

Theorem

Either the homogeneous BVP (14) *has only the trivial solution* $v \equiv 0$, *in which case*

> *the inhomogeneous BVP* (13) *has a unique solution* u *for every choice of* f, α_1 *and* α_2,

OR else the homogeneous BVP admits non-trivial solutions, in which case

> *the inhomogeneous BVP* (13) *has a solution* u *iff* f, α_1 *and* α_2 *satisfy* (20) *for every solution* v *of the homogeneous BVP* (14).

In the latter case, $u + Cv$ *is also a solution of* (13) *for any constant* C.

Domain of a self-adjoint operator

Definition

Let $\mathcal{D}$ be a subspace of a vector space V with inner product $\langle \cdot, \cdot \rangle$, and let L be a linear operator on V with domain $\mathcal{D}$. We say that L is self-adjoint if

$$\langle Lu, v \rangle = \langle u, Lv \rangle \quad \text{for all } u,\ v \in \mathcal{D}.$$

The identity (18) shows that $Lu = -(pu')' + qu$ (i.e., a formally self-adjoint operator) is self-adjoint if $\mathcal{D}$ is the set of C^2 functions $u : [a, b] \to \mathbb{R}$ satisfying the homogeneous boundary conditions

$$\begin{aligned} B_1 u &= 0 \quad \text{at } x = a, \\ B_2 u &= 0 \quad \text{at } x = b. \end{aligned}$$

Example: Bessel operator

Recall the Bessel operator

$$Lu = -(xu')' + (\nu^2 x^{-1} - x)u \quad \text{for } 1 \le x \le 2,$$

and let

$$\begin{aligned} B_1 u &= u', & R_1 u &= u, & &\text{at } x = 1,\\ B_2 u &= 2u' - u, & R_2 u &= -u, & &\text{at } x = 2, \end{aligned}$$

so that with $\langle f, g\rangle = \int_1^2 f(x)g(x)\,dx$ the Lagrange identity implies

$$\langle Lu, v\rangle - \langle u, Lv\rangle = B_1u\,R_1v - R_1u\,B_1v + B_2u\,R_2v - R_2u\,B_2v.$$

Thus, if we let $\mathcal{D}$ denote the vector space of C^2 functions $u : [1,2] \to \mathbb{R}$ satisfying $B_1u = 0$ at $x = 1$ and $B_2u = 0$ at $x = 2$, then

$$\langle Lu, v\rangle = \langle u, Lv\rangle \quad \text{for all } u,\ v \in \mathcal{D}.$$

Part IV

Generalised Fourier Series

Introduction

We will first see how a solution to the vibrating string problem can be constructed using a Fourier sine series expansion. In the remainder of this part of the course, we will study generalise the concept of Fourier expansions that include the familiar trigonometric Fourier series as just one special case. The connection with differential equations comes about via the theory of Sturm–Liouville eigenproblems.

These generalised Fourier series will allow us to solve a range of partial differential equations by separating variables in curvilinear coordinates.

The vibrating string

Our first example of the application of Fourier series methods is to the partial differential equation describing a vibrating string, such as in a musical instrument like a piano. This example reveals a connection between Fourier series and acoustics.

Notation and assumptions

Consider a stretched string having the following properties:

1. the string is fixed at $x = 0$ and $x = \ell$;
2. the string is uniform, perfectly elastic and offers no resistance to bending;
3. the weight of the string is negligible compared to tension;
4. the motion of the string is purely transverse and the deflection is small compared to the length of the string.

Let

$$u = u(x,t) = \text{transverse deflection at position } x \text{ and time } t.,$$
$$T = T(x,t) = \text{magnitude of tension},$$
$$\theta = \theta(x,t) = \text{tangent angle}.$$

Newton's laws

Let ρ be the linear density (mass per unit length) of the string.

For the piece of string between x and $x + \Delta x$,

$$\text{longitudinal momentum} = 0,$$
$$\text{transverse momentum} = \rho \Delta x \, \frac{\partial u}{\partial t}.$$

No motion in the longitudinal direction so the net longitudinal force is zero, i.e.

$$T(x + \Delta x, t) \cos \theta(x + \Delta x, t) = T(x, t) \cos \theta(x, t), \tag{21}$$

whereas for the transverse motion

$$\frac{\partial}{\partial t}\left(\rho \Delta x \, \frac{\partial u}{\partial t}\right) = T(x + \Delta x, t) \sin \theta(x + \Delta x, t) - T(x, t) \sin \theta(x, t). \tag{22}$$

Wave equation

We deduce from (21) that

$$T \cos\theta = T_0$$

for some T_0 independent of x, and we deduce from (22) that

$$\rho \frac{\partial^2 u}{\partial t^2} = \frac{(T \sin\theta)(x + \Delta x, t) - (T \sin\theta)(x, t)}{\Delta x}.$$

Taking the limit as $\Delta x \to 0$ gives

$$\rho \frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial x}(T \sin\theta) = T_0 \frac{\partial}{\partial x}(\tan\theta).$$

But $\tan\theta = \partial u/\partial x$ so u satisfies the wave equation

$$\rho \frac{\partial^2 u}{\partial t^2} = T_0 \frac{\partial^2 u}{\partial x^2}.$$

Initial-boundary value problem (IBVP)

Put $c = \sqrt{T_0/\rho}$ (which has dimensions of length/time and is called the wave speed). Now suppose that the string is initially at rest with a known deflection $u_0(x)$, then

$$
\begin{aligned}
\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} &= 0, && 0 < x < \ell, t > 0, \\
u(0,t) &= 0, && t > 0, \\
u(\ell,t) &= 0, && t > 0, \\
u(x,0) &= u_0(x), && 0 < x < \ell, \\
\frac{\partial u}{\partial t}(x,0) &= 0, && 0 < x < \ell.
\end{aligned}
\tag{23}
$$

Separation of variables

Look for solutions having a simple form:

$$u(x,t) = X(x)T(t).$$

Such a u satsifies the PDE iff

$$XT'' - c^2X''T = 0.$$

Thus, we want

$$\frac{T''}{c^2T} = \frac{X''}{X}.$$

Since $T''/(c^2T)$ is a function of t only, and X''/X is a function of x only, there must be a separation constant λ such that

$$\frac{T''}{c^2T} = -\lambda = \frac{X''}{X}$$

is true for all x in $(0, l)$ and $t > 0$.

A homogeneous BVP

Since

$$u(0,t) = X(0)T(t), \quad u(\ell,t) = X(\ell)T(t),$$
$$\frac{\partial u}{\partial t}(x,0) = X(x)T'(0),$$

we also want

$$X(0) = 0, \qquad X(\ell) = 0, \qquad T'(0) = 0.$$

Thus, X should satisfy

$$\begin{aligned} X'' + \lambda X &= 0 && \text{for } 0 < x < \ell; \\ X &= 0 && \text{at } x = 0; \\ X &= 0 && \text{at } x = \ell. \end{aligned}$$

One solution is $X \equiv 0$, but non-trivial solutions exist for special choices of λ.

Case $\lambda > 0$

Write $\lambda = \omega^2$ where $\omega > 0$.
Thus, $X'' + \omega^2 X = 0$, which has the general solution

$$X = A\cos\omega x + B\sin\omega x.$$

Since $X(0) = A$ the first boundary condition implies $A = 0$ so $X = B\sin\omega x$.
Since $X(\ell) = B\sin\omega\ell$ the second boundary condition implies

$$B\sin\omega\ell = 0.$$

If $B = 0$, then $X \equiv 0$.
If $B \neq 0$, then $\sin\omega\ell = 0$ so $\omega\ell = n\pi$ for some $n \in \{1, 2, 3, \dots\}$. In this case, $\omega = n\pi/\ell$ so

$$\lambda = \left(\frac{n\pi}{\ell}\right)^2 \quad \text{and} \quad X = B\sin\frac{n\pi x}{\ell}.$$

Case $\lambda = 0$

If $\lambda = 0$ then $X'' = 0$ which has the general solution

$$X = Ax + B.$$

Since $X(0) = B$, the first boundary condition implies $B = 0$ so $X = Ax$.

Since $X(\ell) = A\ell$, the second boundary condition implies

$$A\ell = 0.$$

But $\ell > 0$ so $A = 0$ and thus $X \equiv 0$.

Case $\lambda < 0$

Write $\lambda = -\omega^2$ where $\omega > 0$.

Thus, $X'' - \omega^2 X = 0$, which has the general solution

$$X = A\cosh\omega x + B\sinh\omega x.$$

Since $X(0) = A$ the first boundary condition implies $A = 0$ so $X = B\sinh\omega x$.

Since $X(\ell) = B\sinh\omega\ell$ the second boundary condition implies

$$B\sinh\omega\ell = 0.$$

If $B = 0$, then $X \equiv 0$.

If $B \neq 0$, then $\sinh\omega\ell = 0$ so $\omega\ell = 0$, which is impossible if $\omega > 0$.

Non-trivial solutions

Conclusion: get a useful solution only when $\lambda = \lambda_n$ and $X = BX_n$, where

$$\lambda_n = \left(\frac{n\pi}{\ell}\right)^2 \quad \text{and} \quad X_n(x) = \sin\left(\frac{n\pi}{\ell}\,x\right) \quad \text{for } n = 1,\,2,\,3,\,\ldots.$$

Recall that for $T(t)$, we require $T'' + \lambda c^2 T = 0$ and $T'(0) = 0$. Since $\lambda_n c^2 = (n\pi c/\ell)^2$, the general solution is

$$T(t) = C\cos\left(\frac{n\pi c}{\ell}\,t\right) + E\sin\left(\frac{n\pi c}{\ell}\,t\right),$$

and the initial condition implies that $E = 0$, so $T = CT_n$ where

$$T_n(t) = \cos\left(\frac{n\pi c}{\ell}\,t\right).$$

Modes

We call

$$u_n(x,t) = X_n(x)T_n(t) = \sin\left(\frac{n\pi}{\ell}\,x\right)\cos\left(\frac{n\pi c}{\ell}\,t\right)$$

the nth normal mode of vibration or pure harmonic. The smallest number $\tau_n > 0$ such that $u_n(x, t+\tau_n) = u_n(x,t)$ for all x and t is called the period of the nth normal mode. Since

$$\frac{n\pi c}{\ell}\,\tau_n = 2\pi$$

we see that

$$\tau_n = \frac{2\ell}{nc}.$$

The corresponding frequency is

$$\frac{1}{\tau_n} = \frac{nc}{2\ell}.$$

Summary

Each mode u_n satisfies all four *homogeneous* equations in (23),

$$\begin{aligned} \frac{\partial^2 u_n}{\partial t^2} - c^2 \frac{\partial^2 u_n}{\partial x^2} &= 0, \\ u_n(0,t) &= 0, \\ u_n(\ell,t) &= 0, \\ \frac{\partial u_n}{\partial t}(x,0) &= 0, \end{aligned} \tag{24}$$

but since

$$u_n(x,0) = \sin\left(\frac{n\pi}{\ell}\,x\right) \tag{25}$$

the *in*homogeneous initial condition will not be satisfied unless $u_0(x)$ happens to be $\sin(n\pi x/\ell)$.

Superposition principle

Each equation in (24) is linear and homogeneous, so any superposition of normal modes,

$$u(x,t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi}{\ell}\,x\right)\cos\left(\frac{n\pi c}{\ell}\,t\right), \tag{26}$$

also satisfies (24).

We can now hope that (26) is the general solution of the initial-boundary value problem (23): for any given u_0 we can find A_1, A_2, ... such that the remaining condition holds, i.e.,

$$u(x,0) = u_0(x), \quad 0 \le x \le \ell.$$

But this just means that

$$u_0(x) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi}{\ell}\,x\right), \quad 0 \le x \le \ell. \tag{27}$$

Fourier sine coefficients

Thus, the solution to (23) is given by (26) with

$$A_n = \frac{2}{\ell}\int_0^{\ell} u_0(x)\sin\left(\frac{n\pi x}{\ell}\right)dx. \tag{28}$$

Example

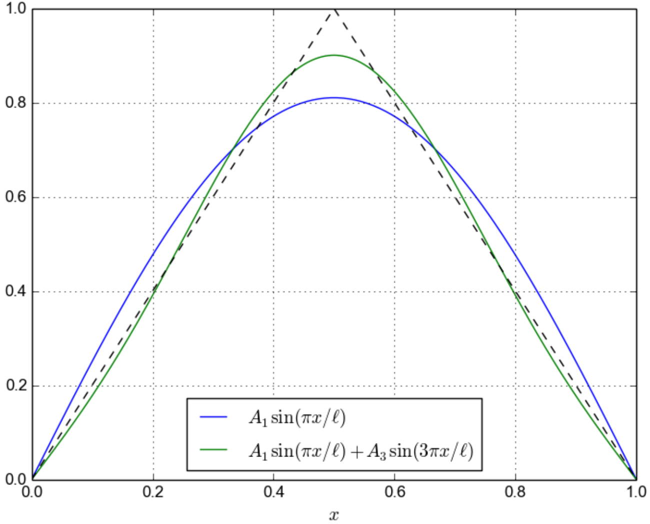
If $\ell = 1$ and the initial deflection is given by (all be it physically implausible) profile

$$u_0(x) = \begin{cases} 2x, & 0 < x \le \frac{1}{2}, \\ 2(1-x), & \frac{1}{2} < x < 1, \end{cases}$$

we construct a *Fourier sine series* of u_0 and find that

$$u(x,t) = \frac{8}{\pi^2}\sum_{k=1}^{\infty}\frac{(-1)^{k-1}}{(2k-1)^2}\sin\big((2k-1)\pi x\big)\cos\big((2k-1)\pi ct\big).$$

Partial sums



Complete orthogonal systems

We now revisit the concept of an inner product and introduce a more general class. We will then investigate the properties of orthogonal functions. Expanding a function as a linear combination of orthogonal function leads naturally to the notion of a generalised Fourier series.

Weighted inner product

If $w : (a, b) \to \mathbb{R}$ satisfies

$$w(x) > 0 \quad \text{for } a < x < b,$$

then we define the inner product with weight function w by

$$\langle f, g\rangle_w = \langle f, gw\rangle = \int_a^b f(x)g(x)w(x)\,dx,$$

and the corresponding norm by

$$\|f\|_w = \sqrt{\langle f, f\rangle_w} = \sqrt{\int_a^b [f(x)]^2 w(x)\,dx}.$$

Two functions f and g are orthogonal with respect to w over the interval (a, b) if $\langle f, g\rangle_w = 0$.

Example: Tchebyshev polynomials

The Tchebyshev polynomials T_0, T_1, T_2, ... are defined recursively by

$$T_0(x) = 1, \qquad T_1(x) = x,$$
$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x) \quad \text{for } n = 1,\, 2,\, 3,\, \ldots .$$

One can verify by induction on n that

$$T_n(\cos\theta) = \cos(n\theta).$$

Hence, using the substitution $x = \cos\theta$ in the following integral we find that if $n \neq j$ then

$$\int_{-1}^{1} \frac{T_n(x)T_j(x)}{\sqrt{1-x^2}}\, dx = 0$$

and so T_n is orthogonal to T_j with respect to the Tchebyshev weight $1/\sqrt{1-x^2}$.

The space L_2

Let $L_2(a, b, w)$ denote the set of all (Lebesgue measurable) functions $f : (a, b) \to \mathbb{R}$ such that $\|f\|_w < \infty$.

The Cauchy–Schwarz inequality states that if f, $g \in L_2(a, b, w)$ then $\langle f, g\rangle_w$ exists and is finite, with

$$|\langle f, g\rangle_w| \leq \|f\|_w \|g\|_w.$$

Furthermore, $L_2(a, b, w)$ is a vector space with

$$\|\lambda f\|_w = |\lambda| \|f\|_w, \quad \|f + g\|_w \leq \|f\|_w + \|g\|_w.$$

Orthogonal set of functions

A set of functions $S \subseteq L_2(a, b, w)$ is said to be orthogonal if every pair of functions in S is orthogonal and if no function is identically zero on (a, b).

We say that S is orthonormal if, in addition, each function has norm 1.

Lemma

If S is orthogonal then S is linearly independent.

Proof

Suppose that ϕ_1, ϕ_2, ..., $\phi_N \in S$ satisfy

$$\sum_{j=1}^{N} C_j \phi_j(x) = 0 \quad \text{for } a < x < b.$$

On the one hand,

$$\left\langle \phi_k, \sum_{j=1}^{N} C_j \phi_j \right\rangle_w = \langle \phi_k, 0 \rangle_w = 0 \quad \text{for } 1 \le k \le N,$$

and on the other hand, because ϕ_k is orthogonal to ϕ_j for $j \ne k$,

$$\left\langle \phi_k, \sum_{j=1}^{N} C_j \phi_j \right\rangle_w = \sum_{j=1}^{N} C_j \langle \phi_k, \phi_j \rangle_w = C_k \langle \phi_k, \phi_k \rangle_w,$$

so $C_k \langle \phi_k, \phi_k \rangle_w = 0$. Since $\langle \phi_k, \phi_k \rangle_w = \|\phi_k\|_w^2 > 0$, it follows that $C_k = 0$ for $1 \le k \le N$. Hence, S is linearly independent. $\square$

Generalised Pythagorus theorem

Lemma

If $\{\phi_1, \dots, \phi_N\}$ *is orthogonal then, for any* $C_1, \dots, C_N \in \mathbb{R}$,

$$\left\| \sum_{j=1}^{N} C_j \phi_j \right\|_w^2 = \sum_{j=1}^{N} C_j^2 \|\phi_j\|_w^2.$$

Proof:

$$\mathit{LHS} = \left\langle \sum_{j=1}^{N} C_j \phi_j, \sum_{k=1}^{N} C_k \phi_k \right\rangle_w = \sum_{j=1}^{N} \sum_{k=1}^{N} C_j C_k \langle \phi_j, \phi_k \rangle_w$$

$$= \sum_{j=k} + \sum_{j \neq k} = \sum_{j=1}^{N} C_j^2 \langle \phi_j, \phi_j \rangle_w + 0 = RHS.$$

Generalised Fourier coefficients

Lemma

If f is in the span of an orthogonal set of functions $\{\phi_1, \phi_2, \ldots, \phi_N\}$ in $L_2(a, b, w)$, then the coefficients in the representation

$$f(x) = \sum_{j=1}^{N} A_j \phi_j(x)$$

are given by

$$A_j = \frac{\langle f, \phi_j \rangle_w}{\|\phi_j\|_w^2} \quad \text{for } 1 \leq j \leq N. \tag{29}$$

We call A_j the jth Fourier coefficient of f with respect to the given orthogonal set of functions.

Proof

Since

$$f(x) = \sum_{j=1}^{N} A_j \phi_j(x),$$

if $1 \le k \le N$ then

$$\begin{aligned}\langle f, \phi_k \rangle_w &= \left\langle \sum_{j=1}^{N} A_j \phi_j, \phi_k \right\rangle_w = \sum_{j=1}^{N} A_j \langle \phi_j, \phi_k \rangle_w \\ &= A_k \langle \phi_k, \phi_k \rangle_w + \sum_{j \ne k} = A_k \|\phi_k\|_w^2 + 0,\end{aligned}$$

so

$$A_k = \frac{\langle f, \phi_k \rangle_w}{\|\phi_k\|_w^2}. \quad \square$$

Classical Fourier sine coefficients

The sequence of functions

$$\phi_n(x) = \sin\frac{n\pi x}{\ell}, \qquad n = 1, 2, 3, \ldots,$$

is orthogonal with respect to the weight $w(x) = 1$ over the interval $(a, b) = (0, \ell)$, that is,

$$\langle \phi_n, \phi_j \rangle = \int_0^\ell \sin\frac{n\pi x}{\ell}\,\sin\frac{j\pi x}{\ell}\,dx = 0 \quad \text{if } n \neq j.$$

Since

$$\|\phi_n\|^2 = \int_0^\ell \sin^2\frac{n\pi x}{\ell}\,dx = \frac{\ell}{2},$$

the Fourier (sine) coefficients of a function $f(x)$ are

$$A_j = \frac{\langle f, \phi_j \rangle}{\|\phi_j\|^2} = \frac{2}{\ell}\int_0^\ell f(x)\sin\frac{j\pi x}{\ell}\,dx.$$

Least-squares approximation

Consider approximating a function $f \in L_2(a, b, w)$ by a function in the span of an orthogonal set $\{\phi_1, \phi_2, \dots, \phi_N\}$, that is, finding coefficients C_j such that

$$f(x) \approx \sum_{j=1}^{N} C_j \phi_j(x) \quad \text{for } a < x < b.$$

We seek to choose the C_j so that the weighted mean-square error

$$\left\| f - \sum_{j=1}^{N} C_j \phi_j \right\|_w^2 = \int_a^b \left(f(x) - \sum_{j=1}^{N} C_j \phi_j(x) \right)^2 w(x)\, dx$$

is as small as possible.

Least-squares approximation

Lemma

For all $C_1, C_2, \ldots, C_N$, the weighted mean-square error satisfies

$$\left\| f - \sum_{j=1}^{N} C_j \phi_j \right\|_w^2 = \|f\|_w^2 - \sum_{j=1}^{N} A_j^2 \|\phi_j\|_w^2 + \sum_{j=1}^{N} (C_j - A_j)^2 \|\phi_j\|_w^2,$$

and so achieves its unique minimum when C_j equals the jth Fourier coefficient of f, that is, when

$$C_j = A_j = \frac{\langle f, \phi_j \rangle_w}{\|\phi_j\|_w^2} \quad \text{for } 1 \leq j \leq N.$$

Proof

$$
\begin{aligned}
\left\| f - \sum_{j=1}^{N} C_j \phi_j \right\|_w^2 &= \left\langle f - \sum_{j=1}^{N} C_j \phi_j , f - \sum_{k=1}^{N} C_k \phi_k \right\rangle_w \\
&= \langle f, f \rangle_w - \left\langle f, \sum_{k=1}^{N} C_k \phi_k \right\rangle_w - \left\langle \sum_{j=1}^{N} C_j \phi_j , f \right\rangle_w \\
&\quad + \left\langle \sum_{j=1}^{N} C_j \phi_j , \sum_{k=1}^{N} C_k \phi_k \right\rangle_w \\
&= \|f\|_w^2 - \sum_{k=1}^{N} C_k \langle f, \phi_k \rangle_w - \sum_{j=1}^{N} C_j \langle \phi_j , f \rangle_w \\
&\quad + \sum_{j=1}^{N} \sum_{k=1}^{N} C_j C_k \langle \phi_j , \phi_k \rangle_w .
\end{aligned}
$$

Since

$$\sum_{j=1}^{N}\sum_{k=1}^{N} C_j C_k \langle \phi_j, \phi_k \rangle_w = \sum_{j=k} + \sum_{j \neq k} = \sum_{j=1}^{N} C_j^2 \|\phi_j\|_w^2 + 0$$

and $\langle f, \phi_j \rangle_w = A_j \|\phi_j\|_w^2$, we can complete the square as follows:

$$\begin{aligned}
\left\| f - \sum_{j=1}^{N} C_j \phi_j \right\|_w^2 &= \|f\|_w^2 - 2\sum_{j=1}^{N} C_j \langle f, \phi_j \rangle_w + \sum_{j=1}^{N} C_j^2 \|\phi_j\|_w^2 \\
&= \|f\|_w^2 - 2\sum_{j=1}^{N} C_j A_j \|\phi_j\|_w^2 + \sum_{j=1}^{N} C_j^2 \|\phi_j\|_w^2 \\
&= \|f\|_2^2 + \sum_{j=1}^{N} \left(C_j^2 - 2C_j A_j + A_j^2 - A_j^2 \right) \|\phi_j\|_w^2 \\
&= \|f\|_2^2 - \sum_{j=1}^{N} A_j^2 \|\phi_j\|_w^2 + \sum_{j=1}^{N} \left(C_j - A_j \right)^2 \|\phi_j\|_w^2. \qquad \square
\end{aligned}$$

Conclusion: Let $f \in L_2(a, b, w)$ and put

$$A_j = \frac{\langle f, \phi_j \rangle_w}{\|\phi_j\|_w^2}.$$

If f belongs to the span of $S = \{\phi_1, \ldots, \phi_N\}$, then

$$f(x) = \sum_{j=1}^{N} A_j \phi_j(x), \qquad a < x < b,$$

but if f does not belong to the span of S then

$$f(x) \approx \sum_{j=1}^{N} A_j \phi_j(x), \qquad a < x < b,$$

is the best approximation (in the weighted L_2-norm) to $f(x)$ by a function in the span of S.

Example of least-squares approximation

Define $f : [0, 1] \to \mathbb{R}$ by

$$f(x) = \begin{cases} 1, & 0 < x < 1/2, \\ 0, & 1/2 < x < 1, \end{cases}$$

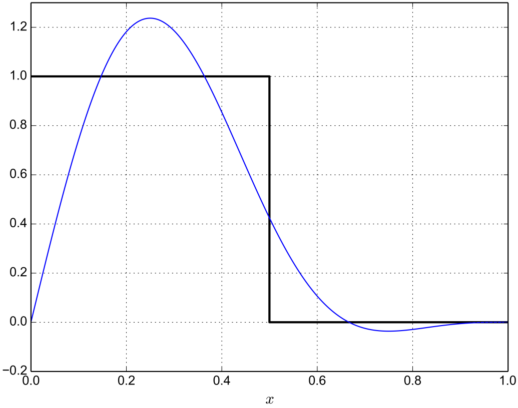
and let $\phi_n(x) = \sin n\pi x$. Then

$$\begin{aligned} A_j = \frac{\langle f, \phi_j \rangle}{\|\phi_j\|^2} &= 2 \int_0^1 f(x) \sin j\pi x \, dx \\ &= 2 \int_0^{1/2} \sin j\pi x \, dx = \frac{2B_j}{j\pi}, \end{aligned}$$

where

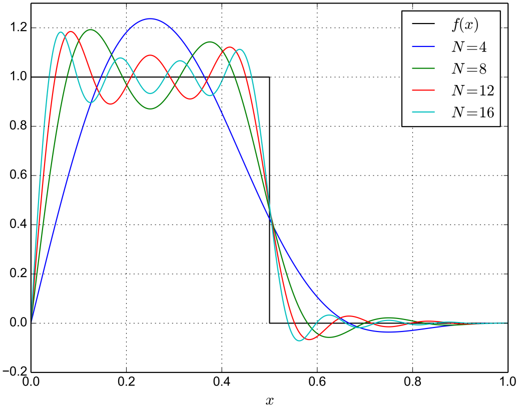
$$B_j = 1 - \cos \frac{j\pi}{2} = \begin{cases} 0, & j \equiv 0 \pmod 4, \\ 1, & j \equiv 1 \text{ or } 3 \pmod 4, \\ 2, & j \equiv 2 \pmod 4, \end{cases}$$

$$f(x) \approx \sum_{j=1}^{4} A_j \phi_j(x) = \frac{2}{\pi}\left(\sin \pi x + \sin 2\pi x + \frac{1}{3}\sin 3\pi x\right).$$



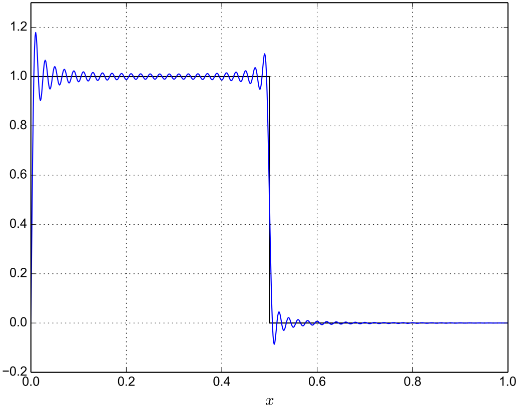
Convergence of least-squares approximation

$$f(x) \approx \sum_{j=1}^{N} A_j \phi_j(x)$$



Gibbs phenomenon

$$f(x) \approx \sum_{j=1}^{100} A_j \phi_j(x)$$



Legendre polynomials

Recall that

$$P_0(z) = 1, \quad P_1(z) = z, \quad P_2(z) = \tfrac{1}{2}(3z^2 - 1),$$
$$P_3(z) = \tfrac{1}{2}(5z^3 - 3z).$$

We will see later that these functions are orthgonal with respect to

$$\langle f, g \rangle = \int_{-1}^{1} f(x)g(x)\,dx,$$

that is,

$$\langle P_n, P_j \rangle = 0 \quad \text{if } n \neq j.$$

It can also be shown that

$$\|P_n\|^2 = \int_{-1}^{1} P_n(x)^2\,dx = \frac{2}{2n+1}.$$

Example of polynomial least-squares approximation

Let

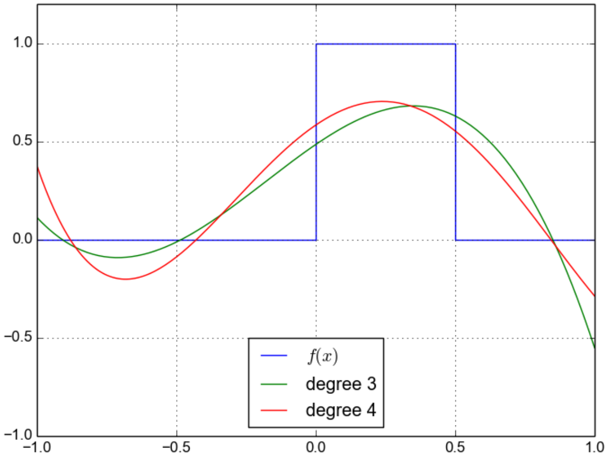
$$f(x) = \begin{cases} 1, & 0 < x < 1/2, \\ 0, & -1 < x < 0 \text{ or } 1/2 < x < 1. \end{cases}$$

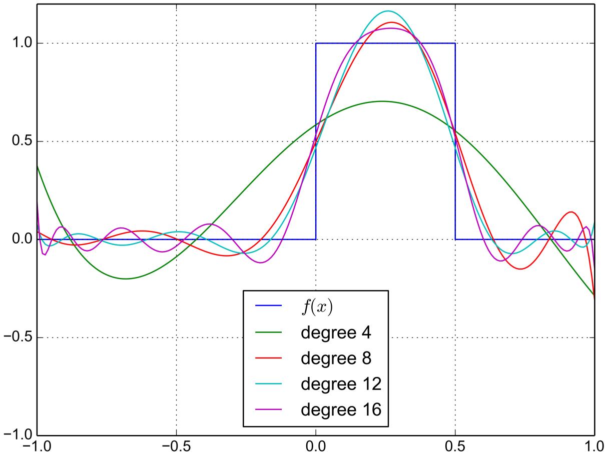
We find that

j	0	1	2	3	4
$\langle f, P_j \rangle$	$1/2$	$1/8$	$-3/16$	$-19/128$	$15/256$
$\Vert P_j \Vert^2$	2	$2/3$	$2/5$	$2/7$	$2/9$
A_j	$1/4$	$3/16$	$-15/32$	$-133/256$	$135/512$

so

$$\begin{aligned} f(x) &\approx \sum_{j=0}^{4} A_j P_j(x) = \sum_{j=0}^{4} \frac{\langle f, P_j \rangle}{\Vert P_j \Vert^2} P_j(x) \\ &= \frac{1}{4} P_0(x) + \frac{3}{16} P_1(x) - \frac{15}{32} P_2(x) - \frac{133}{256} P_3(x) + \frac{135}{512} P_4(x). \end{aligned}$$


1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
$f(x)$
degree 3
degree 4



1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
$f(x)$
degree 4
degree 8
degree 12
degree 16

Orthogonal sequences of functions

Now let $S = \{\phi_1, \phi_2, \phi_3, \ldots\}$ be a countably infinite orthogonal set in $L_2(a, b, w)$. Given any $f \in L_2(a, b, w)$ we write

$$f(x) \approx \sum_{j=1}^{\infty} A_j \phi_j(x)$$

to indicate that A_j is the jth Fourier coefficient of f, i.e.,

$$A_j = \frac{\langle f, \phi_j \rangle_w}{\|\phi_j\|_w^2} \quad \text{for all } j \geq 1.$$

Example

Fourier sine series: for $f \in L_2(0,\pi)$,

$$f(x) \approx \sum_{j=1}^{\infty} A_j \sin jx, \qquad A_j = \frac{2}{\pi} \int_0^{\pi} f(x) \sin jx \, dx.$$

Example

Fourier–Legendre series: for $f \in L_2(-1,1)$,

$$f(x) \approx \sum_{j=0}^{\infty} A_j P_j(x), \qquad A_j = \frac{2j+1}{2} \int_{-1}^{1} f(x) P_j(x) \, dx.$$

Example

Fourier–Tchebyshev series: recall that

$$T_0(x) = 1, \quad T_1(x) = x, \quad T_2(x) = 2x - 1, \quad T_3(x) = 4x^3 - 3x, \quad \ldots$$

are orthogonal over $(-1, 1)$ with respect to the weigth function $w(x) = 1/\sqrt{1 - x^2}$. The substitution $x = \cos\theta$ implies that

$$\|T_j\|_w^2 = \int_{-1}^{1} \frac{T_j(x)^2\,dx}{\sqrt{1 - x^2}} = \begin{cases} \pi, & j = 0, \\ \pi/2, & j \geq 1, \end{cases}$$

so for $f \in L_2(-1, 1, w)$,

$$f(x) \approx \frac{A_0}{2} + \sum_{j=1}^{\infty} A_j T_j(x), \qquad A_j = \frac{\langle f, T_j \rangle}{\|T_j\|_w^2} = \frac{2}{\pi} \int_{-1}^{1} \frac{f(x) T_j(x)\,dx}{\sqrt{1 - x^2}}.$$

Example

Trigonometric Fourier series: the functions

$$1, \quad \cos x, \quad \sin x, \quad \cos 2x, \quad \sin 2x, \quad \cos 3x, \quad \sin 3x, \quad \ldots$$

are orthogonal over $(-\pi, \pi)$ with

$$\int_{-\pi}^{\pi} 1^2\,dx = 2\pi \quad \text{and} \quad \int_{-\pi}^{\pi} \cos^2 jx\,dx = \pi = \int_{-\pi}^{\pi} \sin^2 jx\,dx,$$

so if $f \in L_2(-\pi, \pi)$ then

$$f(x) \approx \frac{A_0}{2} + \sum_{j=1}^{\infty} \bigl(A_j \cos jx + B_j \sin jx\bigr)$$

where

$$A_j = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos jx\,dx \quad \text{and} \quad B_j = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin jx\,dx.$$

Bessel inequality

For every N,

$$0 \le \left\| f - \sum_{j=1}^{N} A_j \phi_j \right\|_w^2 = \|f\|_w^2 - \sum_{j=1}^{N} A_j^2 \|\phi_j\|_w^2$$

so

$$\sum_{j=1}^{N} A_j^2 \|\phi_j\|_w^2 \le \|f\|_w^2,$$

and therefore the Fourier coefficients satisfy Bessel's inequality

$$\sum_{j=1}^{\infty} A_j^2 \|\phi_j\|_w^2 \le \|f\|_w^2.$$

Example

The Fourier sine coefficients of $f \in L_2(0,\pi)$ satisfy

$$\sum_{j=1}^{\infty} A_j^2 \le \frac{2}{\pi} \int_0^{\pi} f(x)^2 \, dx.$$

Example

The Fourier–Legendre coefficients of $f \in L_2(-1,1)$ satisfy

$$\sum_{j=0}^{\infty} \frac{A_j^2}{2j+1} \le \frac{1}{2} \int_{-1}^{1} f(x)^2 \, dx.$$

Example

For $w(x) = 1/\sqrt{1-x^2}$, the Fourier–Tchebyshev coefficients of $f \in L_2(-1,1,w)$ satisfy

$$\frac{A_0^2}{2} + \sum_{j=1}^{\infty} A_j^2 \le \frac{1}{\pi} \int_{-1}^{1} \frac{f(x)^2\,dx}{\sqrt{1-x^2}}.$$

Example

The trigonometric Fourier coefficients of $f \in L_2(-\pi,\pi)$ satisfy

$$\frac{A_0^2}{2} + \sum_{j=1}^{\infty} (A_j^2 + B_j^2) \le \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)^2\,dx.$$

Completeness

An orthogonal set S is complete if there is no non-trivial function in $L_2(a, b, w)$ orthogonal to every function in S, i.e., if the condition

$$\langle f, \phi \rangle_w = 0 \quad \text{for every } \phi \in S$$

implies that

$$\|f\|_w = 0.$$

In particular, if S is a complete orthogonal set, then every proper subset of S fails to be complete.

Example

The set $S = \{ \sin jx : j \geq 1 \text{ and } j \neq 7 \}$ is not complete in $L_2(0, \pi)$ because $\sin 7x$ is orthogonal to every function in S.

Equivalent definitions of completeness

Theorem

If $S = \{\phi_1, \phi_2, \dots\}$ is orthogonal in $L_2(a, b, w)$, then the following properties are equivalent:

1. *S is complete;*
2. *for each $f \in L_2(a, b, w)$, if A_j denotes the jth Fourier coefficient of f then*
$$\left\| f - \sum_{j=1}^{N} A_j \phi_j \right\|_w \to 0 \quad \text{as } N \to \infty;$$
3. *each function $f \in L_2(a, b, w)$ satisfies Parseval's identity:*
$$\|f\|_w^2 = \sum_{j=1}^{\infty} A_j^2 \|\phi_j\|_w^2.$$

Proof that $2 \iff 3$

We saw already that

$$\left\| f - \sum_{j=1}^{N} A_j \phi_j \right\|_w^2 = \|f\|_w^2 - \sum_{j=1}^{N} A_j^2 \|\phi_j\|_w^2,$$

so

$$\text{LHS} \to 0 \qquad \text{iff} \qquad \text{RHS} \to 0.$$

Now observe that RHS $\to 0$ means the same thing as

$$\|f\|_w^2 = \sum_{j=1}^{\infty} A_j^2 \|\phi_j\|_w^2.$$

Proof that $3 \implies 1$

Assume that Parseval's identity holds for every $f \in L_2(a, b, w)$. If

$$\langle f, \phi_j \rangle_w = 0 \quad \text{for all } j \geq 1,$$

then $A_j = 0$ for all $j \geq 1$, so

$$\|f\|_w^2 = \sum_{j=1}^{\infty} A_j^2 \|\phi_j\|_w^2 = 0,$$

showing that $f = 0$. Therefore S is complete.

Proof that $1 \implies 2$

Assume that S is complete. Given $f \in L_2(a,b,w)$, Bessel's inequality implies that the Fourier series of f converges in $L_2(a,b,w)$, so we can define

$$g = f - \sum_{k=1}^{\infty} A_k \phi_k.$$

Since

$$\begin{aligned}\langle g, \phi_j \rangle_w &= \langle f, \phi_j \rangle_w - \sum_{k=1}^{\infty} A_k \langle \phi_j, \phi_k \rangle_w \\ &= \langle f, \phi_j \rangle_w - A_j \|\phi_j\|_w^2 = 0\end{aligned}$$

for every $j \geq 1$, we conclude that $g = 0$ and therefore 2 holds:

$$f = \sum_{j=1}^{\infty} A_j \phi_j. \quad \square$$

Least-squares error

Consider

$$f(x) \approx \sum_{j=1}^{N} A_j \phi_j(x) \quad \text{for } a < x < b,$$

where $A_j = \langle f, \phi_j \rangle_w / \|\phi_j\|_w^2$ is the jth Fourier coefficient of f. We have seen that this choice of A_j minimises the norm in $L_2(a, b, w)$ of the error

$$e_N(x) = f(x) - \sum_{j=1}^{N} A_j \phi_j(x).$$

Corollary

If $S = \{\phi_1, \phi_2, \phi_3, \dots\}$ is a complete orthogonal sequence in $L_2(a, b, w)$, then for any $f \in L_2(a, b, w)$,

$$\|e_N\|^2 = \sum_{j=N+1}^{\infty} A_j \|\phi_j\|_w^2.$$

Proof

We saw in the proof of the theorem ($2 \iff 3$) that

$$\|e_N\|_w^2 = \|f\|_w^2 - \sum_{j=1}^{N} A_j^2 \|\phi_j\|_w^2.$$

Since Parseval's identity holds,

$$\|f\|_w^2 = \sum_{j=1}^{\infty} A_j^2 \|\phi_j\|_w^2,$$

and therefore

$$\begin{aligned}\|e_N\|_w^2 &= \sum_{j=1}^{\infty} A_j^2 \|\phi_j\|_w^2 - \sum_{j=1}^{N} A_j^2 \|\phi_j\|_w^2 \\ &= \sum_{j=N+1}^{\infty} A_j^2 \|\phi_j\|_w^2. \quad \square\end{aligned}$$

Parseval identity

Each of our four examples of orthgonal sequences is in fact complete.

Example

The Fourier sine coefficients of $f \in L_2(0, \pi)$ satisfy

$$\sum_{j=1}^{\infty} A_j^2 = \frac{2}{\pi} \int_0^{\pi} f(x)^2 \, dx.$$

Example

The Fourier–Legendre coefficients of $f \in L_2(-1, 1)$ satisfy

$$\sum_{j=0}^{\infty} \frac{A_j^2}{2j+1} = \frac{1}{2} \int_{-1}^{1} f(x)^2 \, dx.$$

Example

For $w(x) = 1/\sqrt{1-x^2}$, the Fourier–Tchebyshev coefficients of $f \in L_2(-1,1,w)$ satisfy

$$\frac{A_0^2}{2} + \sum_{j=1}^{\infty} A_j^2 = \frac{1}{\pi} \int_{-1}^{1} \frac{f(x)^2\,dx}{\sqrt{1-x^2}}.$$

Example

The trigonometric Fourier coefficients of $f \in L_2(-\pi,\pi)$ satisfy

$$\frac{A_0^2}{2} + \sum_{j=1}^{\infty} (A_j^2 + B_j^2) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)^2\,dx.$$

Estimating the least-squares error

Consider

$$f(x) = x \quad \text{for } 0 < x < \pi.$$

This function has the Fourier sine series

$$f(x) \approx 2\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx$$

so

$$f(x) = 2\sum_{n=1}^{N} \frac{(-1)^{n+1}}{n} \sin nx + e_N(x),$$

and the error term $e_N(x)$ satisfies

$$\int_0^{\pi} e_N(x)^2\,dx = \|e_N\|^2 = \sum_{n=N+1}^{\infty} \underbrace{\left(\frac{2(-1)^{n+1}}{n}\right)^2}_{A_n^2} \underbrace{\frac{\pi}{2}}_{\|\phi_n\|^2}.$$

Thus,

$$\begin{aligned}\|e_N\|^2 &= 2\pi \sum_{n=N+1}^{\infty} \frac{1}{n^2} \\ &\leq 2\pi \int_N^{\infty} \frac{dt}{t^2} = \frac{2\pi}{N},\end{aligned}$$

and so

$$\|e_N\| = O(N^{-1/2}).$$

In other words, the root mean square error for the approximation

$$x \approx 2\left(\sin x - \frac{\sin 2x}{2} + \cdots + (-1)^{N+1}\frac{\sin Nx}{N}\right), \quad 0 < x < \pi,$$

tends to zero like $N^{-1/2}$.

Sturm–Liouville problems

Recall that the eigenvectors of a real $N \times N$ symmetric matrix form a orthogonal basis for $\mathbb{R}^N$. In the same way, we will see that a type of homogeneous boundary-value problem generates a sequence of functions that are orthgonal with respect to an associated weight function.

The sequence of orthogonal functions turns out to be complete, but the proof of this fact relies on results from functional analysis (Math5605) about a compact linear operator on a Hilbert space.

Sturm–Liouville operator

An ODE of the form

$$[p(x)u']' + [\lambda r(x) - q(x)]u = 0, \quad a < x < b, \tag{30}$$

is called a Sturm–Liouville equation. The coefficients p, q, r must all be *real*-valued with

$$p(x) > 0 \quad \text{and} \quad r(x) > 0 \quad \text{for } a < x < b.$$

Defining the formally self-adjoint differential operator

$$Lu = -[p(x)u']' + q(x)u, \tag{31}$$

we can write (30) as

$$Lu = \lambda r u \quad \text{on } (a, b).$$

Eigenfunctions and eigenvalues

Any non-trivial (possibly complex-valued) solution u satisfying $Lu = \lambda r u$ on (a, b) (plus appropriate boundary conditions) is said to be an eigenfunction of L with eigenvalue λ. In this case, we refer to (ϕ, λ) as an eigenpair.

Example

Legendre's equation

$$(1 - x^2)u'' - 2xu' + \nu(\nu + 1)u = 0$$

is equivalent to

$$[(1 - x^2)u']' + \nu(\nu + 1)u = 0$$

which is of the form (30) with

$$p(x) = 1 - x^2, \quad q(x) = 0, \quad r(x) = 1, \quad \lambda = \nu(\nu + 1).$$

Boundary conditions

Assume as before that p, q, r are real-valued with $p(x) > 0$ and $r(x) > 0$ for $a < x < b$. A regular Sturm–Liouville eigenproblem is of the form

$$\begin{aligned} Lu &= \lambda r u && \text{for } a < x < b, \\ B_1 u = b_{11}u' + b_{10}u &= 0 && \text{at } x = a, \\ B_2 u = b_{21}u' + b_{20}u &= 0 && \text{at } x = b, \end{aligned} \tag{32}$$

where a and b are *finite* with

$$p(a) \neq 0 \quad \text{and} \quad p(b) \neq 0,$$

and where b_{10}, b_{11}, b_{20}, b_{21} are real with

$$|b_{10}| + |b_{11}| \neq 0 \quad \text{and} \quad |b_{20}| + |b_{21}| \neq 0.$$

Eigenfunctions are orthogonal

We now describe how a Sturm–Liouville problem leads to an associated complete orthogonal system, and hence to a generalised Fourier expansion.

Theorem

Let L be a Sturm–Liouville differential operator (31)*. If u, $v : [a, b] \to \mathbb{C}$ satisfy*

$$Lu = \lambda r u \quad \text{on } (a, b), \quad \text{with } B_1 u = 0 = B_2 u,$$

and

$$Lv = \mu r v \quad \text{on } (a, b), \quad \text{with } B_1 v = 0 = B_2 v,$$

and if $\lambda \neq \mu$, then u and v are orthogonal on the interval (a, b) with respect to the weight function $r(x)$, i.e.,

$$\langle u, v \rangle_r = \int_a^b u(x) v(x) r(x)\, dx = 0.$$

Proof

Using the Lagrange identity,

$$
\begin{aligned}
(\lambda - \mu)\langle u, v\rangle_r &= (\lambda - \mu)\int_a^b u(x)v(x)r(x)\,dx \\
&= \int_a^b \big[\lambda r(x)u(x)\big]v(x)\,dx - \int_a^b u(x)\big[\mu r(x)v(x)\big]\,dx \\
&= \int_a^b Lu(x)\,v(x)\,dx - \int_a^b u(x)\,Lv(x)\,dx \\
&= \sum_{j=1}^{2}\big(B_j u R_j v - R_j u B_j v\big) = 0,
\end{aligned}
$$

since $B_j u = 0 = B_j v$ for $j \in \{1, 2\}$. By hypothesis, $\lambda \neq \mu$ so $\lambda - \mu \neq 0$, and we conclude that $\langle u, v\rangle_r = 0$. $\square$

Eigenvalues are real

Theorem

Let L be a Sturm–Liouville differential operator (31)*. If $u : [a, b] \to \mathbb{C}$ is not identically zero and satisfies*

$$Lu = \lambda r u \quad \text{on } (a, b), \quad \text{with } B_1 u = 0 = B_2 u,$$

then λ is real.

In the same way, we can prove that for a real, symmetric (or complex, Hermitian) matrix:

- eigenvectors with distinct eigenvalues are orthogonal;
- every eigenvalue is real.

Proof

Since $p(x)$, $q(x)$, $r(x)$ are each real,

$$L\bar{u} = -(p\bar{u}')' + q\bar{u} = \overline{-(pu')' + qu} = \overline{Lu} = \overline{\lambda r u} = \bar{\lambda} r \bar{u},$$

so $\bar{u}$ is an eigenfunction with eigenvalue $\bar{\lambda}$. Thus, if $\lambda \neq \bar{\lambda}$ then

$$0 = \langle u, \bar{u} \rangle_r = \int_a^b u(x)\bar{u}(x)r(x)\,dx = \int_a^b |u|^2 r\,dx,$$

implying $u \equiv 0$, a contradiction.
We conclude $\lambda = \bar{\lambda}$, that is, λ is real. $\square$

Completeness of the eigenfunctions

Theorem

The regular Sturm–Liouville problem (32) *has an infinite sequence of eigenfunctions* ϕ_1, ϕ_2, ϕ_3, ... *with corresponding eigenvalues* λ_1, λ_2, λ_3, ... *and moreover:*

1. *the eigenfunctions* ϕ_1, ϕ_2, ϕ_3, ... *form a complete orthogonal system on the interval* (a, b) *with respect to the weight function* $r(x)$*;*
2. *the eigenvalues satisfy* $\lambda_1 < \lambda_2 < \lambda_3 < \cdots$ *with* $\lambda_j \to \infty$ *as* $j \to \infty$.

In the same way, for every real symmetric (or complex, Hermitian) $n \times n$ matrix A there is an orthgonal basis for $\mathbb{R}^n$ consisting of eigenvectors of A.

Real trigonometric Fourier series

The various trigonometric Fourier series arise from the the Sturm–Liouville ODE

$$u'' + \lambda u = 0.$$

For the interval $[-\ell, \ell]$, periodic boundary conditions

$$u(\ell) = u(-\ell) \quad \text{and} \quad u'(\ell) = u'(-\ell)$$

lead to the eigenfunctions

$$\begin{aligned} \phi_{2j}(x) &= \cos\frac{j\pi x}{\ell} \qquad &&\text{for } j = 0,\ 1,\ 2,\ \ldots, \\ \phi_{2j-1}(x) &= \sin\frac{j\pi x}{\ell} \qquad &&\text{for } j = 1,\ 2,\ \ldots, \end{aligned}$$

and (non-distinct!) eigenvalues

$$\lambda_0 = 0, \qquad \lambda_{2j} = \lambda_{2j-1} = \left(\frac{j\pi}{\ell}\right)^2, \quad j \geq 1.$$

Complex trigonometric Fourier series

Again taking

$$u'' + \lambda u = 0, \quad u(\ell) = u(-\ell), \quad u'(\ell) = u'(-\ell),$$

we can instead take

$$\phi_j(x) = e^{i\pi jx/\ell} \quad \text{and} \quad \lambda_j = \left(\frac{j\pi}{\ell}\right)^2 \quad \text{for } j = 0,\ \pm1,\ \pm2,\ \ldots.$$

Here, the ϕ_j are orthogonal with respect to the complex inner product

$$\langle f, g\rangle = \int_{-\ell}^{\ell} f(x)\overline{g(x)}\,dx,$$

so

$$f(x) \approx \sum_{j=-\infty}^{\infty} A_j e^{ij\pi x/\ell}, \qquad A_j = \frac{1}{2\ell}\int_{-\ell}^{\ell} f(x)e^{-ij\pi x/\ell}\,dx.$$

Half-range trigonometric Fourier series

Consider

$$u'' + \lambda u = 0 \quad \text{for } 0 < x < \ell.$$

Homogeneous Dirichlet boundary conditions

$$u(0) = u(\ell) = 0$$

lead to

$$\phi_j(x) = \sin\frac{j\pi x}{\ell} \quad \text{and} \quad \lambda_j = \left(\frac{j\pi}{\ell}\right)^2 \quad \text{for } j = 1, 2, 3, \ldots,$$

whereas homogeneous Neumann boundary conditions

$$u'(0) = u'(\ell) = 0$$

lead to

$$\phi_j(x) = \cos\frac{j\pi x}{\ell} \quad \text{and} \quad \lambda_j = \left(\frac{j\pi}{\ell}\right)^2 \quad \text{for } j = 0, 1, 2, \ldots.$$

Least-squares error for cosine expansion

Consider once again the function

$$f(x) = x \quad \text{for } 0 < x < \pi.$$

This time, instead of the sine series use the cosine series

$$\begin{aligned} f(x) &\approx \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi n^2} (\cos n\pi - 1) \cos nx \\ &\approx \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\cos(2k-1)x}{(2k-1)^2}, \end{aligned}$$

and estimate the error term $e_N(x)$ where

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{N} \frac{2}{\pi n^2} (\cos n\pi - 1) \cos nx + e_N(x).$$

With $N = 2K - 1$,

$$\begin{aligned}
\|e_N\|^2 &= \sum_{n=N+1}^{\infty} \underbrace{\left(\frac{2}{\pi n^2}(\cos n\pi - 1)\right)^2}_{A_n^2} \underbrace{\frac{\pi}{2}}_{\|\phi_n\|^2} \\
&= \sum_{k=K}^{\infty} \frac{4}{\pi^2(2k-1)^4} \frac{\pi}{2} = \frac{1}{8\pi} \sum_{k=K}^{\infty} \frac{1}{(k-\frac{1}{2})^4} \\
&\leq \frac{1}{8\pi} \int_K^{\infty} \frac{dt}{(t-\frac{1}{2})^4} = \frac{1}{24\pi} \frac{1}{(K-\frac{1}{2})^3} = \frac{2}{3\pi} \frac{1}{N^3}.
\end{aligned}$$

Thus, for this choice of f, the root mean square error using the cosine expansion is

$$\|e_N\| = O(N^{-3/2}),$$

versus $O(N^{-1/2})$ for the sine expansion.

Fourier–Bessel series

In this section, we study a singular Sturm–Liouville problem that will arise when we deal with eigenproblems for partial differential equations. The proof that the eigenfunctions are orthogonal needs to be modified because of the different type of boundary condition imposed at $x = 0$.

A singular Sturm–Liouville problem

Consider the parameterised Bessel equation of order $\nu \geq 0$,

$$x^2 u'' + x u' + (k^2 x^2 - \nu^2) u = 0,$$

which can be written in Sturm–Liouville form as

$$(x u')' + \left(k^2 x - \nu^2 x^{-1}\right) u = 0,$$

i.e.,

$$p(x) = x, \qquad r(x) = x, \qquad \lambda = k^2, \qquad q(x) = \frac{\nu^2}{x}.$$

On the interval $[0, \ell]$, we do not have a *regular* Sturm–Liouville problem because

$$p(0) = 0$$

and $q(x)$ is unbounded as $x \to 0^+$.

A condition ensuring $\lambda \geq 0$

Lemma

Consider the ODE

$$(xu')' + (\lambda x - \nu^2 x^{-1})u = 0 \quad \text{for } 0 < x < \ell, \tag{33}$$

with boundary conditions

$$\begin{gathered} u(x) \text{ bounded and } xu'(x) \to 0 \text{ as } x \to 0^+, \\ c_1 u' + c_0 u = 0 \text{ at } x = \ell, \end{gathered} \tag{34}$$

and suppose that $c_0 c_1 \geq 0$.

1. *If a non-trivial solution u exists, then $\lambda \geq 0$.*
2. *A non-trivial solution u exists for $\lambda = 0$ only when $\nu = 0$ and $c_0 = 0$, in which case u is a constant.*

Proof

Multiply (33) by u and integrate to obtain

$$\int_0^\ell \left[(xu')'u + (\lambda x - \nu^2 x^{-1})u^2\right] dx = 0.$$

Integration by parts gives

$$\int_0^\ell (xu')'u\,dx = \ell u'(\ell)u(\ell) - \int_0^\ell x(u')^2\,dx,$$

since the first boundary condition ensures $xu'(x)u(x) \to 0$ as $x \to 0$, so

$$\lambda \int_0^\ell xu^2\,dx = \int_0^\ell x(u')^2\,dx + \nu^2 \int_0^\ell x^{-1}u^2\,dx - \ell u'(\ell)u(\ell).$$

The boundary condition at $x = \ell$ gives

$$-u'(\ell)u(\ell) = \frac{u'(\ell)}{c_0}\big[-c_0 u(\ell)\big] = \frac{u'(\ell)}{c_0}\big[c_1 u'(\ell)\big] = \frac{c_1}{c_0}\,\big[u'(\ell)\big]^2,$$

if $c_0 \neq 0$, and

$$-u'(\ell)u(\ell) = \big[-c_1 u'(\ell)\big]\frac{u(\ell)}{c_1} = \big[c_0 u(\ell)\big]\frac{u(\ell)}{c_1} = \frac{c_0}{c_1}\,\big[u(\ell)\big]^2,$$

if $c_1 \neq 0$. In either case, $-u'(\ell)u(\ell) \geq 0$ because $c_0 c_1 \geq 0$. Hence, $\lambda \geq 0$, proving part 1.
If $\lambda = 0$ then u satisfies

$$\int_0^\ell x(u')^2\,dx + \nu^2 \int_0^\ell x^{-1}u^2\,dx = \ell u'(\ell)u(\ell) \leq 0.$$

For a non-trivial u, it follows that $u' \equiv 0$ and $\nu^2 = 0$, so u is constant and $\nu = 0$, proving part 2. □

Bessel functions

Theorem

Assume that $c_0 c_1 \geq 0$*. If* $c_0 \neq 0$ *or* $\nu > 0$*, then the eigenvalues* λ_1*,* λ_2*, ... and eigenfunctions* ϕ_1*,* ϕ_2*, ... of the Bessel equation* (33) *with boundary conditions* (34) *are given by*

$$\lambda_j = k_j^2 \quad \text{and} \quad \phi_j(x) = J_\nu(k_j x) \quad \text{for } j \geq 1,$$

where k_j *is the* j*th positive solution of the equation*

$$c_0 J_\nu(k\ell) + c_1 k J_\nu'(k\ell) = 0. \tag{35}$$

If $c_0 = 0 = \nu$*, so that* $u'(\ell) = 0$ *by* (34)*, then we have an additional eigenvalue and eigenfunction*

$$\lambda_0 = 0 \quad \text{and} \quad \phi_0(x) = 1.$$

Proof

First suppose that $\lambda = k^2 > 0$. Then (33) is equivalent to the parameterised Bessel equation

$$x^2 u'' + x u' + (k^2 x^2 - \nu^2) u = 0$$

which has the general solution $u = AJ_\nu(kx) + BY_\nu(kx)$. Since $Y_\nu(kx)$ is unbounded as $x \to 0$, we must have $B = 0$, so $u = AJ_\nu(kx)$ and $u' = AkJ'_\nu(kx)$. Thus, at $x = \ell$,

$$c_1 u' + c_0 u = A\big(c_1 k J'_\nu(k\ell) + c_0 J_\nu(k\ell)\big),$$

showing that the boundary condition holds iff k is a solution of (35), in which case $J_\nu(kx)$ is an eigenfunction.

When $\lambda = 0$, we already know that $c_0 = 0 = \nu$ and $u = A$ for some constant A.

Orthogonality and normalization

Theorem

Assume that $c_0 c_1 \geq 0$. *The Bessel eigenfunctions, satisfying* (33) *and* (34), *have the orthogonality property*

$$\int_0^\ell \phi_i(x)\phi_j(x)x\,dx = 0 \quad \text{if } i \neq j.$$

If $c_1 \neq 0$, *then*

$$\int_0^\ell \phi_j(x)^2 x\,dx = \frac{1}{2k_j^2}\left[\left(\frac{\ell c_0}{c_1}\right)^2 + (k_j\ell)^2 - \nu^2\right] J_\nu(k_j\ell)^2 \quad \text{for } j \geq 1,$$

whereas if $c_1 = 0$, *so that* $\phi_j(\ell) = 0$ *by* (34), *then*

$$\int_0^\ell \phi_j(x)^2 x\,dx = \frac{\ell^2}{2} J_{\nu+1}(k_j\ell)^2 \quad \text{for } j \geq 1.$$

In the case $c_0 = 0 = \nu$, *we have* $\int_0^\ell \phi_0(x)^2 x\,dx = \ell^2/2$.

Proof of orthogonality

Put $Lu = -(xu')' + \nu^2 x^{-1} u$ so that $L\phi_j = x\lambda_j\phi_j$. It suffices to show that

$$\langle Lu, v\rangle = \langle u, Lv\rangle \quad \text{for all } u,\ v \in \mathcal{D},$$

where $\mathcal{D}$ is the space of C^2 functions satisfying the boundary conditions (34).
In fact, if u, $v \in \mathcal{D}$ then $x(u'v - uv') \to 0$ as $x \to 0$ so the Lagrange identity gives

$$\langle Lu, v\rangle - \langle u, Lv\rangle = -\ell\big[u'(\ell)v(\ell) - u(\ell)v'(\ell)\big]$$

and it suffices to prove that $u'v - uv' = 0$ at $x = \ell$. Using the boundary conditions $c_1u' + c_0u = 0 = c_1v' + c_0v$, we have
$c_0u'v = u'(c_0v) = u'(-c_1v') = (-c_1u')v' = (c_0u)v' = c_0uv'$ and
$c_1u'v = (-c_0u)v = u(-c_0v) = u(c_1v') = c_1uv'$, so $c_j(u'v - uv') = 0$ for $j \in \{0, 1\}$. At least one of c_0 and c_1 is not zero, and thus $u'v - uv' = 0$ at $x = \ell$. $\square$

Fourier–Bessel expansion

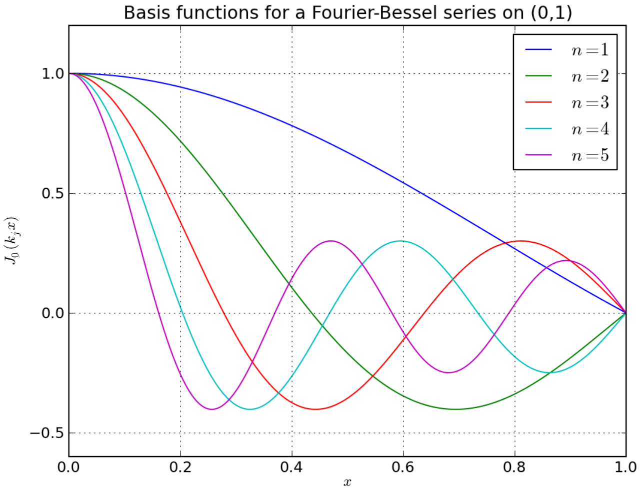
For instance, when $c_1 = 0$ we have

$$f(x) \approx \sum_{j=1}^{\infty} A_j J_\nu(k_j x), \quad 0 < x < \ell,$$

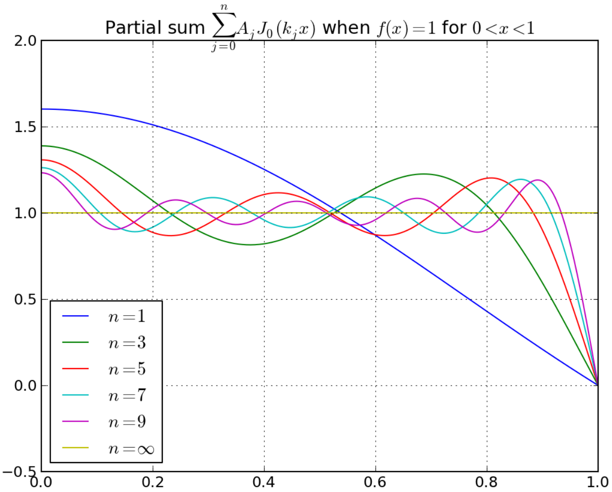
with

$$\begin{aligned} A_j &= \int_0^\ell f(x) J_\nu(k_j x) x\,dx \Big/ \int_0^\ell J_\nu(k_j x)^2 x\,dx \\ &= \frac{2}{\ell^2 J_{\nu+1}(k_j \ell)^2} \int_0^\ell f(x) J_\nu(k_j x) x\,dx. \end{aligned}$$

Plots when $\ell = 1$, $\nu = 0$, $c_1 = 0$



Partial sums



Another singular Sturm–Liouville problem

Consider the Legendre equation with parameter ν,

$$(1-x^2)u'' - 2xu' + \nu(\nu+1)u = 0 \quad \text{for } -1 < x < 1,$$

which can be written in Sturm–Liouville form as

$$[(1-x^2)u']' + \nu(\nu+1)u = 0,$$

i.e.,

$$p(x) = 1 - x^2, \qquad r(x) = 1, \qquad \lambda = \nu(\nu+1), \qquad q(x) = 0.$$

On the interval $[-1, 1]$ we do not have a regular Sturm–Liouville problem because

$$p(-1) = 0 = p(1).$$

We assume without loss of generality that $\nu \geq -\frac{1}{2}$ because

$$\nu(\nu+1) = \mu(\mu+1) \quad \text{if } \mu = -\nu - 1.$$

Eigenvalues are positive

Consider the ODE

$$[(1-x^2)u']' + \lambda u = 0 \tag{36}$$

with boundary conditions

$$u(x) \text{ bounded and } (1 \mp x)u'(x) \to 0 \text{ as } x \to \pm 1. \tag{37}$$

Muliplying the ODE by u and integrating (by parts) we find

$$\lambda \int_{-1}^{1} u^2 \, dx = \int_{-1}^{1} (1-x^2)(u')^2 \, dx,$$

so if a non-trivial solution u exists then $\lambda \geq 0$.

Legendre polynomials are orthogonal

If ν is not an integer then the boundary-value problem (36)–(37) has only the trivial solution $u \equiv 0$ (see the tutorial problems), whereas if $\nu = n$ is a non-negative integer, then all solutions have the form

$$u = CP_n(x),$$

where P_n is the Legendre polynomial of degree n, so

$$\phi_n(x) = P_n(x) \quad \text{and} \quad \lambda_n = n(n+1)$$

for $n = 0$, 1, 2, 3, ...

Theorem

The Legendre polynomials are orthogonal over the interval $[-1, 1]$,

$$\langle P_n, P_m \rangle = \int_{-1}^{1} P_n(x) P_m(x)\, dx = 0 \quad \textit{if } m \neq n.$$

Proof

Since $r(x) = 1$, it suffices to show that the Legendre differential operator $Lu = -[(1-x^2)u']'$ is self-adjoint,

$$\langle Lu, v\rangle = \langle u, Lv\rangle \quad \text{for all } u,\ v \in \mathcal{D},$$

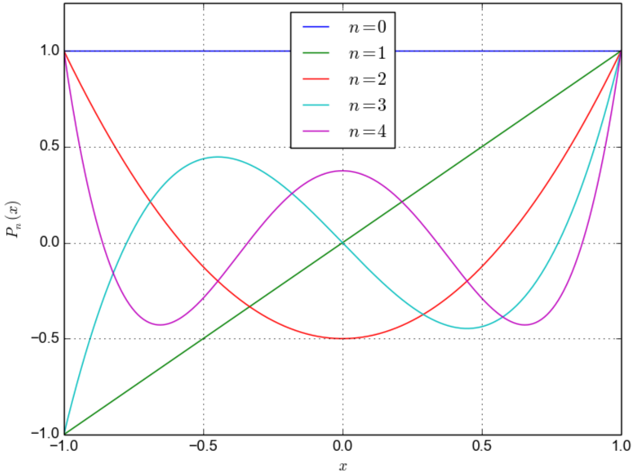
where $\mathcal{D}$ is the space of C^2 functions on $(-1, 1)$ that satisfy the boundary conditions (37).

In fact, the Lagrange identity shows that

$$\langle Lu, v\rangle - \langle u, Lv\rangle = \left[-(1-x^2)(u'v - uv')\right]_{-1}^{1},$$

which is zero by (37). $\square$

Plots



Schrödinger equation

To conclude this part of the course, we use separation of variables to solve a partial differential equation that arises in quantum theory, and has a similar structure to the heat equation but with the imaginary unit i multiplying the time derivative term. As a consequence, the modes that arise do not decay exponentially but instead have a complex phase factor.

Quantum wave function

The state of a quantum mechanical particle in 1D can be represented by a complex-valued wave function $\psi(x, t)$. We can interpret $|\psi(x, t)|^2$ as the probability density function for the position x of the particle, so that

$$\int_a^b |\psi(x, t)|^2 \, dx$$

is the probability that the particle lies in the interval $[a, b]$ at time t. Thus, the wave function must have the normalisation

$$\int_{-\infty}^{\infty} |\psi(x, t)|^2 \, dx = 1.$$

Time-dependent Schrödinger equation

In 1925, Erwin Schrödinger postulated that ψ obeys the PDE

$$i\hbar \frac{\partial \psi}{\partial t} + \frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} - V(x,t)\psi = 0. \tag{38}$$

Here, m is the mass of the particle, $V(x,t)$ is the potential of the applied force and $\hbar = h/(2\pi)$ where

$$h = 6.626\,070\,040 \times 10^{-34} \text{ Joule-seconds}$$

is Planck's constant.

Separable solutions

Suppose that V does not depend on t, and look for separable solutions

$$\psi(x,t) = u(x)T(t).$$

We find that

$$\frac{i\hbar T'}{T} = E = -\frac{\hbar^2}{2m}\,\frac{u''}{u} + V(x)$$

where the separation constant E has the physical dimensions of energy.

Thus,

$$T' = -\frac{iE}{\hbar}\,T$$

and we may take $T(t) = e^{-iEt/\hbar}$, so that

$$\psi(x,t) = e^{-iEt/\hbar}u(x).$$

Time-independent Schrödinger equation

The spatial dependence of the wave function satisfies

$$\frac{\hbar^2}{2m}\,u'' + \big[E - V(x)\big]u = 0, \tag{39}$$

which has the form of a Sturm–Liouville ODE (30) with

$$p(x) = \frac{\hbar^2}{2m}, \qquad \lambda = E, \qquad r(x) = 1, \qquad q(x) = V(x).$$

If (ϕ_n, E_n) is the nth eigenpair, then ϕ_n and E_n are real, and

$$\psi_n(x,t) = e^{-iE_n t/\hbar}\phi_n(x)$$

is a solution of the time-dependent Schrödinger equation (38).

Eigenstates

Since $|\psi_n(x,t)|^2 = \phi_n(x)^2$ the eigenfunction ϕ_n must be normalised to satisfy

$$\int_{-\infty}^{\infty} \phi_n(x)^2 \, dx = 1. \tag{40}$$

In this case, ψ_n is called the nth eigenstate. Physically, the eigenvalue E_n is the energy of the particle in this eigenstate.

The general solution is then a superposition

$$\psi(x,t) = \sum_{n=0}^{\infty} A_n \psi_n(x,t) = \sum_{n=0}^{\infty} A_n e^{-iE_n t/\hbar} \phi_n(x).$$

Define the complex inner product and norm with weight function $r(x) = 1$,

$$\langle f, g\rangle = \int_{-\infty}^{\infty} f(x)\overline{g(x)}\,dx \quad \text{and} \quad \|f\|^2 = \int_{-\infty}^{\infty} |f(x)|^2\,dx.$$

The ϕ_n are orthonormal ($\langle \phi_n, \phi_j\rangle = \delta_{nj}$) so the amplitudes A_n satisfy Parseval's identity:

$$\begin{aligned}
1 &= \int_{-\infty}^{\infty} |\psi(x,t)|^2\,dx = \langle \psi(\cdot,t), \psi(\cdot,t)\rangle \\
&= \left\langle \sum_{n=0}^{\infty} A_n e^{-iE_n t/\hbar}\phi_n, \sum_{k=0}^{\infty} A_k e^{-iE_k t/\hbar}\phi_k \right\rangle \\
&= \sum_{n=0}^{\infty}\sum_{k=0}^{\infty} \left(A_n e^{-iE_n t/\hbar}\right)\left(\overline{A}_k e^{+iE_k t/\hbar}\right)\langle \phi_n, \phi_k\rangle \\
&= \sum_{n=0}^{\infty} |A_n|^2.
\end{aligned}$$

Quantum mechanical harmonic oscillator

For the classical harmonic oscillator, $m\ddot{x} + kx = 0$, the restoring force $-kx$ has the potential $V(x) = \frac{1}{2}kx^2$, and the motion of the particle is given by

$$x_{\text{Classical}} = A\cos\omega t + B\sin\omega t \quad \text{where} \quad \omega = \sqrt{k/m}.$$

Putting $V(x) = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2 x^2$ in the time-independent Schrödinger equation (39) we have

$$\frac{\hbar^2}{2m}u'' + \left[E - \tfrac{1}{2}m\omega^2 x^2\right]u = 0. \tag{41}$$

To simplify the coefficients, let

$$v(y) = u(x) \quad \text{where} \quad y = (m\omega/\hbar)^{1/2}x,$$

and obtain

$$v'' + (2\alpha + 1 - y^2)v = 0, \qquad \alpha = \frac{E}{\hbar\omega} - \frac{1}{2}.$$

Series solutions

Next we put

$$v(y) = e^{-y^2/2} w(y)$$

and find that w satisfies Hermite's equation

$$w'' - 2yw' + 2\alpha w = 0,$$

which admits power series solutions

$$w(y) = \sum_{k=0}^{\infty} B_k y^k.$$

The recurrence relation for the coefficients is

$$B_{k+2} = \frac{2k - 2\alpha}{(k+2)(k+1)} B_k \quad \text{for all } k \geq 0.$$

Series solutions

Thus, $w(y) = B_0 w_0(y) + B_1 w_1(y)$ where

$$w_0 = 1 - \frac{2\alpha}{2!}\,y^2 - \frac{2^2(2-\alpha)\alpha}{4!}\,y^4 - \frac{2^3(4-\alpha)(2-\alpha)\alpha}{6!}\,y^6 - \cdots\,,$$
$$w_1 = y + \frac{2(1-\alpha)}{3!}\,y^3 + \frac{2^2(3-\alpha)(1-\alpha)}{5!}\,y^5 + \cdots\,.$$

It can be shown that if α is not a non-negative integer then w_0 and w_1 are both $O(e^{y^2})$ so $v(y)$ is $O(e^{y^2/2})$, so the normalisation condition (40) is impossible to satisfy.

If $\alpha = n$ is even, then w_0 is a polynomial of degree n but w_1 is again $O(e^{y^2})$.

If $\alpha = n$ is odd, then w_1 is a polynomial of degree n but w_0 is again $O(e^{y^2})$.

Hermite polynomials

The Hermite polynomial H_n of degree n is the polynomial solution of

$$w'' - 2yw' + 2nw = 0, \tag{42}$$

normalised by the condition that the coefficient of y^n equals 2^n. We conclude that the eigenpairs (ϕ_n, E_n) of (41) are given by

$$\phi_n(x) = C_n e^{-y^2/2} H_n(y) \quad \text{and} \quad E_n = (n + \tfrac{1}{2})\hbar\omega$$

for $n \in \{0, 1, 2, \ldots\}$ and suitable C_n.

Writing the Hermite ODE (42) in self-adjoint form,

$$(e^{-y^2} w')' + 2n e^{-y^2} w = 0,$$

we see that

$$\int_{-\infty}^{\infty} H_n(y) H_k(y) e^{-y^2}\, dy = 0, \qquad n \neq k.$$

Normalisation

It can be shown that

$$\int_{-\infty}^{\infty} H_n(y)^2 e^{-y^2}\,dy = 2^n n!\sqrt{\pi},$$

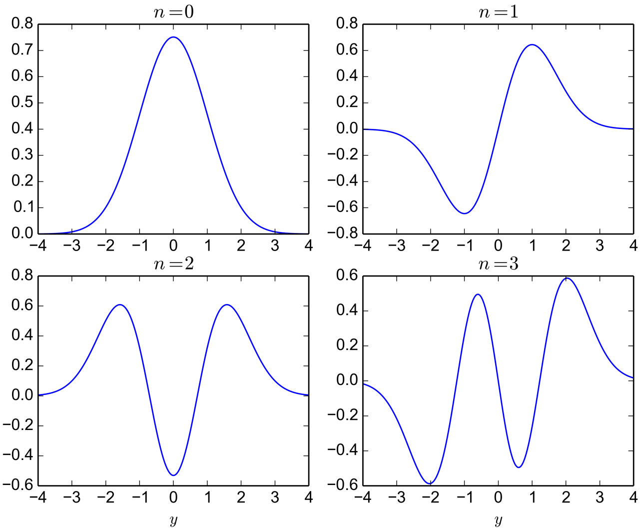
so

$$\begin{aligned}
\int_{-\infty}^{\infty} \phi_n(x)^2\,dx &= \int_{-\infty}^{\infty} \left[C_n e^{-y^2/2} H_n(y)\right]^2 \left(\frac{\hbar}{m\omega}\right)^{1/2} dy \\
&= C_n^2 \left(\frac{\hbar}{m\omega}\right)^{1/2} \int_{-\infty}^{\infty} H_n(y)^2 e^{-y^2}\,dy \\
&= C_n^2 \left(\frac{\hbar}{m\omega}\right)^{1/2} 2^n n!\sqrt{\pi}
\end{aligned}$$

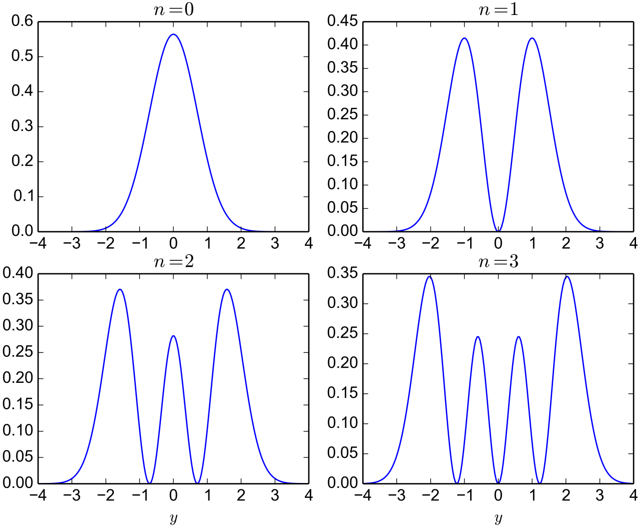
and we therefore set

$$C_n = \left(\frac{m\omega}{\hbar}\right)^{1/4} \left(\frac{2^{-n}}{n!\sqrt{\pi}}\right)^{1/2}.$$

Standing waves ϕ_n



Probability densities $|\psi_n|^2$



Part V

Initial-Boundary Value Problems in 2D

Introduction

In this final part of the course, we use separation of variables to solve partial differential equations involving two space variables. First we see how the Fredholm alternative applies to boundary-value problems in 2D (or 3D) for a class of self-adjoint partial differential equations. Here, the second Green identity plays the role of the Lagrange identity in 1D. Next we generalise the Sturm–Liouville theory to obtain complete orthogonal systems of eigenfunctions in 2D, allowing us to construct solutions to initial-boundary value problems in 2D as generalised Fourier series.

The significance of Bessel's equation will finally be revealed.

Elliptic differential operators

We begin by defining a class of second-order, linear partial differential operators that are well-behaved in certain key respects. These operators appear as the spatial part of 3 key partial differential equations, each of which has many important applications.

Vector calculus notation

Partial derivative operator $\partial_j = \partial/\partial x_j$.

For a scalar field $u : \mathbb{R}^d \to \mathbb{R}$, the gradient is the vector field $\operatorname{grad} u : \mathbb{R}^d \to \mathbb{R}^d$ defined by

$$\operatorname{grad} u = \nabla u = \sum_{j=1}^{d} \partial_j u \, \boldsymbol{e}_j = \begin{bmatrix} \partial_1 u \\ \partial_2 u \\ \vdots \\ \partial_d u \end{bmatrix}$$

For a vector field $\boldsymbol{F} : \mathbb{R}^d \to \mathbb{R}^d$, the divergence is the scalar field $\operatorname{div} \boldsymbol{F} : \mathbb{R}^d \to \mathbb{R}$ defined by

$$\operatorname{div} \boldsymbol{F} = \nabla \cdot \boldsymbol{F} = \sum_{j=1}^{d} \partial_j F_j = \partial_1 F_1 + \partial_2 F_2 + \cdots + \partial_d F_d.$$

Second-order linear PDEs in $\mathbb{R}^d$

The most general second-order linear partial differential operator in $\mathbb{R}^d$ has the form

$$Lu = -\sum_{j=1}^{d}\sum_{k=1}^{d} a_{jk}(\boldsymbol{x})\partial_j\partial_k u + \sum_{k=1}^{d} b_k(\boldsymbol{x})\partial_k u + c(\boldsymbol{x})u. \tag{43}$$

Example

The Laplacian is defined by $\nabla^2 u = \nabla\cdot(\nabla u) = \mathrm{div}(\mathrm{grad}\, u)$, that is,

$$\nabla^2 u = \sum_{j=1}^{d} \partial_j^2 u = \partial_1^2 u + \partial_2^2 u + \cdots + \partial_d^2 u.$$

Thus, $-\nabla^2 u$ has the form (43) with

$$a_{jk}(\boldsymbol{x}) = \delta_{jk}, \qquad b_k(\boldsymbol{x}) = 0, \qquad c(\boldsymbol{x}) = 0.$$

Principal part

We call

$$L_0 u = -\sum_{j=1}^{d}\sum_{k=1}^{d} a_{jk}(\boldsymbol{x})\partial_j\partial_k u \tag{44}$$

the principal part of the partial differential operator (43).

In many applications, L arises in the divergence form

$$Lu = -\operatorname{div}(A\,\operatorname{grad} u) = -\sum_{j=1}^{d}\partial_j\sum_{k=1}^{d} a_{jk}(\boldsymbol{x})\partial_k u,$$

where $A = [a_{ij}]$. We can still write such an L in the general form (43) with the same principal part (44) and with lower-order coefficients

$$b_k(\boldsymbol{x}) = -\sum_{j=1}^{d}\partial_j a_{jk}(\boldsymbol{x}) \quad \text{and} \quad c(\boldsymbol{x}) = 0.$$

Ellipticity

Definition

A second-order linear partial differential operator (43) is (uniformly) elliptic in a subset $\Omega \subseteq \mathbb{R}^d$ if there exists a positive constant c such that

$$\boldsymbol{\xi}^T A(\boldsymbol{x})\boldsymbol{\xi} \geq c\|\boldsymbol{\xi}\|^2 \quad \text{for all } \boldsymbol{x} \in \Omega \text{ and } \boldsymbol{\xi} \in \mathbb{R}^d.$$

Written out explicitly, the ellipticity condition looks like

$$\sum_{j=1}^{d}\sum_{k=1}^{d} a_{jk}(x)\xi_j\xi_k \geq c\sum_{k=1}^{d}\xi_k^2.$$

Example

The operator $L = -\nabla^2$ is elliptic (with $c = 1$) on any $\Omega \subseteq \mathbb{R}^d$, since

$$\sum_{j=1}^{d} \sum_{k=1}^{d} \delta_{jk} \xi_j \xi_k = \sum_{k=1}^{d} \xi_k^2 = \|\boldsymbol{\xi}\|^2.$$

Example

The operator $L = -(\partial_1^2 + 2\partial_2^2 - \partial_3^2)$ is not elliptic in $\mathbb{R}^3$ since in this case the quadratic form

$$\boldsymbol{\xi}^T A \boldsymbol{\xi} = \begin{bmatrix} \xi_1 & \xi_2 & \xi_3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{bmatrix} = \xi_1^2 + 2\xi_2^2 - \xi_3^2$$

is negative if $\xi_1 = \xi_2 = 0$ and $\xi_3 \neq 0$.

Symmetry and skew-symmetry

Put

$$a^{\mathrm{sy}}_{jk} = \frac{1}{2}\big(a_{jk} + a_{kj}\big) = \text{symmetric part of } a_{jk},$$
$$a^{\mathrm{sk}}_{jk} = \frac{1}{2}\big(a_{jk} - a_{kj}\big) = \text{skew-symmetric part of } a_{jk},$$

so that

$$a_{jk} = a^{\mathrm{sy}}_{jk} + a^{\mathrm{sk}}_{jk}, \qquad a^{\mathrm{sy}}_{kj} = a^{\mathrm{sy}}_{jk}, \qquad a^{\mathrm{sk}}_{kj} = -a^{\mathrm{sk}}_{jk}$$

When investigating if L is elliptic, it suffices to look at a^{sy}_{jk}.

Lemma

$$\sum_{j=1}^{d}\sum_{k=1}^{d} a_{jk}(\boldsymbol{x})\xi_j\xi_k = \sum_{j=1}^{d}\sum_{k=1}^{d} a^{\mathrm{sy}}_{jk}(\boldsymbol{x})\xi_j\xi_k.$$

Proof

$$
\begin{aligned}
\sum_{j=1}^{d}\sum_{k=1}^{d} a^{\mathrm{sk}}_{jk}(\boldsymbol{x})\xi_j\xi_k &= \sum_{j=1}^{d}\sum_{k=1}^{d}\big(-a^{\mathrm{sk}}_{kj}(\boldsymbol{x})\big)\xi_j\xi_k \\
&= -\sum_{k=1}^{d}\sum_{j=1}^{d} a^{\mathrm{sk}}_{jk}(\boldsymbol{x})\xi_k\xi_j \\
&= -\sum_{j=1}^{d}\sum_{k=1}^{d} a^{\mathrm{sk}}_{jk}(\boldsymbol{x})\xi_k\xi_j \\
&= -\sum_{j=1}^{d}\sum_{k=1}^{d} a^{\mathrm{sk}}_{jk}(\boldsymbol{x})\xi_j\xi_k
\end{aligned}
$$

so

$$
2\sum_{j=1}^{d}\sum_{k=1}^{d} a^{\mathrm{sk}}_{jk}(\boldsymbol{x})\xi_j\xi_k = 0. \qquad \square
$$

Theorem

Denote the eigenvalues of the real symmetric matrix $[a^{\mathrm{sy}}_{jk}(\boldsymbol{x})]$ *by* $\lambda_j(x)$ *for* $1 \le j \le d$*. The operator* (43) *is elliptic on* Ω *if and only if there exists a positive constant* c *such that*

$$\lambda_j(\boldsymbol{x}) \ge c \quad \text{for } 1 \le j \le d \text{ and all } \boldsymbol{x} \in \Omega.$$

Example

Show that the $L = -(3\partial_1^2 + 2\partial_1\partial_2 + 2\partial_2^2)$ is elliptic.

Example

Show that $L = -(\partial_1^2 - 4\partial_1\partial_2 + \partial_2^2)$ is not elliptic.

Proof

Fix x, then the real symmetric matrix $A = [a_{jk}^{\mathrm{sy}}]$ is diagonalisable, that is, $\mathbb{R}^d$ has an orthonormal basis of eigenvectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_d$ and so

$$Q^T A Q = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_d)$$

where the matrix $Q = [\boldsymbol{v}_1 \quad \boldsymbol{v}_2 \quad \cdots \quad \boldsymbol{v}_d]$ is orthogonal: $Q^T = Q^{-1}$. Put $\boldsymbol{\eta} = Q^T \boldsymbol{\xi}$, then $\boldsymbol{\xi} = Q\boldsymbol{\eta}$ so

$$\boldsymbol{\xi}^T A \boldsymbol{\xi} = (Q\boldsymbol{\eta})^T A (Q\boldsymbol{\eta}) = \boldsymbol{\eta}^T (Q^T A Q) \boldsymbol{\eta} = \sum_{j=1}^{d} \lambda_j \eta_j^2,$$

and $\|\boldsymbol{\xi}\|^2 = (Q\boldsymbol{\eta})^T (Q\boldsymbol{\eta}) = \boldsymbol{\eta}^T (Q^T Q) \boldsymbol{\eta} = \boldsymbol{\eta}^T (I) \boldsymbol{\eta} = \|\boldsymbol{\eta}\|^2$, so

$$\boldsymbol{\xi}^T A \boldsymbol{\xi} \geq c \|\boldsymbol{\xi}\|^2 \iff \sum_{j=1}^{d} \lambda_j \eta_j^2 \geq c\|\eta\|^2. \quad \square$$

Three important PDEs

In the remainder of this course, we mainly focus on three PDEs.

1. Poisson equation (elliptic):

$$-\nabla^2 u = f.$$

2. Diffusion equation or heat equation (parabolic):

$$\frac{\partial u}{\partial t} - \nabla^2 u = f.$$

3. Wave equation (hyperbolic):

$$\frac{\partial^2 u}{\partial t^2} - \nabla^2 u = f.$$

Here, $u = u(x, t)$ for $t > 0$ and $x \in \Omega$ where Ω is a bounded, open subset of $\mathbb{R}^d$ with a piecewise smooth boundary.

Green identities and boundary-value problems

We saw how the Lagrange identity leads to the concept of self-adjointness, and allowed us to understand the solvability of boundary-value problems in 1D. In higher dimensions, the same role is played by the second Green identity, which we use to prove (the easy half of) the Fredholm alternative.

Inner products

We denote the (real) inner product in $L_2(\Omega)$ by

$$\langle f, g\rangle = \langle f, g\rangle_\Omega = \int_\Omega f\,g\,dV = \begin{cases} \iint_\Omega f\,g\,dx\,dy, & d = 2, \\ \iiint_\Omega f\,g\,dx\,dy\,dz, & d = 3. \end{cases}$$

Sometimes we will require a weighted inner product

$$\langle f, g\rangle_r = \langle f, g\rangle_{r,\Omega} = \int_\Omega f\,g r\,dV.$$

We will also require an inner product on a part of the boundary, $\Gamma \subseteq \partial\Omega$,

$$\langle f, g\rangle_\Gamma = \int_\Gamma f\,g\,dS,$$

or a weighted version

$$\langle f, g\rangle_{r,\Gamma} = \int_\Gamma f\,g r\,dS.$$

A class of second-order elliptic operators

To generalise the Sturm–Liouville theory, we consider a second-order, linear partial differential operator of the form

$$Lu = -\nabla \cdot (p\nabla u) + qu, \tag{45}$$

with real-valued coefficients $p(x)$ and $q(x)$.

Example

If $d = 1$ then $\nabla = d/dx$ so $Lu = -(pu')' + qu$.

Example

If $d = 2$ then

$$\begin{aligned} Lu &= -(pu_x)_x - (pu_y)_y + qu \\ &= -\frac{\partial}{\partial x}\left(p(x,y)\,\frac{\partial u}{\partial x}\right) - \frac{\partial}{\partial y}\left(p(x,y)\,\frac{\partial u}{\partial y}\right) + qu. \end{aligned}$$

Ellipticity

The principal part of L is in divergence form $-\nabla \cdot (A\nabla u)$ with $A = [p(\boldsymbol{x})\delta_{jk}] = p(\boldsymbol{x})I$ so

$$\boldsymbol{\xi}^T A\boldsymbol{\xi} = p(\boldsymbol{x})\|\xi\|^2$$

and therefore L is elliptic on Ω iff

$$p(\boldsymbol{x}) \geq c > 0 \quad \text{for all } \boldsymbol{x} \in \Omega.$$

Example

If $p(x) = 1$ and $q(x) = 0$ then $L = -\nabla^2$.

Divergence theorem (Math2111)

Notation: $\overline{\Omega} = \Omega \cup \partial\Omega$ is the closure of Ω. Since Ω is bounded, the set $\overline{\Omega}$ is compact and hence, by the Extreme Value Theorem, any continuous function $f : \overline{\Omega} \to \mathbb{R}$ must be bounded.

Theorem

If a vector field $\boldsymbol{F} : \overline{\Omega} \to \mathbb{R}^d$ is continuously differentiable then

$$\int_\Omega \nabla \cdot \boldsymbol{F}\, dV = \oint_{\partial\Omega} \boldsymbol{F} \cdot \boldsymbol{n}\, dS,$$

where $\boldsymbol{n}$ is the outward unit normal to Ω.

Notation: $\partial u/\partial n = \boldsymbol{n} \cdot \nabla u$.

First Green identity

Theorem

If u is C^2 and v is C^1 on $\overline{\Omega}$, then

$$\langle Lu, v\rangle_\Omega = \int_\Omega \bigl(p\nabla u \cdot \nabla v + quv\bigr)\, dV - \oint_{\partial\Omega} p\,\frac{\partial u}{\partial n}\, v\, dS. \tag{46}$$

Proof.

Since

$$\nabla \cdot [(p\nabla u)v] = [\nabla \cdot (p\nabla u)]v + p\nabla u \cdot \nabla v$$

we have

$$(Lu)v = p\nabla u \cdot \nabla v + quv - \nabla \cdot [(p\nabla u)v]$$

and the result follows after applying the divergence theorem to $\boldsymbol{F} = (p\nabla u)v$, because $\boldsymbol{F} \cdot \boldsymbol{n} = p(\partial u/\partial n)v$. $\square$

Second Green identity

Theorem

If u and v are C^2 on $\overline{\Omega}$, then

$$\langle Lu, v\rangle_\Omega - \langle u, Lv\rangle_\Omega = \oint_{\partial\Omega} p\left(u\,\frac{\partial v}{\partial n} - v\,\frac{\partial u}{\partial n}\right) dS. \tag{47}$$

Proof.

Since

$$(Lu)v = p\nabla u \cdot \nabla v + quv - \nabla \cdot [(p\nabla u)v]$$

and

$$u\,Lv = p\nabla v \cdot \nabla u + qvu - \nabla \cdot [(p\nabla v)u],$$

the Lagrange identity generalizes to

$$(Lu)v - u\,Lv = \nabla \cdot \big[p(u\nabla v - v\nabla u)\big],$$

and the result follows after applying the divergence theorem. □

Boundary conditions and self-adjointness

Suppose that $\partial\Omega$ is divided into two parts, Γ_{D} and Γ_{N}, where the D and N stand for Dirichlet and Neumann boundary conditions.

The second Green identity (47) shows that

$$\langle Lu, v\rangle_\Omega = \langle u, Lv\rangle_\Omega \quad \text{for all } u,\ v \in \mathcal{D}, \tag{48}$$

if we choose $\mathcal{D}$, the domain of the partial differential operator L, to be the space of C^2 functions on $\overline{\Omega}$ satisfying the homogeneous boundary conditions

$$\begin{aligned} u &= 0 \quad \text{on } \Gamma_{\mathrm{D}}, \\ \frac{\partial u}{\partial n} &= 0 \quad \text{on } \Gamma_{\mathrm{N}}. \end{aligned}$$

Elliptic boundary value problem

Recall $L = -\nabla \cdot (p\nabla u) + qu$ and assume

$$p(x) \geq c > 0 \quad \text{for all } x \in \Omega,$$

so that L is elliptic.

Given a source term f, Dirichlet data g_{D} and Neumann data g_{N}, we seek a solution u to the BVP

$$\begin{aligned} Lu &= f && \text{in } \Omega, \\ u &= g_{\mathrm{D}} && \text{on } \Gamma_{\mathrm{D}}, \\ p\,\frac{\partial u}{\partial n} &= g_{\mathrm{N}} && \text{on } \Gamma_{\mathrm{N}}. \end{aligned} \tag{49}$$

Technical requirement: for u to count as a solution it must have finite energy:

$$\int_\Omega \left(|\nabla u|^2 + |u|^2\right) dV < \infty.$$

Fredholm alternative

Either the homogeneous problem

$$Lv = 0 \text{ in } \Omega, \qquad v = 0 \text{ on } \Gamma_{\mathrm{D}}, \qquad p\,\frac{\partial v}{\partial n} = 0 \text{ on } \Gamma_{\mathrm{N}}, \tag{50}$$

has only the trivial solution $v \equiv 0$, in which case

> *the inhomogeneous problem* (49) *has a unique solution* u *for "every" choice of* f, g_{D} *and* g_{N}*;*

or else the homogeneous problem admits finitely many, linearly independent solutions $v_1, v_2, \ldots, v_m$, in which case

> *the inhomogeneous problem* (49) *has a solution* u *iff the data* f, g_{D} *and* g_{N} *satisfy*
>
> $$\int_\Omega f v_k\, dV = \int_{\Gamma_{\mathrm{D}}} g_{\mathrm{D}}\, p\,\frac{\partial v_k}{\partial n}\, dS - \int_{\Gamma_{\mathrm{N}}} g_{\mathrm{N}}\, v_k\, dS$$
>
> *for* $k = 1$, *2*, $\ldots$, m.

Partial proof

Write the second Green identity (47) as

$$\langle Lu, v\rangle_\Omega - \langle u, Lv\rangle_\Omega = \oint_{\partial\Omega}\left(u\,p\,\frac{\partial v}{\partial n} - p\,\frac{\partial u}{\partial n}\,v\right) dS.$$

If u satisfies (49) and v satisfies (50) then, since $\partial\Omega = \Gamma_{\mathrm{D}} \cup \Gamma_{\mathrm{N}}$,

$$\langle f, v\rangle_\Omega - \langle u, 0\rangle_\Omega = \int_{\Gamma_{\mathrm{D}}}\left(g_{\mathrm{D}}\,p\,\frac{\partial v}{\partial n} - p\,\frac{\partial u}{\partial n} \times 0\right) dS + \int_{\Gamma_{\mathrm{N}}}\left(u \times 0 - g_{\mathrm{N}}\,v\right) dS,$$

and so

$$\int_\Omega f\,v\,dV = \int_{\Gamma_{\mathrm{D}}} g_{\mathrm{D}}\,p\,\frac{\partial v}{\partial n}\,dS - \int_{\Gamma_{\mathrm{N}}} g_{\mathrm{N}} v\,dS. \quad \square$$

A pure Neumann problem

Consider the BVP

$$-\nabla^2 u = f \text{ in } \Omega, \qquad \frac{\partial u}{\partial n} = g \text{ on } \partial\Omega. \tag{51}$$

If v is a solution of the homogeneous problem,

$$-\nabla^2 v = 0 \text{ in } \Omega, \qquad \frac{\partial v}{\partial n} = 0 \text{ on } \partial\Omega,$$

then by the first Green identity,

$$\int_\Omega \underbrace{(-\nabla^2 v)}_{0} v \, dV = \int_\Omega \|\nabla v\|^2 \, dV - \oint_{\partial\Omega} \underbrace{\frac{\partial v}{\partial n}}_{0} v \, dS.$$

so $\int_\Omega \|\nabla v\|^2 \, dV = 0$. If Ω is connected then v must be constant, so the inhomogeneous problem (51) is solvable iff the data satisfy

$$\int_\Omega f \, dV + \oint_{\partial\Omega} g \, dS = 0.$$

Elliptic eigenproblems

We now study second-order, self-adjoint elliptic eigenproblems. After looking at a few simple, concrete examples that can be solved by separation of variables, we show some key properties of the eigenfunctions that generalise earlier results for Sturm–Liouville problems in 1D. We then see how eigenfunctions can be used to construct a series solution to a boundary-value problem.

By analogy with the Sturm–Liouville problem in 1D, we now consider the PDE

$$\nabla \cdot (p\nabla u) + (\lambda r - q)u = 0 \quad \text{in } \Omega. \tag{52}$$

Here, the coefficients p, q, r are given real-valued, continuous functions on $\overline{\Omega}$, with

$$p(x) \geq \mathrm{const} > 0 \quad \text{and} \quad r(x) \geq \mathrm{const} > 0 \quad \text{for } x \in \overline{\Omega},$$

with $r(x)$ not identically zero.

Writing

$$Lu = -\nabla \cdot (p\nabla u) + qu$$

as before, the PDE (52) can be written in the form

$$Lu = \lambda ru \quad \text{in } \Omega.$$

Boundary conditions

Again suppose that $\partial\Omega$ is divided into two parts, Γ_{D} and Γ_{N}, where the D and N stand for Dirichlet and Neumann boundary conditions.

If $u : \Omega \to \mathbb{C}$ is not identically zero and if

$$\begin{aligned} Lu &= \lambda r u && \text{in } \Omega, \\ u &= 0 && \text{on } \Gamma_{\mathrm{D}}, \\ p\,\frac{\partial u}{\partial n} &= 0 && \text{on } \Gamma_{\mathrm{N}}, \end{aligned} \tag{53}$$

then u is said to be an eigenfunction with eigenvalue λ, and we call (u, λ) an eigenpair.

The results shown earlier for Sturm–Liouville problems and Fourier series in 1D generalise in a natural way to this 2D or 3D setting.

Example: eigenproblem on the unit square

The eigenvalues λ_{kn} and eigenfunctions $u = \phi_{kn}(x, y)$ for

$$\nabla^2 u + \lambda u = 0, \quad 0 < x < 1, \quad 0 < y < 1,$$

with boundary conditions

$$u(x, 0) = u(x, 1) = 0, \quad 0 < x < 1,$$

and

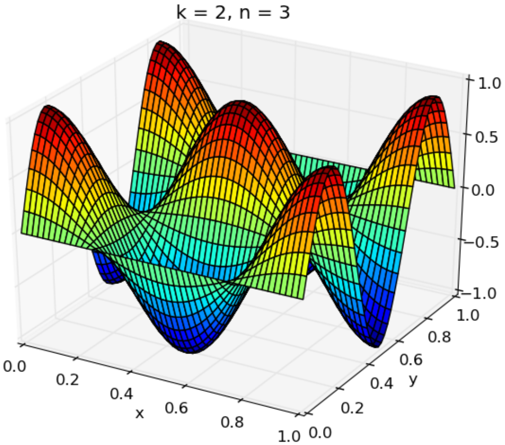
$$\frac{\partial u}{\partial x}(0, y) = \frac{\partial u}{\partial x}(1, y) = 0, \quad 0 < y < 1,$$

are given by

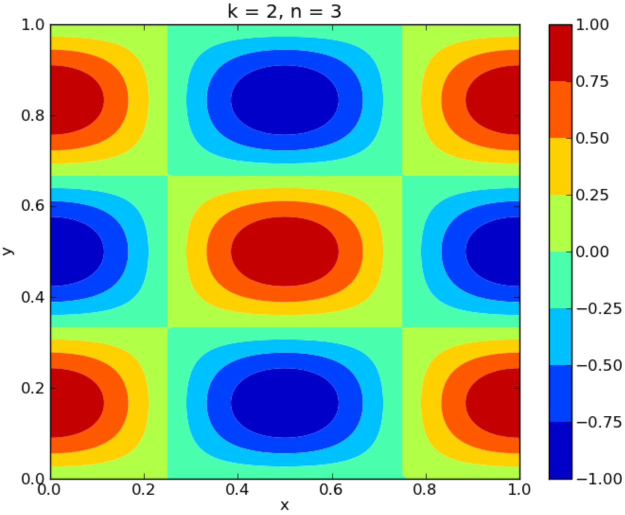
$$\lambda_{kn} = (k^2 + n^2)\pi^2 \qquad \text{and} \qquad \phi_{kn}(x, y) = \cos(k\pi x)\sin(n\pi y)$$

for $k = 0$, 1, 2, ... and $n = 1$, 2, 3, ...

The eigenfunction $\phi_{kn}(x, y) = \cos(k\pi x)\sin(n\pi y)$



Contours of ϕ_{kn}



Example: eigenproblem on the unit disk

Introduce polar coordinates in $\mathbb{R}^2$,

$$x = r\cos\theta \quad \text{and} \quad y = r\sin\theta,$$

and recall that

$$\nabla^2 u = \frac{1}{r}\frac{\partial}{\partial r}\left(r\,\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\,\frac{\partial^2 u}{\partial\theta^2}.$$

Consider

$$\nabla^2 u + \lambda u = 0, \quad 0 \le r < 1, \quad -\pi < \theta \le \pi,$$

with the boundary condition

$$u(1,\theta) = 0, \quad -\pi \le \theta < \pi.$$

The eigenvalues $\lambda = \lambda_{nj}$ are

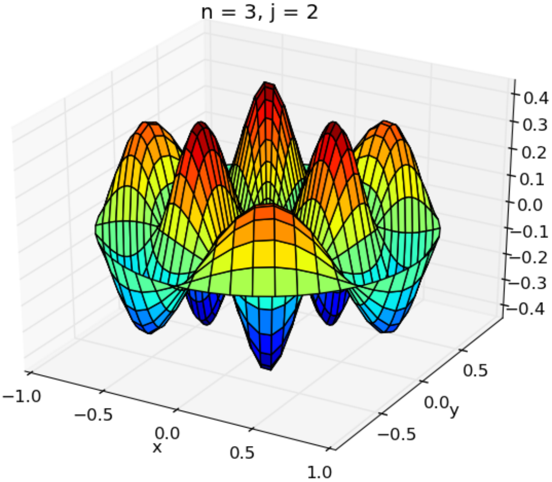
$$\lambda_{nj} = k_{nj}^2 \quad \text{where} \quad J_n(k_{nj}) = 0,$$

for $n = 0$, 1, 2, ... and $j = 1$, 2, 3, ...

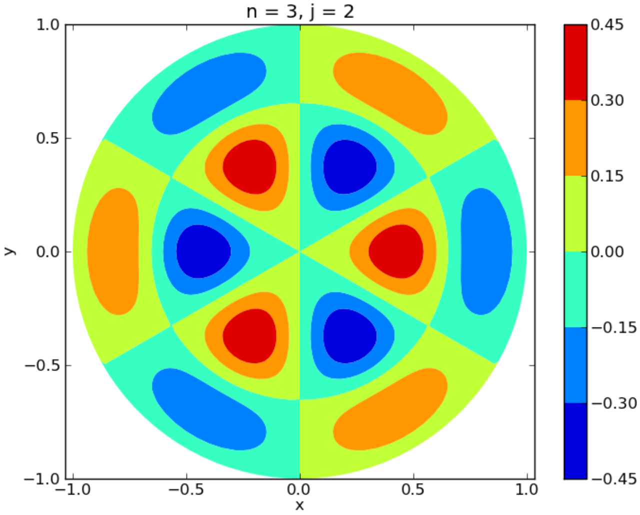
The corresponding eigenfunctions $u = \phi_{nj}$ are

$$\begin{aligned}
\phi_{0j}(r, \theta) &= J_0(k_{0j} r), \\
\phi_{nj}^{\mathsf{C}}(r, \theta) &= J_n(k_{nj} r) \cos(n\theta), \quad n \geq 1, \\
\phi_{nj}^{\mathsf{S}}(r, \theta) &= J_n(k_{nj} r) \sin(n\theta), \quad n \geq 1.
\end{aligned}$$

The eigenfunction $\phi^{\mathrm{C}}_{nj} = J_n(k_{nj} r)\cos(n\theta)$



Contours of ϕ_{nj}



Eigenfunctions are orthogonal

Theorem

Let L be the partial differential operator (45)*. If u, v are (possibly complex-valued) functions satisfying*

$$\begin{aligned} Lu &= \lambda r u, & Lv &= \mu r v, & &\text{in } \Omega, \\ u &= 0, & v &= 0, & &\text{on } \Gamma_{\mathrm{D}}, \\ p\,\frac{\partial u}{\partial n} &= 0, & p\,\frac{\partial v}{\partial n} &= 0, & &\text{on } \Gamma_{\mathrm{N}}, \end{aligned}$$

and if $\lambda \neq \mu$, then u and v are orthogonal on Ω with respect to the weight function r, i.e.,

$$\langle u, v \rangle_r = \int_\Omega u v r \, dV = 0.$$

Eigenvalues are real

Theorem

If u is a nontrivial solution of the elliptic eigenproblem (53) *then λ is real. Moreover, both $\mathrm{Re}\, u$ and $\mathrm{Im}\, u$ are also solutions of* (53).

Thus, it is always possible to choose purely real-valued eigenfunctions for L.

The proofs proceed exactly as in the 1D case.

Proofs

The self-adjointness (48) of L implies that

$$\begin{aligned}(\lambda - \mu)\langle u, v\rangle_r &= \langle \lambda r u, v\rangle - \langle u, \mu r v\rangle \\ &= \langle Lu, v\rangle - \langle u, Lv\rangle = 0,\end{aligned}$$

so if $\lambda \neq \mu$ then $\langle u, v\rangle_r = 0$.
The coefficents p, q, r are real-valued so we easily see that $(\bar{u}, \bar{\lambda})$ is an eigenpair and therefore

$$(\lambda - \bar{\lambda})\langle u, \bar{u}\rangle_r = 0.$$

Since $\langle u, \bar{u}\rangle_r = \int_\Omega |u|^2 r \, dV > 0$, we conclude that $\lambda = \bar{\lambda}$, or in other words λ is real.
Finally, because u and $\bar{u}$ are eigenfunctions with eigenvalue λ, so are

$$\operatorname{Re} u = \frac{u + \bar{u}}{2} \quad \text{and} \quad \operatorname{Im} u = \frac{u - \bar{u}}{2i}. \quad \square$$

[Since the Green's Identity holds the above proofs are valid. However, note that the real inner product is used and this does not form an inner product space for complex functions.]

Completeness of the eigenfunctions

If several eigenfunctions share the same eigenvalue then they are not necessarily orthogonal. In this case, however, we can always orthogonalise them using the Gram–Schmidt procedure (Math2601).

In general, the elliptic eigenproblem (53) has a sequence of eigenvalues $\lambda = \lambda_j$ and corresponding eigenfunctions $u = \phi_j$ such that

1. the eigenvalues satisfy $\lambda_1 \le \lambda_2 \le \lambda_3 \le \cdots$ with $\lambda_j \to \infty$ as $j \to \infty$;
2. the eigenfunctions ϕ_1, ϕ_2, ϕ_3, ... form a complete orthogonal system on Ω with respect to the weight function r.

Steklov eigenproblem

As before, let

$$Lu = -\nabla \cdot (p\nabla u) + qu,$$

but suppose now that $\partial\Omega$ is divided into three parts: Γ, Γ_{D} and Γ_{N}. If $u : \Omega \to \mathbb{C}$ is not identically zero and if

$$\begin{aligned} Lu &= 0 && \text{in } \Omega, \\ p\,\frac{\partial u}{\partial n} &= \lambda r u && \text{on } \Gamma, \\ u &= 0 && \text{on } \Gamma_{\mathrm{D}}, \\ p\,\frac{\partial u}{\partial n} &= 0 && \text{on } \Gamma_{\mathrm{N}}, \end{aligned} \tag{54}$$

then we say that (u, λ) is a Steklov eigenpair.

Example: a Steklov eigenproblem for the unit square

The eigenvalues λ_n and eigenfunctions $u = \phi_n(x, y)$ for

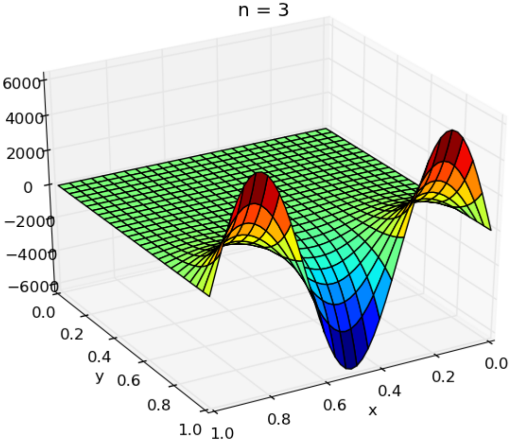
$$\begin{aligned}
\nabla^2 u = 0, \qquad & 0 < x < 1, \quad 0 < y < 1, \\
\frac{\partial u}{\partial y}(x, 1) = \lambda u(x, 1), \qquad & 0 < x < 1, \\
u(0, y) = u(1, y) = 0, \qquad & 0 < y < 1, \\
\frac{\partial u}{\partial y}(x, 0) = 0, \qquad & 0 < x < 1,
\end{aligned}$$

are

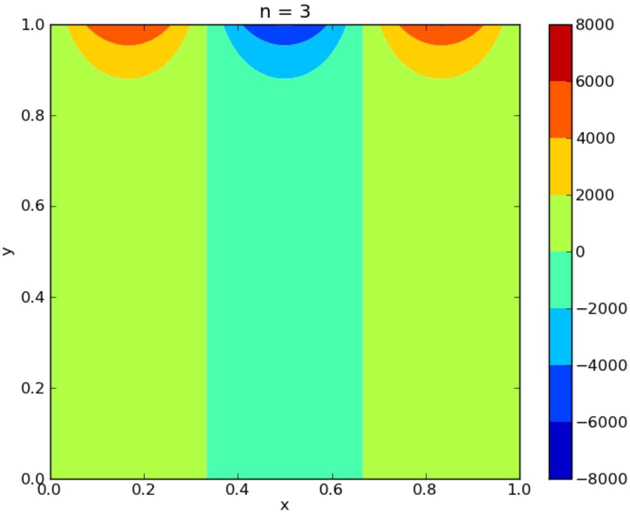
$$\lambda_n = n\pi \tanh(n\pi) \quad \text{and} \quad \phi_n(x, y) = \sin n\pi x \cosh n\pi y$$

for $n = 1$, 2, $\ldots$.

The eigenfunction $\phi_n(x, y) = \sin(n\pi x)\cosh(n\pi y)$



Contours of ϕ_n



Steklov eigenvalues are real

The first Green identity with $v = \bar{u}$ gives

$$\langle Lu, \bar{u} \rangle_\Omega = \int_\Omega \left(p|\nabla u|^2 + q|u|^2\right) dV - \oint_{\partial\Omega} p \, \frac{\partial u}{\partial n} \, \bar{u} \, dS.$$

If u satisfies (54), then $Lu = 0$ in Ω so

$$\int_\Omega \left(p|\nabla u|^2 + q|u|^2\right) dV = \oint_{\partial\Omega} p \, \frac{\partial u}{\partial n} \, \bar{u} \, dS,$$

and since $\partial\Omega = \Gamma \cup \Gamma_{\mathrm{D}} \cup \Gamma_{\mathrm{N}}$ the boundary conditions imply that

$$\oint_{\partial\Omega} p \, \frac{\partial u}{\partial n} \, \bar{u} \, dS = \int_\Gamma (\lambda r u)\bar{u} \, dS + \int_{\Gamma_{\mathrm{D}}} \left(p \, \frac{\partial u}{\partial n}\right) (\bar{0}) \, dS + \int_{\Gamma_{\mathrm{N}}} (0)\bar{u} \, dS.$$

We conclude that λ is real, because

$$\int_\Omega \left(p|\nabla u|^2 + q|u|^2\right) dV = \lambda \int_\Gamma r|u|^2 \, dS.$$

Steklov eigenfunctions are orthogonal over Γ

If u and v are eigenfunctions with eigenvalues λ and μ, then by the second Green identity,

$$
\begin{aligned}
(\lambda-\mu)\int_\Gamma uvr\,dS &= \int_\Gamma (\lambda ru)v\,dS - \int_\Gamma u(\mu rv)\,dS \\
&= \int_\Gamma p\left(\frac{\partial u}{\partial n}\,v - u\,\frac{\partial v}{\partial n}\right)dS \\
&= \int_{\partial\Omega} p\left(\frac{\partial u}{\partial n}\,v - u\,\frac{\partial v}{\partial n}\right)dS \\
&= \langle u, Lv\rangle_\Omega - \langle Lu, v\rangle_\Omega \\
&= \langle u, 0\rangle_\Omega - \langle 0, v\rangle_\Omega = 0,
\end{aligned}
$$

so

$$\langle u, v\rangle_{r,\Gamma} = \int_\Gamma uvr\,dS = 0 \quad \text{if } \lambda \neq \mu.$$

Completeness?

The second type of eigenproblem (54) has a sequence of eigenvalues $\lambda = \lambda_j$ and corresponding eigenfunctions $u = \phi_j$ such that

1. the eigenfunctions ϕ_1, ϕ_2, ϕ_3, $\ldots$ are orthogonal with respect to r on Γ;
2. the eigenvalues satisfy $\lambda_1 \le \lambda_2 \le \lambda_3 \le \cdots$ with $\lambda_j \to \infty$ as $j \to \infty$.

The eigenfunctions are complete in $L_2(\Gamma, r)$ iff the homogeneous BVP

$$\begin{aligned} Lv &= 0 && \text{in } \Omega, \\ v &= 0 && \text{on } \Gamma \cup \Gamma_{\mathrm{D}}, \\ p\,\frac{\partial v}{\partial n} &= 0 && \text{on } \Gamma_{\mathrm{N}}, \end{aligned} \tag{55}$$

has only the trivial solution $v \equiv 0$ in Ω.

Exceptional case

If (55) has a nontrivial solution v, then by the second Green identity

$$\begin{aligned}\int_\Gamma \phi_j p \,\frac{\partial v}{\partial n}\,dS &= \oint_{\partial\Omega} p\left(\phi_j \,\frac{\partial v}{\partial n} - v\,\frac{\partial \phi_j}{\partial n}\right)dS \\ &= \langle L\phi_j, v\rangle_\Omega - \langle \phi_j, Lv\rangle_\Omega,\end{aligned}$$

and so, since $L\phi_j = 0 = Lv$ in Ω,

$$\int_\Gamma \phi_j p \,\frac{\partial v}{\partial n}\,dS = 0.$$

Thus, the function

$$w = \frac{p}{r}\,\frac{\partial v}{\partial n}$$

satisfies

$$\langle \phi_j, w\rangle_{r,\Gamma} = 0 \quad \text{for all } j.$$

Boundary-value problem with $g_{\mathrm{D}} = 0$ and $g_{\mathrm{N}} = 0$

Suppose that we have homogeneous boundary data,

$$Lu = f \text{ in } \Omega, \qquad u = 0 \text{ on } \Gamma_{\mathrm{D}}, \qquad p\,\frac{\partial u}{\partial n} = 0 \text{ on } \Gamma_{\mathrm{N}}.$$

To construct u, find the eigenpairs (ϕ_j, λ_j) satisfying

$$L\phi_j = \lambda_j r \phi_j \text{ in } \Omega, \qquad \phi_j = 0 \text{ on } \Gamma_{\mathrm{D}}, \qquad p\,\frac{\partial \phi_j}{\partial n} = 0 \text{ on } \Gamma_{\mathrm{N}}.$$

Then u exists iff

$$\langle f, \phi_j \rangle_\Omega = 0 \quad \text{whenever} \quad \lambda_j = 0,$$

in which case (with arbitrary constants C_j),

$$u(\boldsymbol{x}) \sim \sum_{\lambda_j = 0} C_j \phi_j + \sum_{\lambda_j \neq 0} \frac{\langle f, \phi_j \rangle_\Omega}{\lambda_j \|\phi_j\|_{r,\Omega}^2}\, \phi_j(\boldsymbol{x}), \quad x \in \Omega.$$

Boundary-value problem with $f = 0$ and $g_{\mathrm{N}} = 0$

Consider

$$Lu = 0 \text{ in } \Omega, \qquad u = g_{\mathrm{D}} \text{ on } \Gamma_{\mathrm{D}}, \qquad p\,\frac{\partial u}{\partial n} = 0 \text{ on } \Gamma_{\mathrm{N}}.$$

Find the Steklov eigenpairs (ϕ_j, λ_j) satisfying

$$L\phi_j = 0 \text{ in } \Omega, \qquad p\,\frac{\partial \phi_j}{\partial n} = \lambda_j r \phi_j \text{ on } \Gamma_{\mathrm{D}}, \qquad p\,\frac{\partial \phi_j}{\partial n} = 0 \text{ on } \Gamma_{\mathrm{N}}.$$

If the homogeneous problem

$$Lv = 0 \text{ in } \Omega, \qquad v = 0 \text{ on } \Gamma_{\mathrm{D}}, \qquad p\,\frac{\partial v}{\partial n} = 0 \text{ on } \Gamma_{\mathrm{N}},$$

has only the trivial solution $v \equiv 0$, then

$$u(\boldsymbol{x}) \sim \sum_{j=1}^{\infty} \frac{\langle g_{\mathrm{D}}, \phi_j \rangle_{r,\Gamma_{\mathrm{D}}}}{\|\phi_j\|^2_{r,\Gamma_{\mathrm{D}}}}\,\phi_j(\boldsymbol{x}), \quad \boldsymbol{x} \in \Omega.$$

Boundary-value problem with $f = 0$ and $g_\mathrm{D} = 0$

Consider

$$Lu = 0 \text{ in } \Omega, \qquad u = 0 \text{ on } \Gamma_\mathrm{D}, \qquad p\,\frac{\partial u}{\partial n} = g_\mathrm{N} \text{ on } \Gamma_\mathrm{N}.$$

Find the Steklov eigenpairs (ϕ_j, λ_j) satisfying

$$L\phi_j = 0 \text{ in } \Omega, \qquad \phi_j = 0 \text{ on } \Gamma_\mathrm{D}, \qquad p\,\frac{\partial \phi_j}{\partial n} = \lambda_j r \phi_j \text{ on } \Gamma_\mathrm{N}.$$

Then u exists iff

$$\langle g_\mathrm{N}, \phi_j\rangle_{\Gamma_\mathrm{N}} = 0 \quad \text{whenever} \quad \lambda_j = 0,$$

in which case

$$u(\boldsymbol{x}) \sim \sum_{\lambda_j = 0} C_j \phi_j(\boldsymbol{x}) + \sum_{\lambda_j \neq 0} \frac{\langle g_\mathrm{N}, \phi_j\rangle_{\Gamma_\mathrm{N}}}{\lambda_j \|\phi_j\|^2_{r,\Gamma_\mathrm{N}}}\,\phi_j(\boldsymbol{x}), \quad \boldsymbol{x} \in \Omega.$$

Example

Consider the following boundary-value problem on the unit square:

$$\begin{aligned} \nabla^2 u &= 0, && 0 < x < 1, \quad 0 < y < 1, \\ \frac{\partial u}{\partial y}(x,1) &= 1, && 0 < x < 1, \\ u(0,y) = u(1,y) &= 0, && 0 < y < 1, \\ \frac{\partial u}{\partial y}(x,0) &= 0, && 0 < x < 1. \end{aligned}$$

Use the associated Steklov eigenproblem (solved previously) to show that

$$u(x,y) \sim \frac{4}{\pi^2} \sum_{k=1}^{\infty} \frac{1}{(2k-1)^2} \sin(2k-1)\pi x \, \frac{\cosh(2k-1)\pi y}{\sinh(2k-1)\pi}.$$

Boundary-value problem with general data

To handle the most general situation

$$Lu = f \text{ in } \Omega, \qquad u = g_{\mathrm{D}} \text{ on } \Gamma_{\mathrm{D}}, \qquad p\,\frac{\partial u}{\partial n} = g_{\mathrm{N}} \text{ on } \Gamma_{\mathrm{N}},$$

we write u as a sum of three terms

$$u = u_1 + u_2 + u_3,$$

where each term solves one of the cases treated above:

$$\begin{aligned}
Lu_1 &= f \text{ in } \Omega, & u_1 &= 0 \text{ on } \Gamma_{\mathrm{D}}, & p\,\frac{\partial u_1}{\partial n} &= 0 \text{ on } \Gamma_{\mathrm{N}},\\
Lu_2 &= 0 \text{ in } \Omega, & u_2 &= g_{\mathrm{D}} \text{ on } \Gamma_{\mathrm{D}}, & p\,\frac{\partial u_2}{\partial n} &= 0 \text{ on } \Gamma_{\mathrm{N}},\\
Lu_3 &= 0 \text{ in } \Omega, & u_3 &= 0 \text{ on } \Gamma_{\mathrm{D}}, & p\,\frac{\partial u_3}{\partial n} &= g_{\mathrm{N}} \text{ on } \Gamma_{\mathrm{N}}.
\end{aligned}$$

An alternative approach

If we can find $w : \Omega \cup \partial\Omega \to \mathbb{R}$ satisfying just the boundary conditions in (49),

$$w = g_{\mathrm{D}} \text{ on } \Gamma_{\mathrm{D}} \qquad \text{and} \qquad p\,\frac{\partial w}{\partial n} = g_{\mathrm{N}} \text{ on } \Gamma_{\mathrm{N}},$$

then we can solve (49) by putting $f^\star = f - Lw$, solving

$$\begin{aligned} Lu^\star &= f^\star && \text{in } \Omega, \\ u^\star &= 0 && \text{on } \Gamma_{\mathrm{D}}, \\ p\,\frac{\partial u^\star}{\partial n} &= 0 && \text{on } \Gamma_{\mathrm{N}}, \end{aligned}$$

and then putting $u = u^\star + w$.

This will often be easier than finding three sets of eigensystems.

Example

Consider the PDE on the unit square,

$$-\nabla^2 u = x(1-x) \quad \text{for } 0 < x < 1 \text{ and } 0 < y < 1,$$

subject to the boundary conditions

$$u(x,0) = 1 \quad \text{and} \quad u(x,1) = 2 \quad \text{for } 0 < x < 1,$$

plus

$$\frac{\partial u}{\partial x}(0,y) = 0 = \frac{\partial u}{\partial x}(1,y) \quad \text{for } 0 < y < 1.$$

Use a previously solved eigenproblem to show that

$$u(x,y) \sim y + 1 + \frac{2}{3\pi^3} \sum_{m=1}^{\infty} \frac{\sin(2m-1)\pi y}{(2m-1)^3} - \frac{4}{\pi^5} \sum_{j=1}^{\infty} \sum_{m=1}^{\infty} \frac{\cos 2j\pi x \, \sin(2m-1)\pi y}{j^2(2m-1)\big[4j^2 + (2m-1)^2\big]}.$$

Wave and diffusion equations

We now consider two time-dependent partial differential equations. Both are solved in the same way as their 1D versions. The only change is that we must use 2D eigenfunctions.

Vibrating membrane

Consider a stretched membrane in the shape of a planar region Ω. Assuming that the membrane is fixed along $\partial\Omega$, has initial displacement $u_0(\boldsymbol{x})$ and is initially at rest, we have the following initial boundary value problem for the transverse displacement $u = u(\boldsymbol{x}, t)$:

$$\begin{aligned} \frac{\partial^2 u}{\partial t^2} - c^2 \nabla^2 u &= 0 \quad \text{in } \Omega\text{, for } t > 0, \\ u = u_0 \text{ and } \frac{\partial u}{\partial t} &= 0 \quad \text{in } \Omega\text{, when } t = 0, \\ u &= 0 \quad \text{on } \partial\Omega\text{, for } t > 0. \end{aligned} \tag{56}$$

This problem is a 2D version of the vibrating string problem. The constant c^2 depends on how tightly the membrane is stretched and on the density of the membrane.

Reduction to a sequence of ODEs

Let λ_1, λ_2, $\ldots$ and ϕ_1, ϕ_2, $\ldots$ be the eigenvalues and eigenfunctions of the Laplacian,

$$-\nabla^2\phi_j = \lambda_j\phi_j \quad \text{in } \Omega, \qquad \text{with } \phi_j = 0 \text{ on } \partial\Omega,$$

then

$$u(\boldsymbol{x},t) \sim \sum_{j=1}^{\infty} u_j(t)\phi_j(\boldsymbol{x}), \qquad u_j(t) = \frac{\langle u(\cdot,t),\phi_j\rangle_\Omega}{\langle \phi_j,\phi_j\rangle_\Omega},$$

and

$$\frac{\partial^2 u}{\partial t^2} - c^2\nabla^2 u = \sum_{j=1}^{\infty}\left(\frac{d^2u_j}{dt^2} + c^2\lambda_j u_j\right)\phi_j(\boldsymbol{x}).$$

Thus, u_j satisfies a second-order ODE,

$$\frac{d^2u_j}{dt^2} + c^2\lambda_j u_j = 0, \quad u_j(0) = u_{0j} = \frac{\langle u_0,\phi_j\rangle_\Omega}{\langle \phi_j,\phi_j\rangle_\Omega}, \quad \frac{du_j}{dt}(0) = 0.$$

Normal modes of vibration

Solving the IVP for u_j, we find that

$$u_j(t) = u_{0j} \cos(\omega_j t) \quad \text{where } \omega_j = c\sqrt{\lambda_j},$$

so

$$u(\boldsymbol{x}, t) \sim \sum_{j=1}^{\infty} u_{0j} \cos(\omega_j t) \phi_j(\boldsymbol{x}). \tag{57}$$

The separable solution

$$\cos(\omega_j t) \phi_j(\boldsymbol{x})$$

is the jth normal mode of vibration. The period (τ_j) and frequency $(\frac{1}{\tau_j})$of this mode are

$$\tau_j = \frac{2\pi}{\omega_j} \qquad \text{and} \qquad \frac{1}{\tau_j} = \frac{\omega_j}{2\pi}.$$

Existence and uniqueness

If we confirm existence and uniqueness of solutions to wave (and heat) IBVPs then a solution derived using separation of variables will represent *the* solution to a problem.

Existence of solutions to (56) is guaranteed for square integrable initial conditions u_0 since ϕ_1, ϕ_2, ... form a complete set. That is, solutions of the form (57) meet all the initial and boundary conditions.

For uniqueness assume solutions u_1 and u_2 to (56) and define $U = u_1 - u_2$. Clearly U satisfies (56) with $U(\boldsymbol{x}, 0) = 0$. Now define

$$E(t) = \frac{1}{2}\int_\Omega c^2\|U\| + (U_t)^2\,dV.$$

Taking the derivative of E in time we have

$$\frac{dE}{dt} = \int_\Omega c^2 \nabla U_t \cdot \nabla(U) + U_t\, U_{tt}\, dV.$$

Exploiting the identity $\nabla \cdot (\phi \nabla \psi) = \nabla\phi \cdot \nabla\psi + \phi \nabla^2 \psi$ we find

$$\begin{aligned}\frac{dE}{dt} &= c^2 \int_\Omega \nabla \cdot (U_t \nabla U) - c^2 U_t \nabla^2 U + U_t\, U_{tt}\, dV \\ &= c^2 \oint_{\partial\Omega} U_t \nabla U\, dA + \int_\Omega U_t (U_{tt} - c^2 \nabla^2 U)\, dV.\end{aligned}$$

On $\partial\Omega$ we have $U = 0 = U_t$ so the first term is zero, as is the second term. Since E is constant and $E(0) = 0$ this implies $E = 0$ for all t and hence $U = 0$ and the solution is unique.

Heat equation

Our second example of an application of Fourier series to an initial-boundary value problem is the application that led, in 1807, to Fourier's original paper that introduced the idea of expanding an arbitrary function in a trigonometric series.

Applications of the heat equation and diffusion in general are ubiquitous in chemistry, biology, finance, optimisation, numerical methods and a plethora of other fields.

Heat conduction in 1D

Let

$$u(x,t) = \text{temperature at position } x \text{ and time } t.$$

Here, we assume that the temperature does not vary in the y and z directions, and that the physical parameters

$$\begin{aligned} K &= \text{thermal conductivity}, \\ \rho &= \text{mass density}, \\ \sigma &= \text{specific heat}, \end{aligned}$$

are all constant (and positive).

Fourier's law of heat conduction asserts that the heat flux q in the x-direction is given by

$$q = -K\,\frac{\partial u}{\partial x}$$

Conservation law for thermal energy

In a (fixed) region $x_1 < x < x_2$, the thermal energy per unit of cross sectional area equals

$$\int_{x_1}^{x_2} \rho\sigma u \, dx.$$

The heat flux entering the region at $x = x_1$ equals $q(x_1)$, and the heat flux entering the region at $x = x_2$ equals $-q(x_2)$.

The rate of change of thermal energy must equal the total heat flux entering the region, so

$$\frac{d}{dt}\int_{x_1}^{x_2} \rho\sigma u \, dx = q(x_1) - q(x_2),$$
$$\int_{x_1}^{x_2} \rho\sigma \frac{\partial u}{\partial t} \, dx = -\int_{x_1}^{x_2} \frac{\partial q}{\partial x} \, dx.$$

Heat equation

$$\int_{x_1}^{x_2} \left(\rho\sigma \frac{\partial u}{\partial t} + \frac{\partial q}{\partial x} \right) dx = 0.$$

By Fourier's law, $q = -K\partial u/\partial x$, so

$$\frac{1}{x_2 - x_1} \int_{x_1}^{x_2} \left(\rho\sigma \frac{\partial u}{\partial t} - K \frac{\partial^2 u}{\partial x^2} \right) dx = 0.$$

The LHS equals the average value of the integrand over the interval (x_1, x_2), which tends to the pointwise value at $x = x_1$ if we let $x_2 \to x_1$. Since the choice of x_1 is arbitrary, we conclude that u must satisfy the heat equation,

$$\rho\sigma \frac{\partial u}{\partial t} - K \frac{\partial^2 u}{\partial x^2} = 0,$$

or equivalently,

$$u_t - a u_{xx} = 0, \qquad \text{where } a = \frac{K}{\rho\sigma}.$$

Heat equation in 3D

If we let

$$
\begin{aligned}
u(\boldsymbol{x},t) &= \text{temperature}, & K &= \text{thermal conductivity},\\
\boldsymbol{q}(\boldsymbol{x},t) &= \text{heat flux vector}, & \rho &= \text{mass density},\\
f(\boldsymbol{x},t) &= \text{vol. density of heat sources}, & \sigma &= \text{specific heat},
\end{aligned}
$$

Fourier's law becomes

$$\boldsymbol{q} = -K\nabla u,$$

and conservation of thermal energy leads to the heat equation,

$$\rho\sigma\,\frac{\partial u}{\partial t} - \nabla\cdot(K\nabla u) = f(\boldsymbol{x},t).$$

An application: the cooling-off problem

A body Ω is initially at a temperature u_0, and then cools down as heat leaves through $\partial\Omega$. A (linearised) model of this process is

$$
\begin{aligned}
\frac{\partial u}{\partial t} - \nabla^2 u &= 0 && \text{in } \Omega, \text{ for } t > 0, \\
c_1 \frac{\partial u}{\partial n} + c_0 u &= 0 && \text{on } \partial\Omega, \text{ for } t > 0, \\
u &= u_0 && \text{on } \Omega, \text{ when } t = 0,
\end{aligned}
\tag{58}
$$

where $b > 0$ is a constant. The associated eigenproblem takes the form

$$
\begin{aligned}
-\nabla^2 \phi &= \lambda \phi && \text{in } \Omega, \\
c_1 \frac{\partial \phi}{\partial n} + c_0 \phi &= 0 && \text{on } \partial\Omega.
\end{aligned}
\tag{59}
$$

We have not previously discussed such a Robin boundary condition.

Eigenvalues are strictly positive

The first Green identity gives

$$\lambda \int_\Omega |\phi|^2 \, dV = \int_\Omega \|\nabla \phi\|^2 \, dV - \int_{\partial\Omega} \frac{\partial \phi}{\partial n} \overline{\phi} \, dS.$$

If $c_1 = 0$ the $\phi|_{\partial\Omega} = 0 = \overline{\phi}|_{\partial\Omega}$. Otherwise

$$\lambda \int_\Omega |\phi|^2 \, dV = \int_\Omega \|\nabla \phi\|^2 \, dV + \frac{c_0}{c_1} \int_{\partial\Omega} |\phi|^2 \, dS,$$

so $\lambda \geq 0$. Moreover, if $\lambda = 0$ then

$$\nabla \phi = 0 \text{ in } \Omega \quad \text{and} \quad \phi = 0 \text{ on } \partial\Omega,$$

implying that ϕ is constant on each component of Ω, and that each of the constant values is zero; that is, $\phi \equiv 0$. Hence, every eigenvalue is strictly positive.

Radially symmetric example

Suppose now that Ω is the unit ball $\rho < 1$, and that $u_0 = u_0(\rho)$ is radially symmetric. In spherical coordinates (Math2111) $\nabla^2 u$ equals

$$\frac{1}{\rho^2}\frac{\partial}{\partial\rho}\left(\rho^2\frac{\partial u}{\partial\rho}\right) + \frac{1}{\rho^2\sin\varphi}\frac{\partial}{\partial\varphi}\left(\sin\varphi\frac{\partial u}{\partial\varphi}\right) + \frac{1}{\rho^2\sin^2\varphi}\frac{\partial^2 u}{\partial\theta^2}$$

but from symmetry, u is independent of the angular variables φ and θ, that is, $u = u(\rho, t)$. Thus, it suffices to consider radially symmetric eigenfunctions $\phi = \phi(\rho)$ satisfying

$$-\frac{1}{\rho^2}\frac{d}{d\rho}\left(\rho^2\frac{d\phi}{d\rho}\right) = \lambda\phi \quad \text{for } 0 < \rho < 1.$$

Putting $\lambda = k^2$ with $k > 0$, we conclude that ϕ satisfies a spherical Bessel equation,

$$(\rho^2\phi')' + k^2\rho^2\phi = 0.$$

A tutorial problem shows that the function $v(\rho) = \rho^{1/2}\phi(\rho)$ satisfies the parameterised Bessel equation of order $1/2$,

$$\rho^2 v'' + \rho v' + (k^2\rho^2 - \tfrac{1}{4})v = 0,$$

so $v = AJ_{1/2}(k\rho) + BY_{1/2}(k\rho)$ and thus

$$\phi = \rho^{-1/2}\big(AJ_{1/2}(k\rho) + BY_{1/2}(k\rho)\big).$$

Here, $B = 0$ because ϕ is bounded at $\rho = 0$, and k must be chosen so that

$$c_1\frac{d\phi}{d\rho} + c_0\phi = 0 \quad \text{at } \rho = 1.$$

Recall the identity

$$\frac{1}{z}\,\frac{d}{dz}\big(z^{-\nu}J_\nu(z)\big) = -z^{-\nu-1}J_{\nu+1}(z)$$

and put $s = k\rho$, then

$$\begin{aligned}\frac{d}{d\rho}\big(\rho^{-1/2}J_{1/2}(k\rho)\big) &= \frac{ds}{d\rho}\,\frac{d}{ds}\big(k^{1/2}s^{-1/2}J_{1/2}(s)\big)\\ &= -k^{3/2}s^{-1/2}J_{3/2}(s) = -k\rho^{-1/2}J_{3/2}(k\rho),\end{aligned}$$

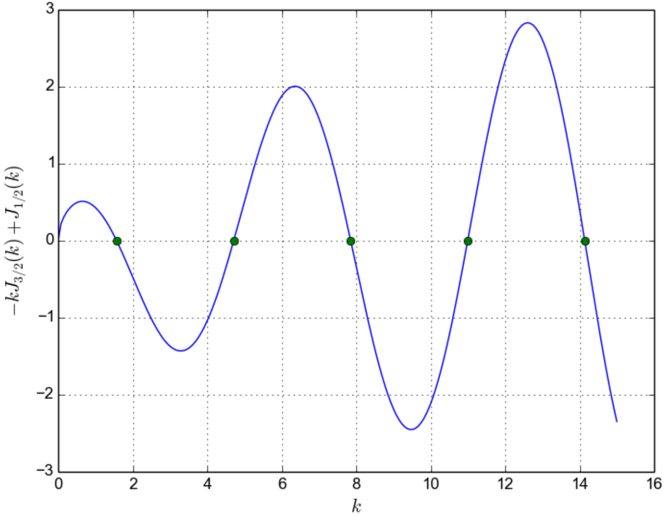
so the equation for $k > 0$ is

$$-c_1kJ_{3/2}(k) + c_0J_{1/2}(k) = 0,$$

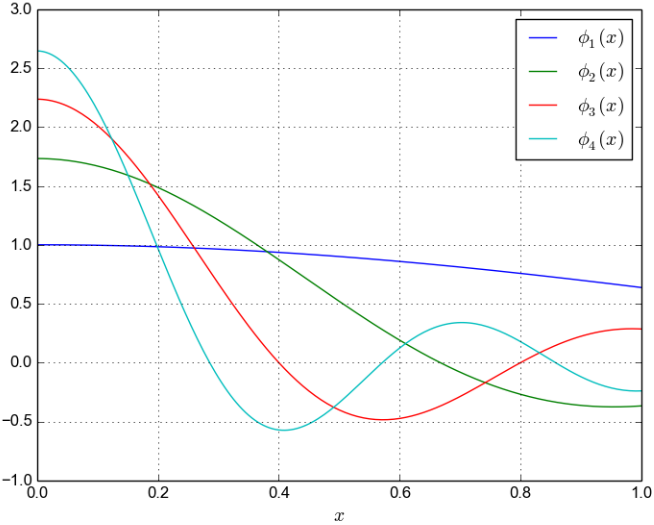
which has a sequence of solutions $0 < k_1 < k_2 < k_3 < \cdots$ tending to ∞, and corresponding eigenfunctions

$$\phi_j(\rho) = \rho^{-1/2}J_{1/2}(k_j\rho).$$

Solutions when $c_1 = 1 = c_0$.



Eigenfunctions



Orthogonality

Put

$$\langle f, g \rangle = \int_0^1 f(\rho) g(\rho)\, \rho^2\, d\rho$$

then

$$\langle \phi_n, \phi_j \rangle = 0 \quad \text{if } n \neq j,$$

since $u = \phi_n$ and $\lambda = k_n^2$ satisfy $(\rho^2 u')' + \lambda \rho^2 u = 0$. Note that $dV = 4\pi\rho^2\, d\rho$ so if we view ϕ_n and ϕ_j as functions on Ω, then

$$\int_\Omega \phi_n \phi_j\, dV = 4\pi \int_0^1 \phi_n \phi_j \rho^2\, d\rho = 0 \quad \text{if } n \neq j.$$

Exercise: prove this orthogonality property using the second Green identity and (59).

Fourier modes

The Fourier expansion

$$u(\rho, t) \sim \sum_{j=1}^{\infty} u_j(t)\phi_j(\rho), \qquad u_j(t) = \frac{\langle u(\rho, t), \phi_j\rangle}{\|\phi_j\|^2},$$

gives

$$u_t - \nabla^2 u \sim \sum_{j=1}^{\infty} (\dot{u}_j + \lambda_j u_j)\phi_j(\rho),$$

so we require that

$$\dot{u}_j + \lambda_j u_j = 0 \quad \text{for } t > 0, \quad \text{with} \quad u_j(0) = u_{0j}.$$

Thus,

$$u_j(t) = u_{0j} e^{-\lambda_j t}.$$

The series

$$u(\rho, t) \sim \sum_{j=1}^{\infty} u_{0j} e^{-\lambda_j t} \phi_j(\rho), \quad 0 < \rho < 1, \quad t > 0,$$

satisfies the PDE and boundary condition. The initial condition gives

$$u_0(\rho) = u(\rho, 0) \sim \sum_{j=1}^{\infty} u_{0j} \phi_j(\rho),$$

so u_{0j} must be the jth Fourier coefficient of the initial data:

$$u_{0j} = \frac{\langle u_0, \phi_j \rangle}{\langle \phi_j, \phi_j \rangle} = \frac{\int_0^1 u_0(\rho) \phi_j(\rho) \rho^2 \, d\rho}{\int_0^1 \phi_j(\rho)^2 \rho^2 \, d\rho}.$$

As expected, $u(\rho, t) \to 0$ as $t \to \infty$.